

Random Forest Feature Importance — Soil Drying Time

Random Forest Feature Importance Analysis
Soil Drying Time Drivers

Event groups: BSC_00XX_Q34, IRRIGATION, IRRIGATION_STRONG

BSC_00XX_Q34 selection:

- BSC bits 1-2 = '00' (SRP area in Q1 & Q2 below USRP)
- Huff quartile 3 or 4 (peak intensity in 2nd half of event)
 - Concentrated late-event input = discrete wetting pulse with clean drying tail and no masking by early rain

Depths: -10 cm / -30 cm / -45 cm

Target: dry_h- $\{depth\}$ — hours from VWC peak to pre-event level

Model: RandomForestRegressor (150 trees, 3-fold CV)

Importance: permutation importance (10 repeats, signed)

SHAP: signed beeswarm (shap available)

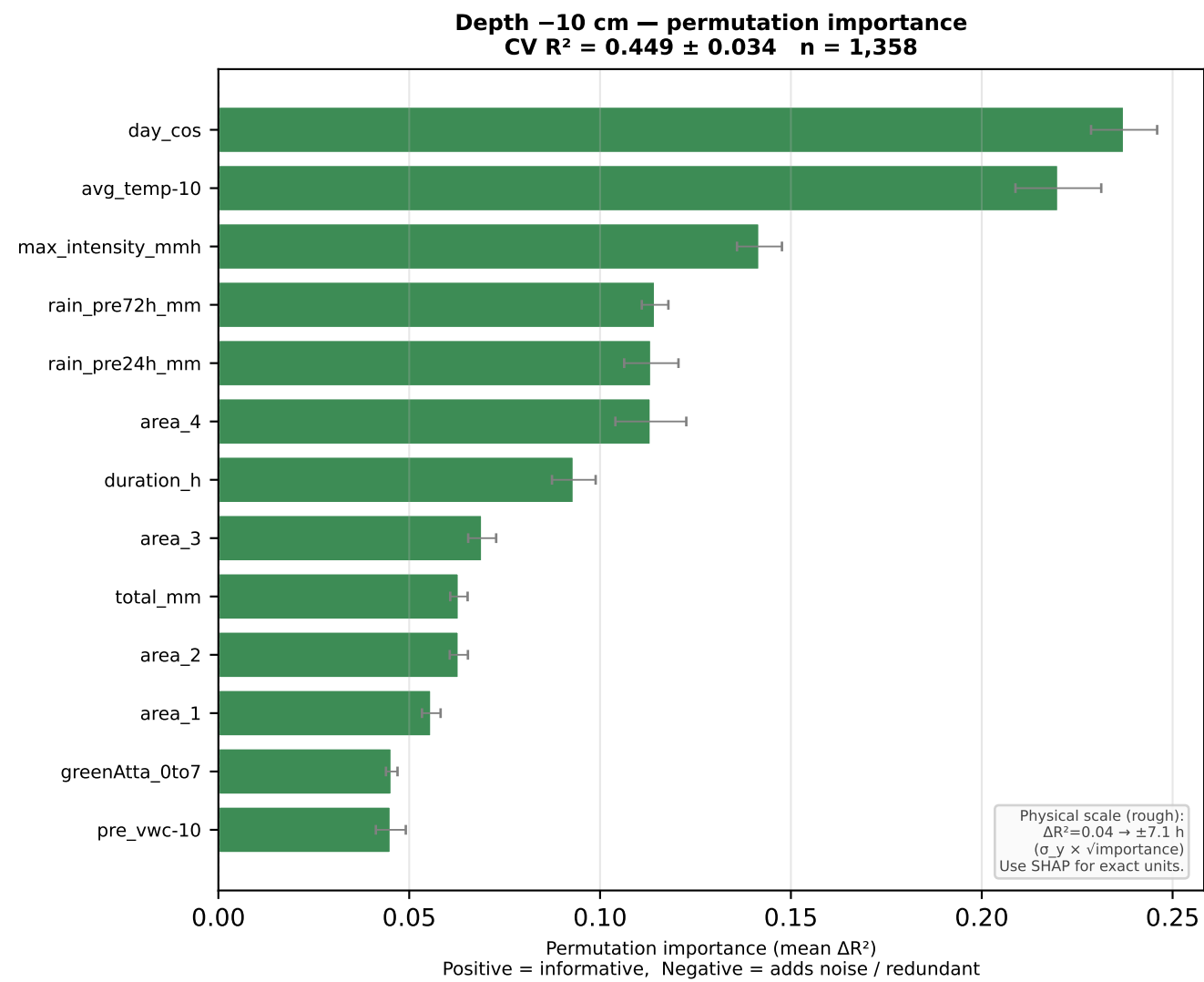
Leakage prevention:

Excluded: delta_vwc, delta_pct_dyn, peak_vwc, t_peak_min at all depths

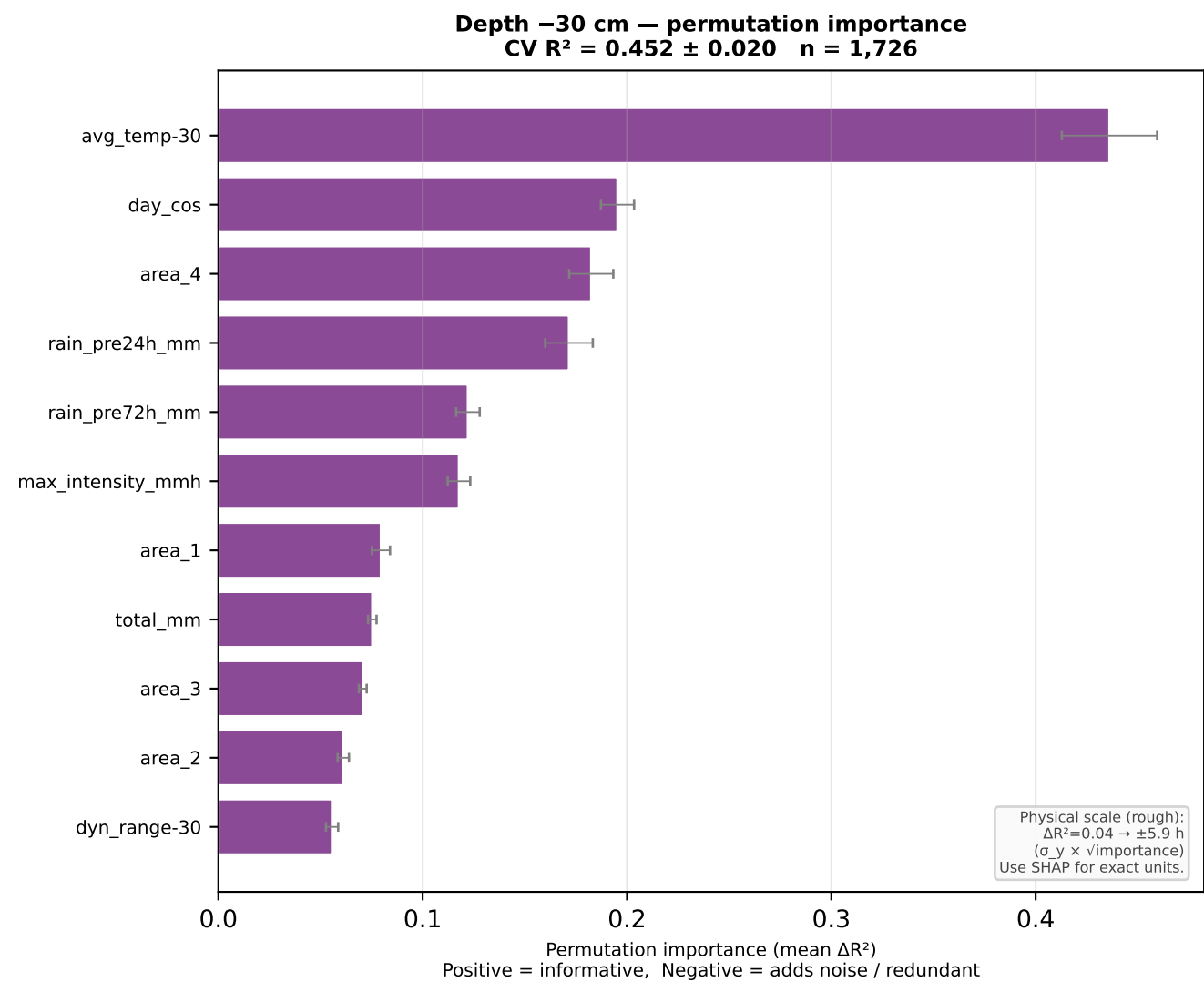
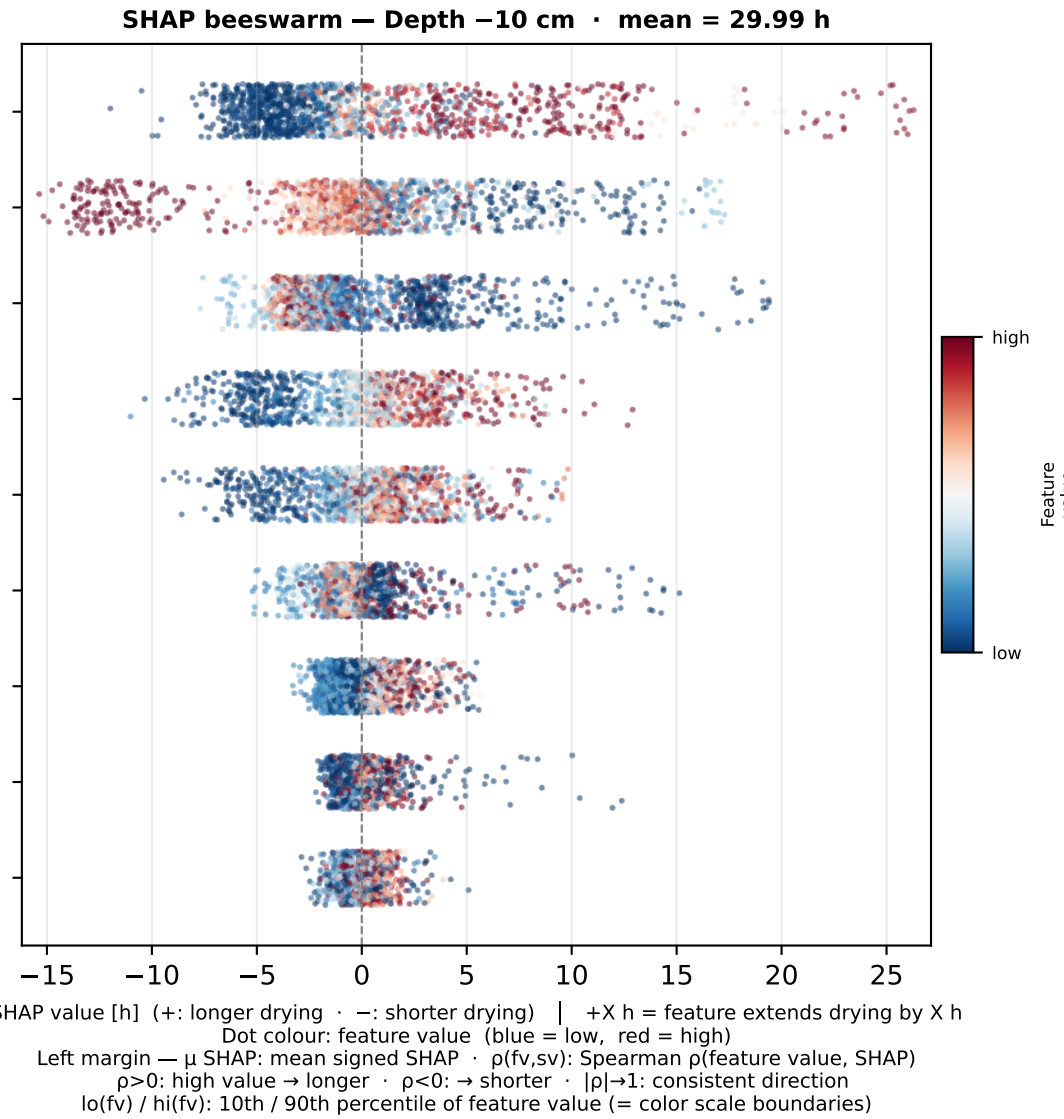
Retained: pre_vwc, dyn_range for the target depth only

Excluded: vwc_p05/p95 (redundant with dyn_range = p95 - p05)

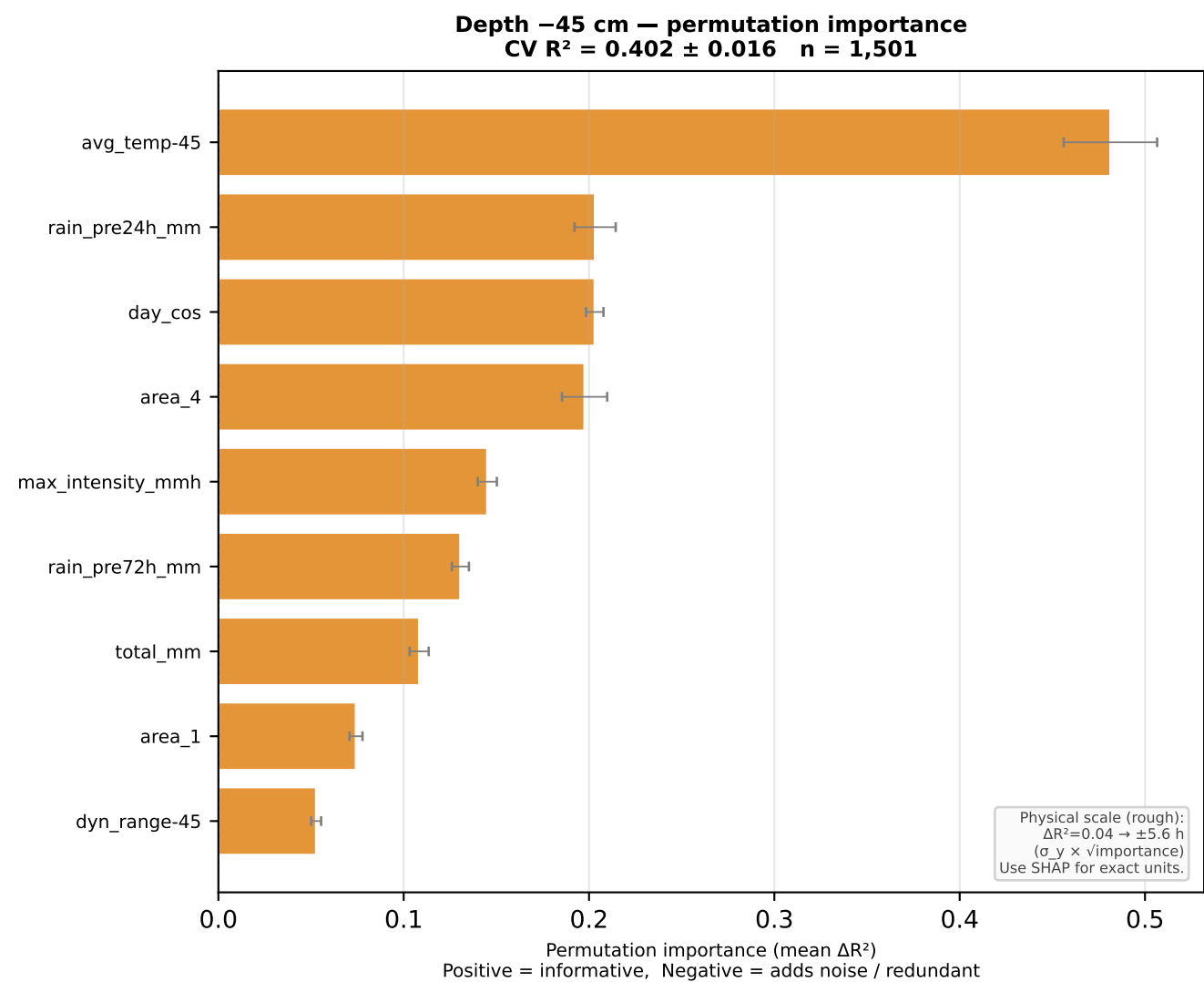
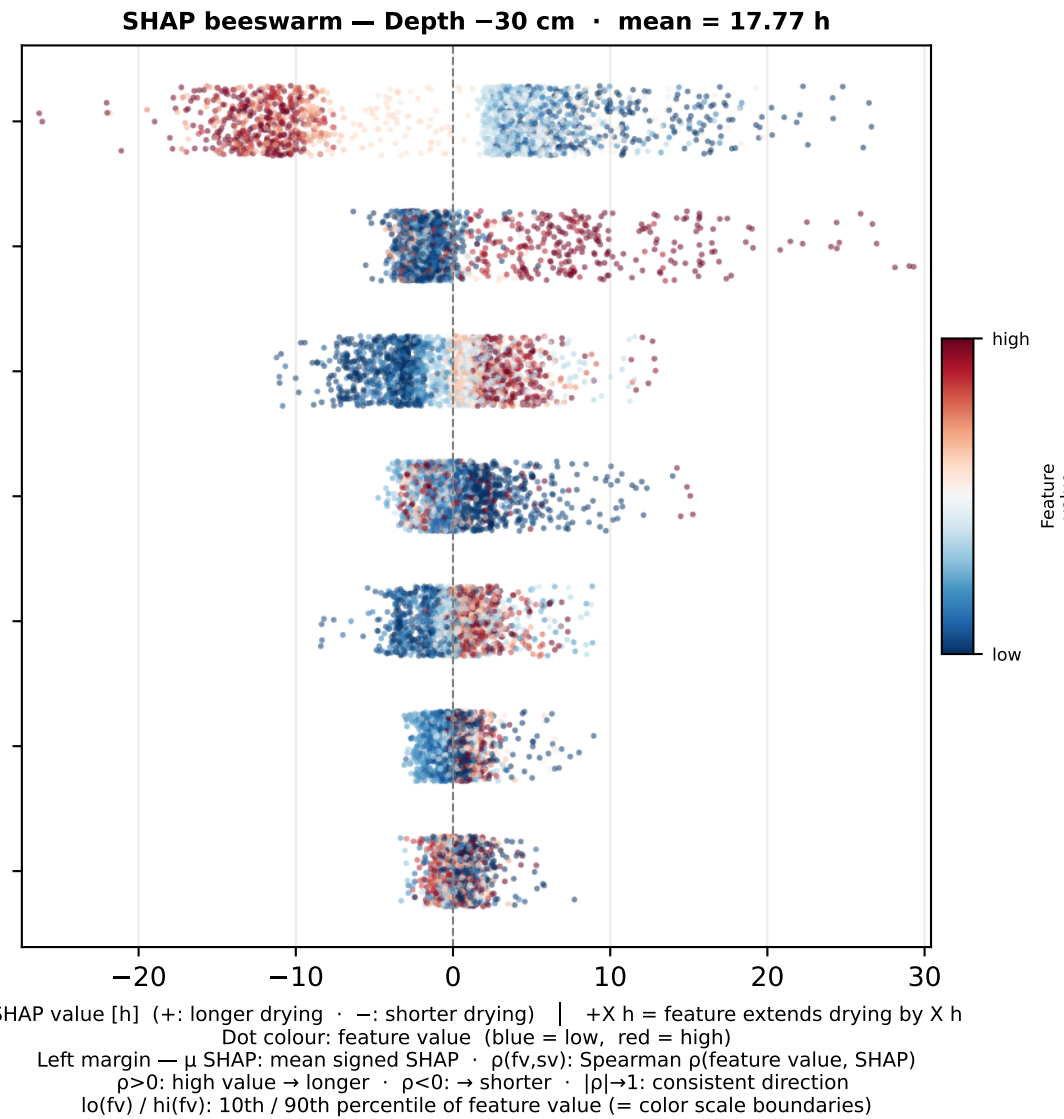
Random Forest Feature Importance — BSC 00XX + Huff Q3/Q4 (back-loaded rain)
Target: Drying time (h) at each depth (5,961 events)



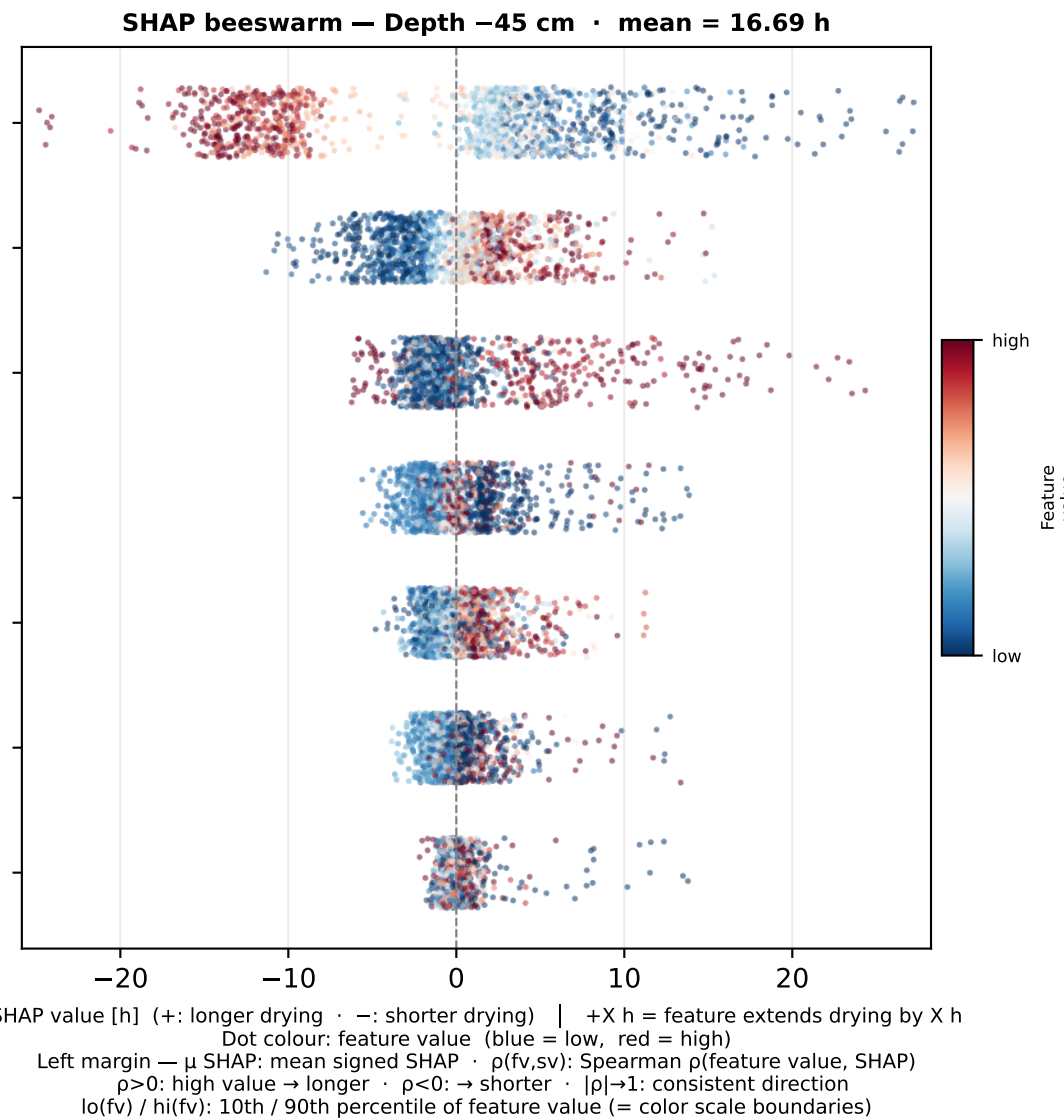
	hi(fv)	lo(fv)	ρ(fv,sv)	μ SHAP
day_cos	0.975	-0.965	+0.76	+0.291
avg_temp-10	22.4 °C	1.48 °C	-0.70	+0.0892
max_intensity_mmh	6.8 mm	1.1 mm	-0.67	-0.155
rain_pre24h_mm	15.7 mm	0.9 mm	+0.80	-0.0298
rain_pre72h_mm	25.3 mm	3.4 mm	+0.65	+0.105
duration_h	28 h	6 h	-0.13	-0.118
total_mm	23 mm	5.6 mm	+0.47	+0.0954
greenAtta_0to7	176	5.51	+0.19	+0.0202
pre_vwc-10	41	10.8	+0.31	+0.0312

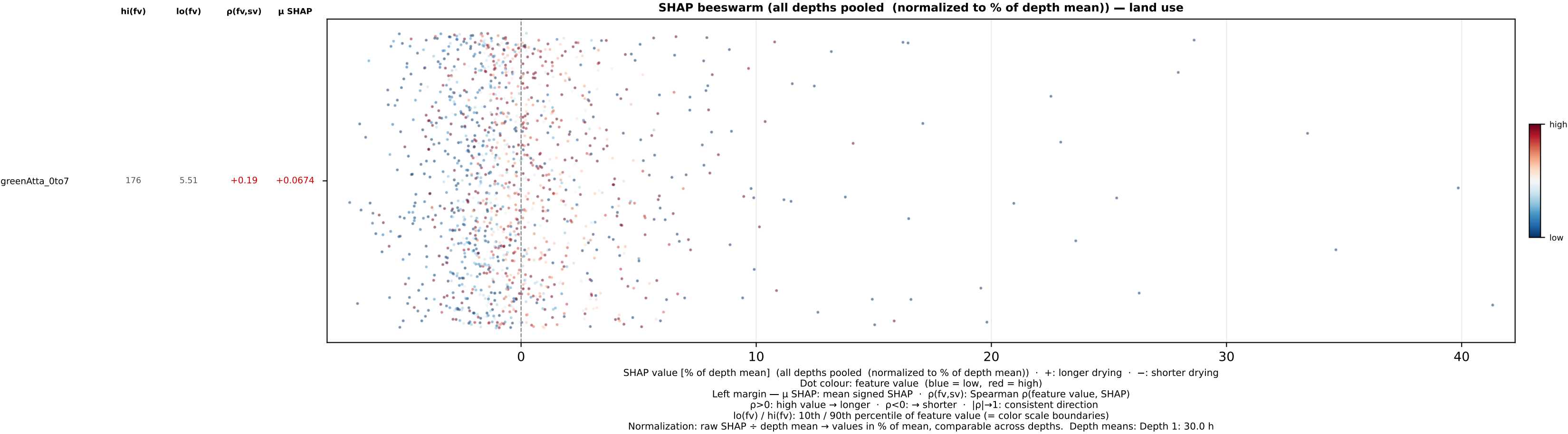
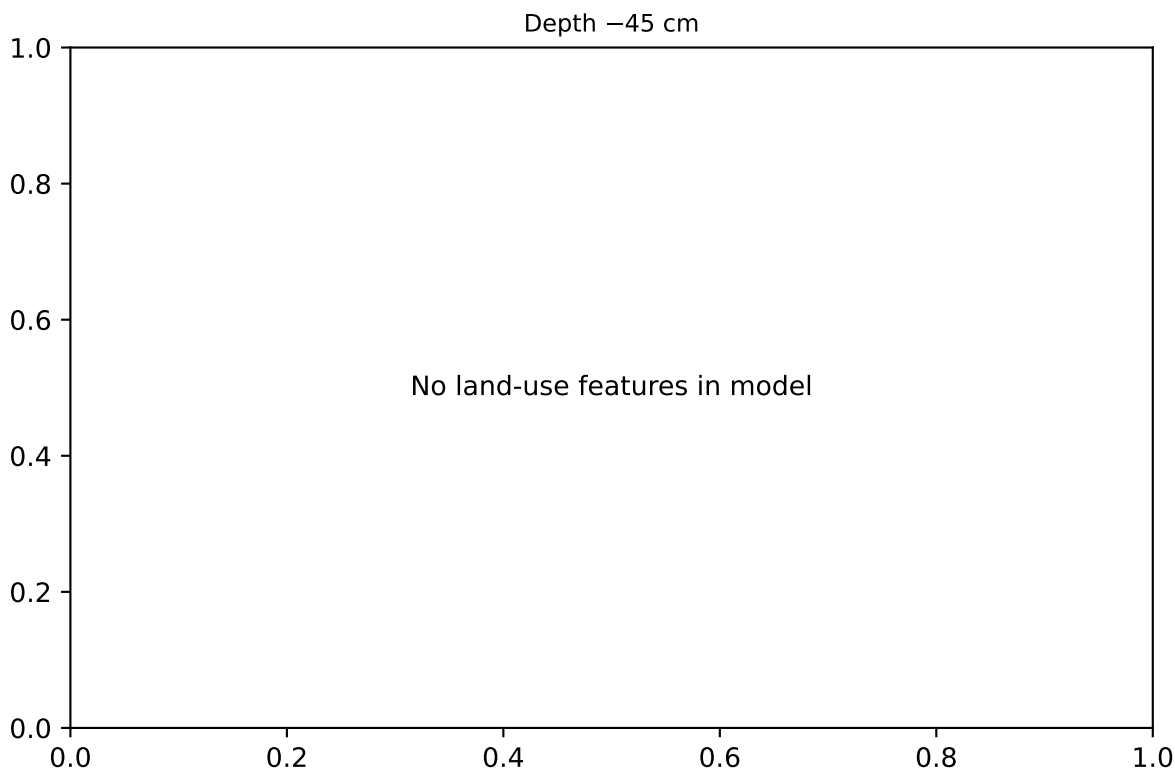
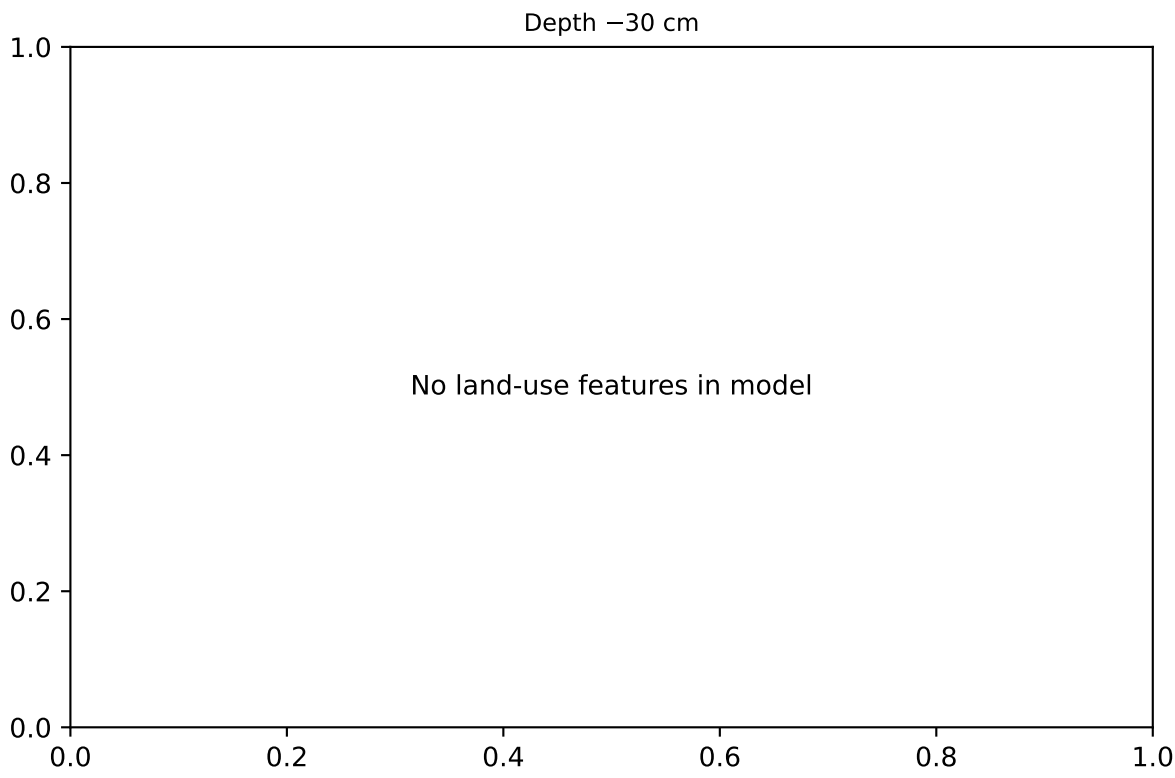
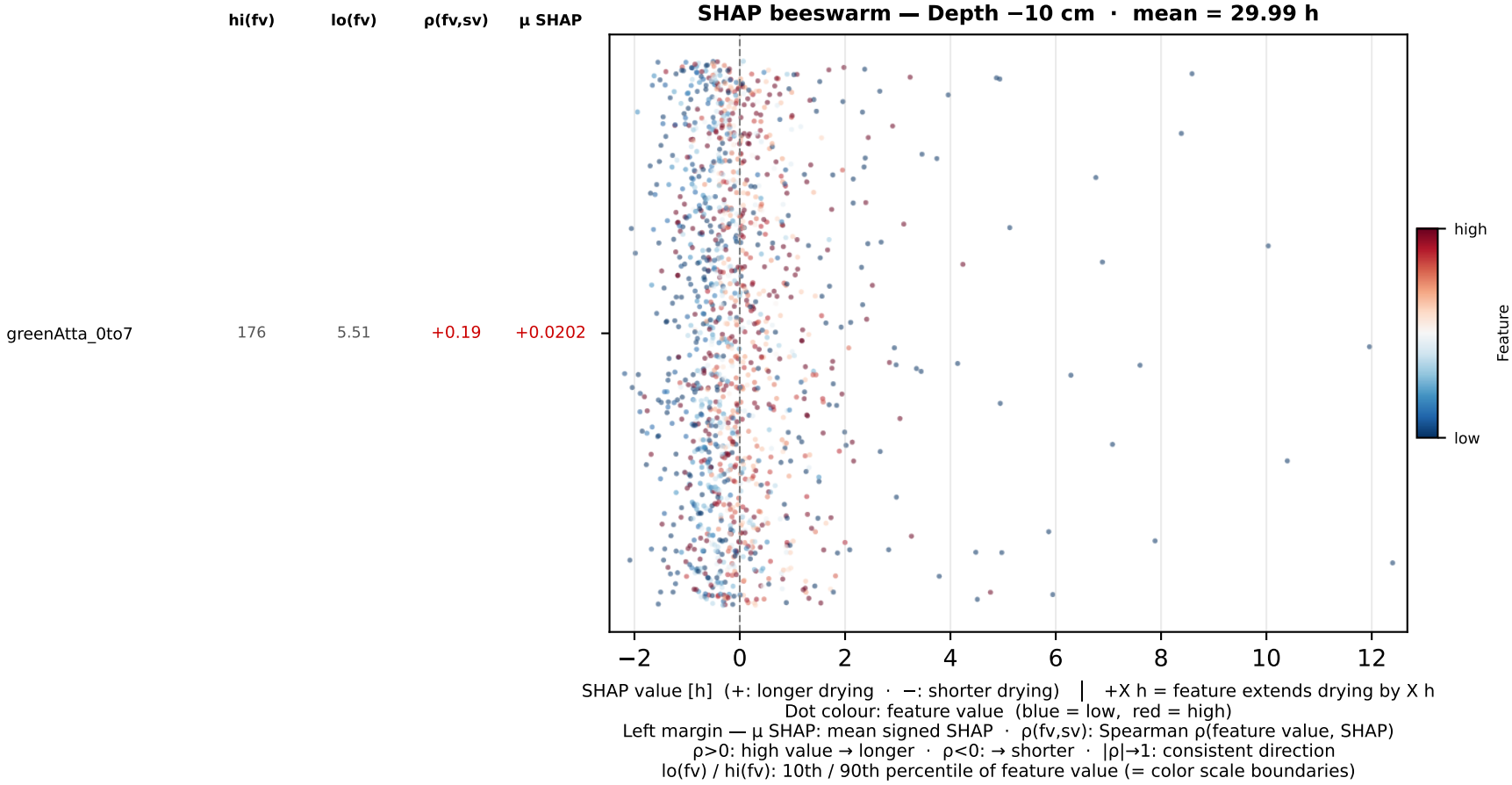
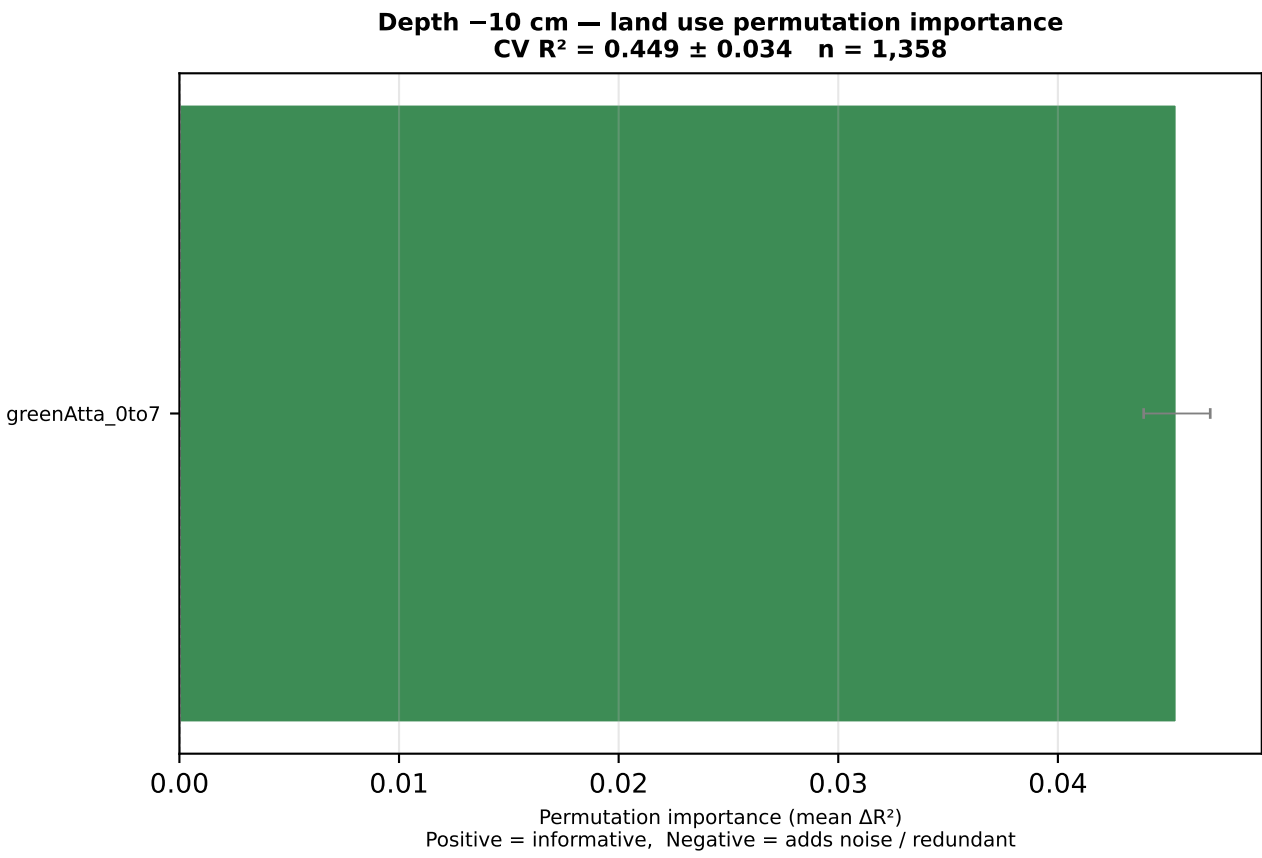


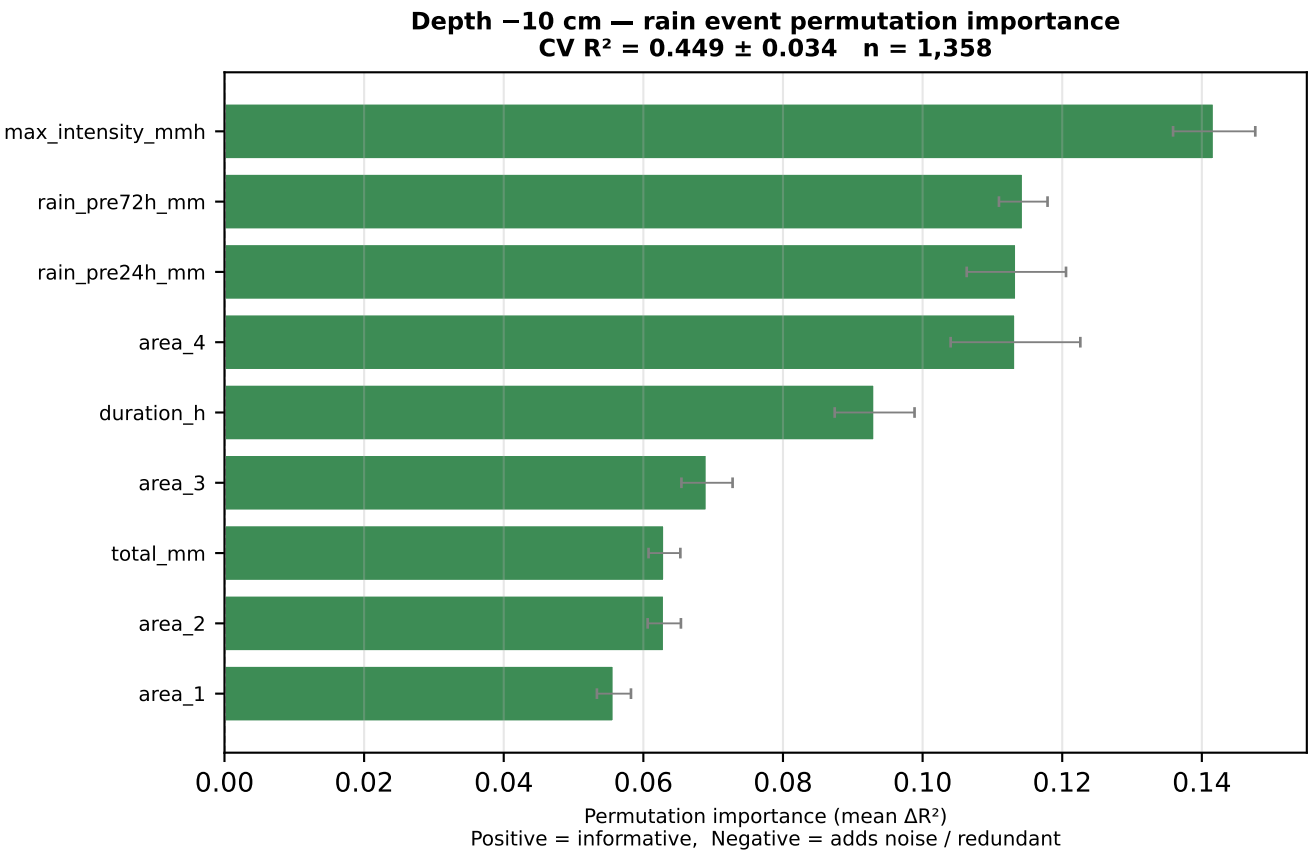
	hi(fv)	lo(fv)	ρ(fv,sv)	μ SHAP
avg_temp-30	34.4 °C	2.49 °C	-0.78	+0.222
day_cos	0.916	-0.969	+0.30	-0.193
rain_pre24h_mm	15 mm	0.4 mm	+0.82	-0.0841
max_intensity_mmh	7.3 mm	1.2 mm	-0.51	+0.148
rain_pre72h_mm	24.6 mm	2.04 mm	+0.66	+0.152
total_mm	24.2 mm	5.5 mm	+0.26	+0.0993
dyn_range-30	24.5	6.05	-0.10	+0.0221



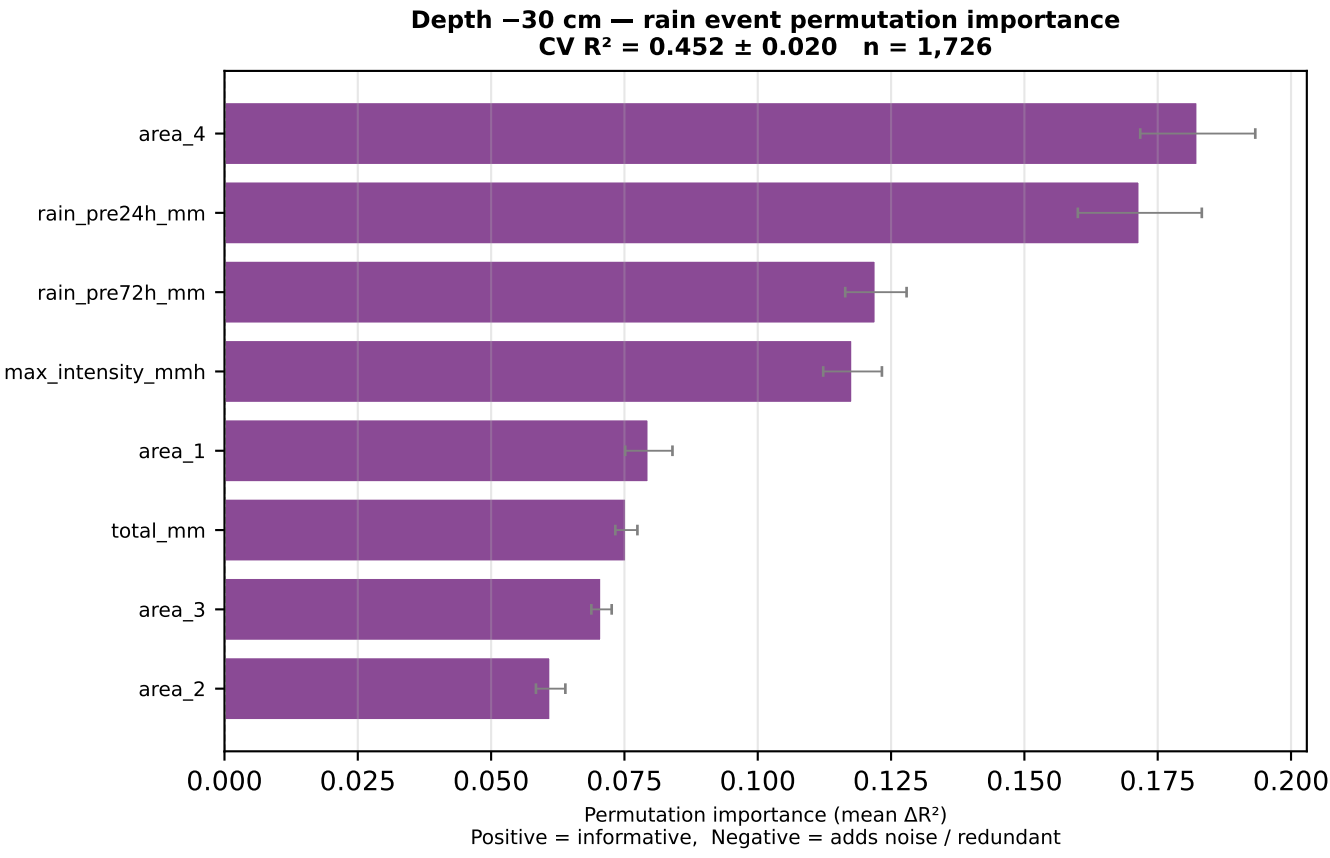
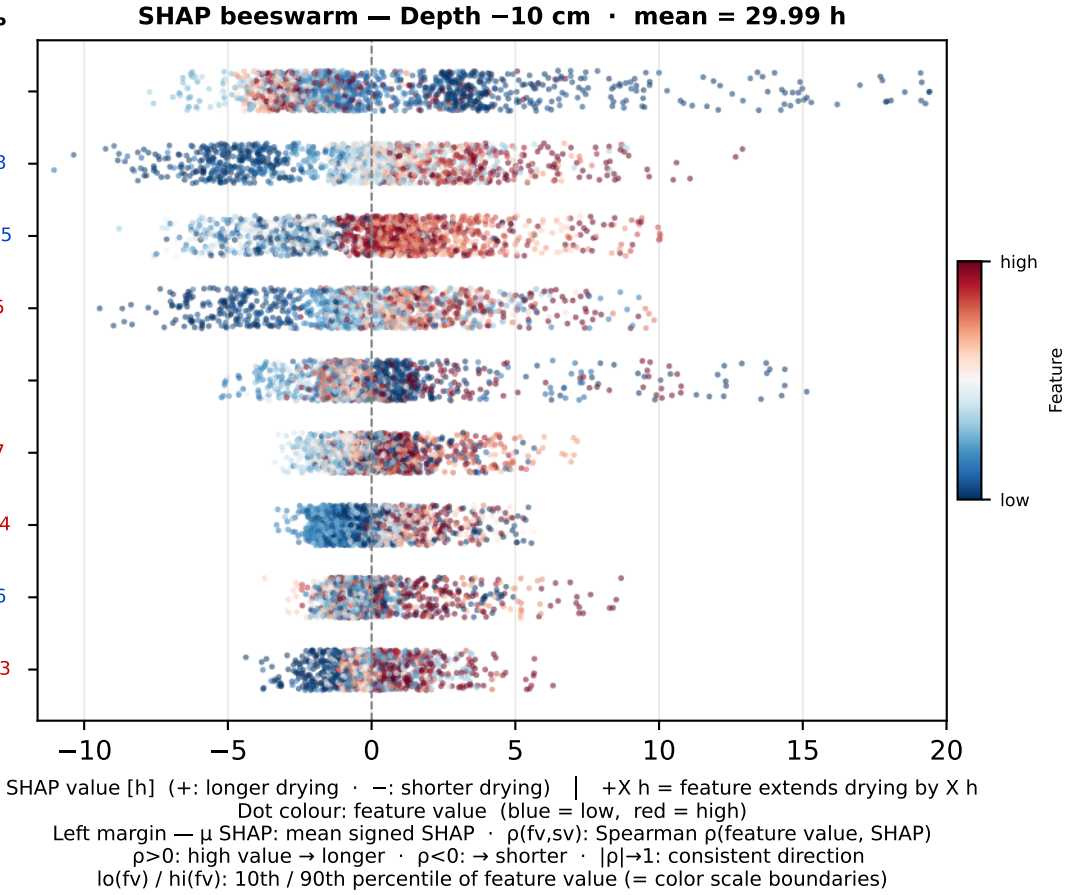
	hi(fv)	lo(fv)	ρ(fv,sv)	μ SHAP
avg_temp-45	32.6 °C	3.1 °C	-0.80	+0.0761
rain_pre24h_mm	14.9 mm	0.4 mm	+0.84	-0.091
day_cos	0.923	-0.965	+0.13	-0.0356
max_intensity_mmh	7 mm	1.2 mm	-0.14	+0.0062
rain_pre72h_mm	24.6 mm	2.4 mm	+0.60	+0.154
total_mm	23.8 mm	5.6 mm	-0.03	+0.0641
dyn_range-45	23	4.61	+0.06	+0.0245



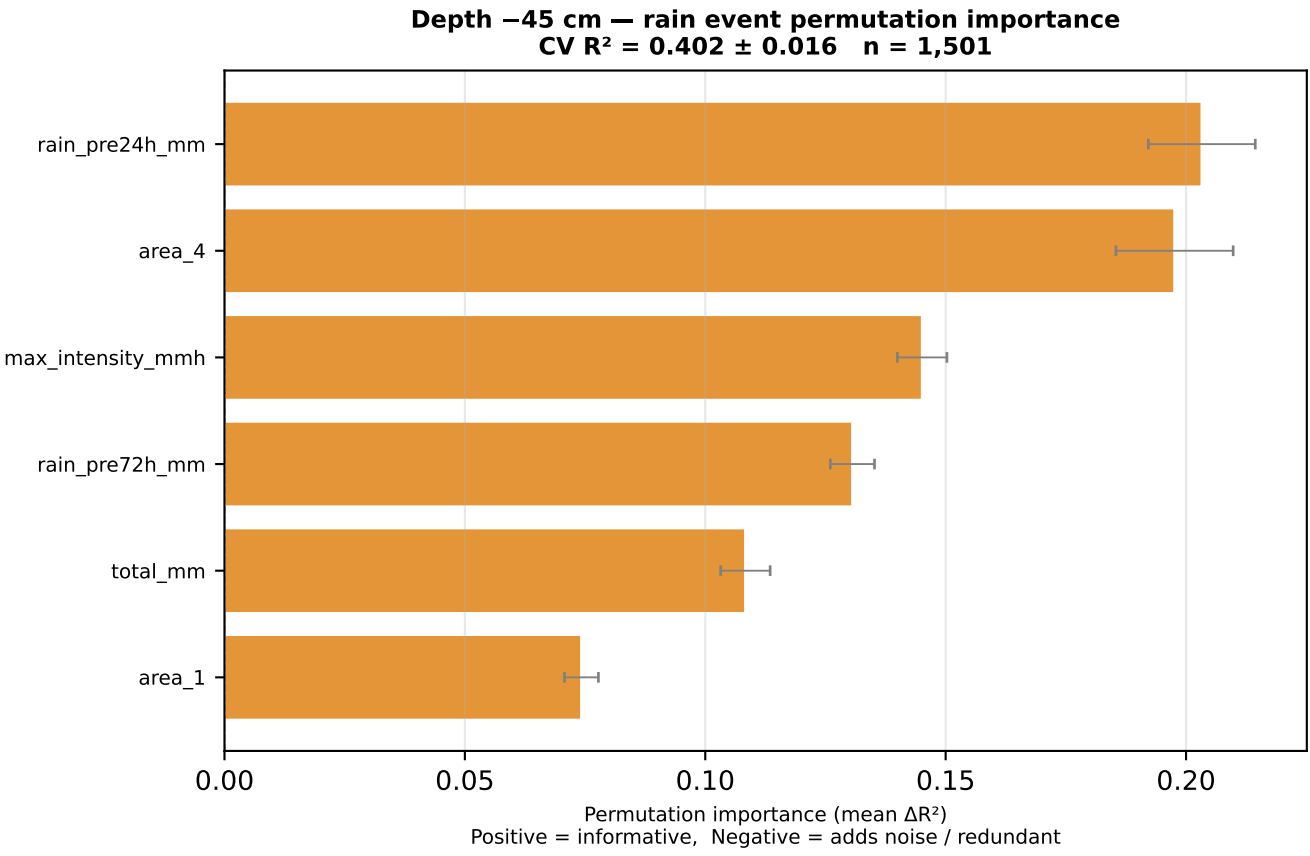
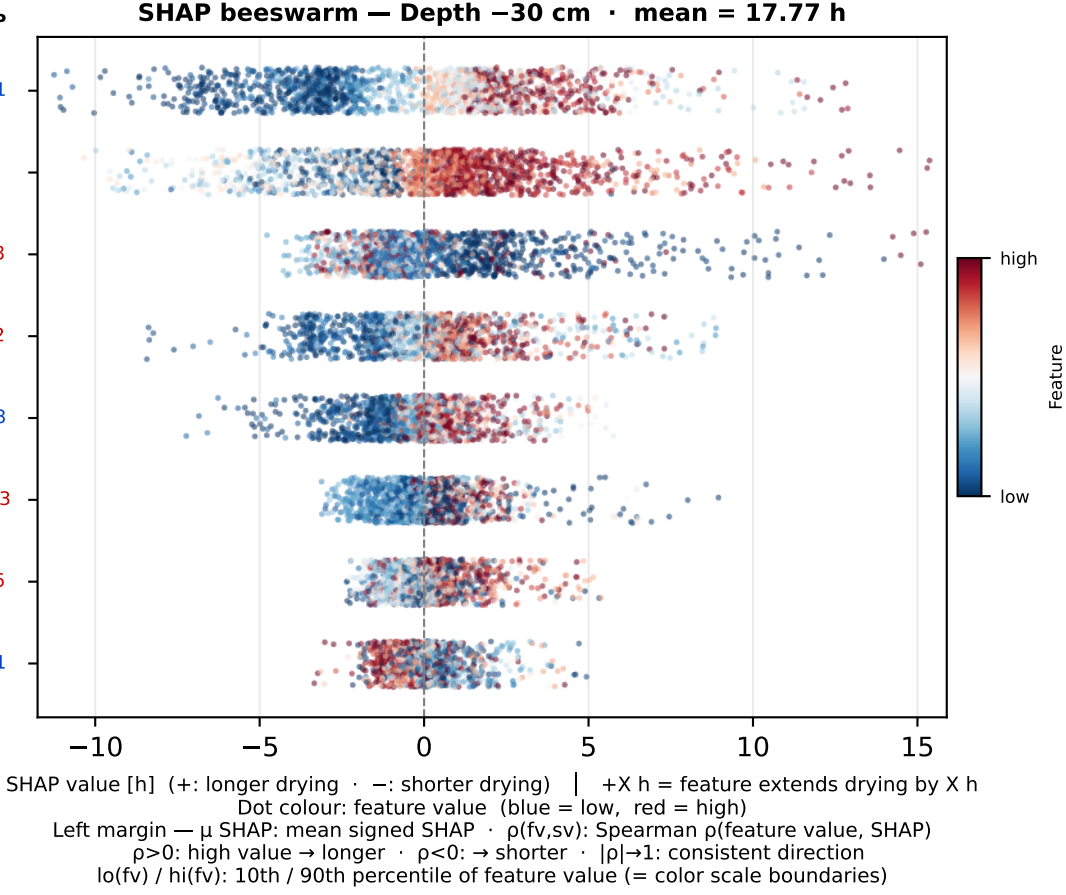




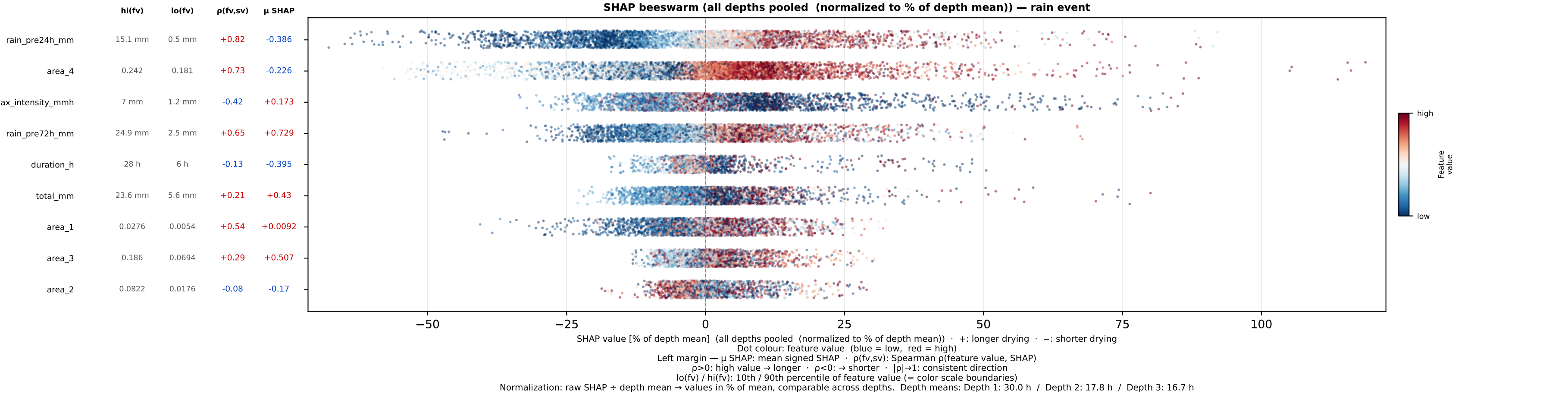
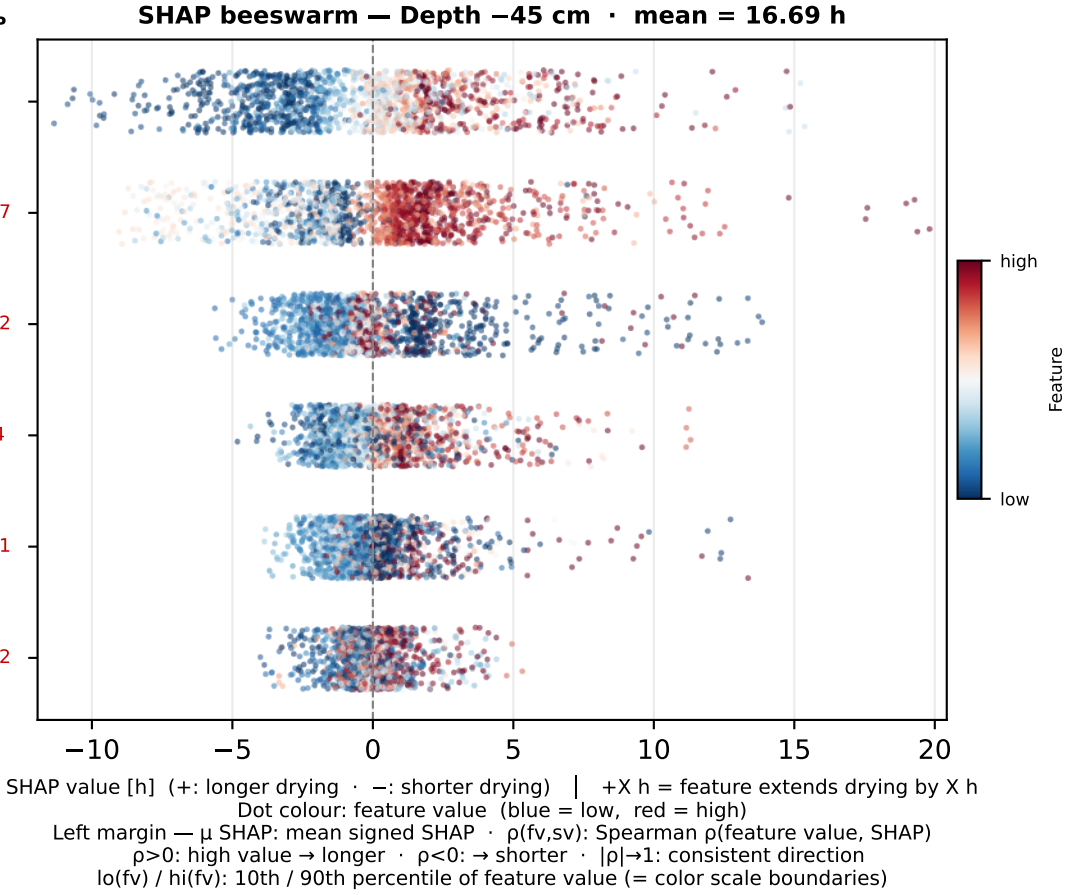
	hi(fv)	lo(fv)	ρ(fv,sv)	μ SHAP
max_intensity_mmh	6.8 mm	1.1 mm	-0.67	-0.155
rain_pre24h_mm	15.7 mm	0.9 mm	+0.80	-0.0298
area_4	0.243	0.185	+0.62	-0.00805
rain_pre72h_mm	25.3 mm	3.4 mm	+0.65	+0.105
duration_h	28 h	6 h	-0.13	-0.118
area_3	0.185	0.0708	+0.29	+0.117
total_mm	23 mm	5.6 mm	+0.47	+0.0954
area_2	0.0819	0.0176	+0.19	-0.0426
area_1	0.0272	0.00558	+0.54	+0.0613



	hi(fv)	lo(fv)	ρ(fv,sv)	μ SHAP
rain_pre24h_mm	15 mm	0.4 mm	+0.82	-0.0841
area_4	0.242	0.179	+0.77	-0.142
max_intensity_mmh	7.3 mm	1.2 mm	-0.51	+0.148
rain_pre72h_mm	24.6 mm	2.04 mm	+0.66	+0.152
area_1	0.0276	0.00541	+0.64	-0.0253
total_mm	24.2 mm	5.5 mm	+0.26	+0.0993
area_3	0.186	0.0691	+0.29	+0.106
area_2	0.0824	0.0176	-0.30	-0.0341

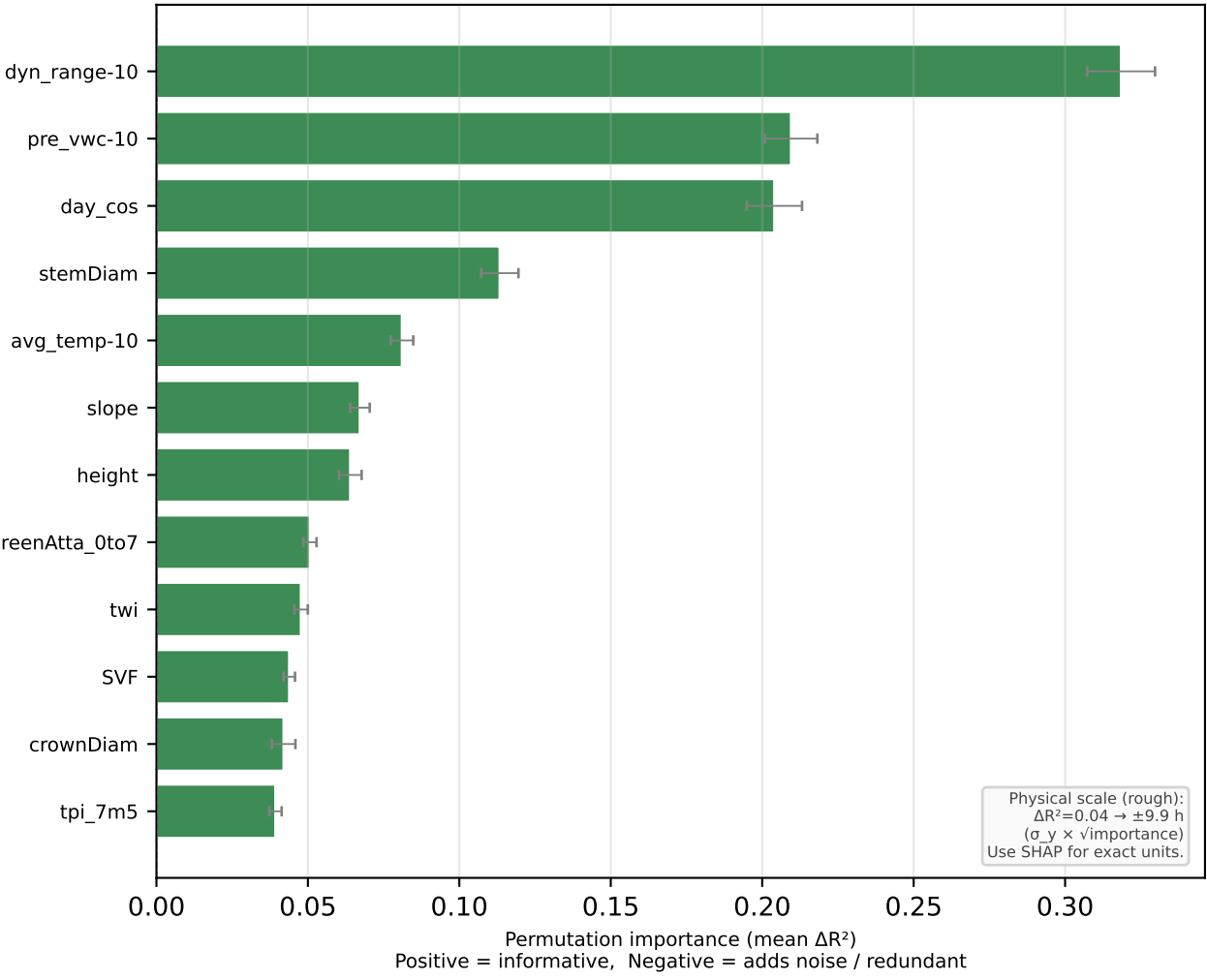


	hi(fv)	lo(fv)	ρ(fv,sv)	μ SHAP
rain_pre24h_mm	14.9 mm	0.4 mm	+0.84	-0.091
area_4	0.242	0.179	+0.73	+0.0427
max_intensity_mmh	7 mm	1.2 mm	-0.14	+0.0062
rain_pre72h_mm	24.6 mm	2.4 mm	+0.60	+0.154
total_mm	23.8 mm	5.6 mm	-0.03	+0.0641
area_1	0.0277	0.00539	+0.38	+0.0012



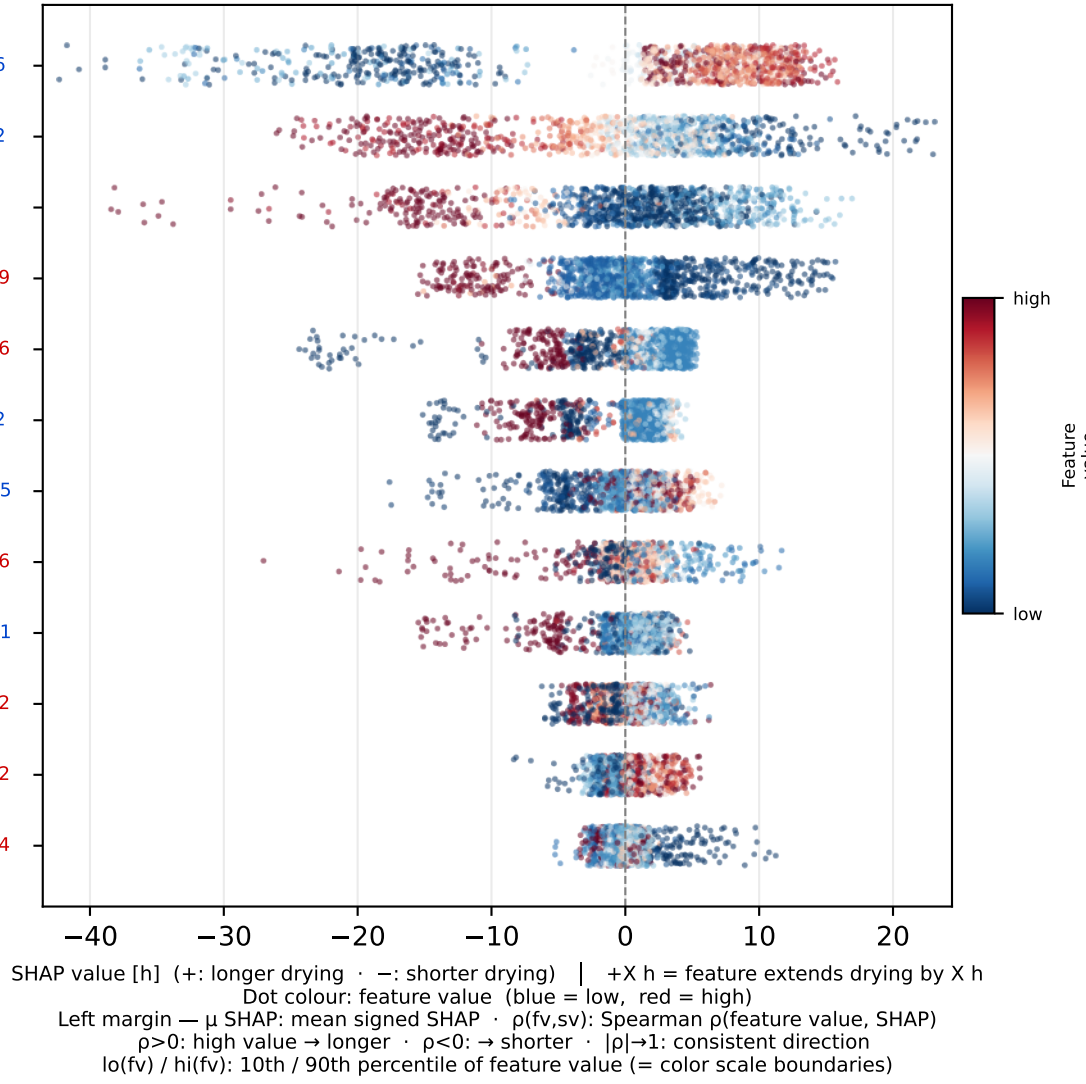
Random Forest Feature Importance — Irrigation
Target: Drying time (h) at each depth (5,242 events)

Depth -10 cm — permutation importance
CV R² = 0.514 ± 0.053 n = 1,131

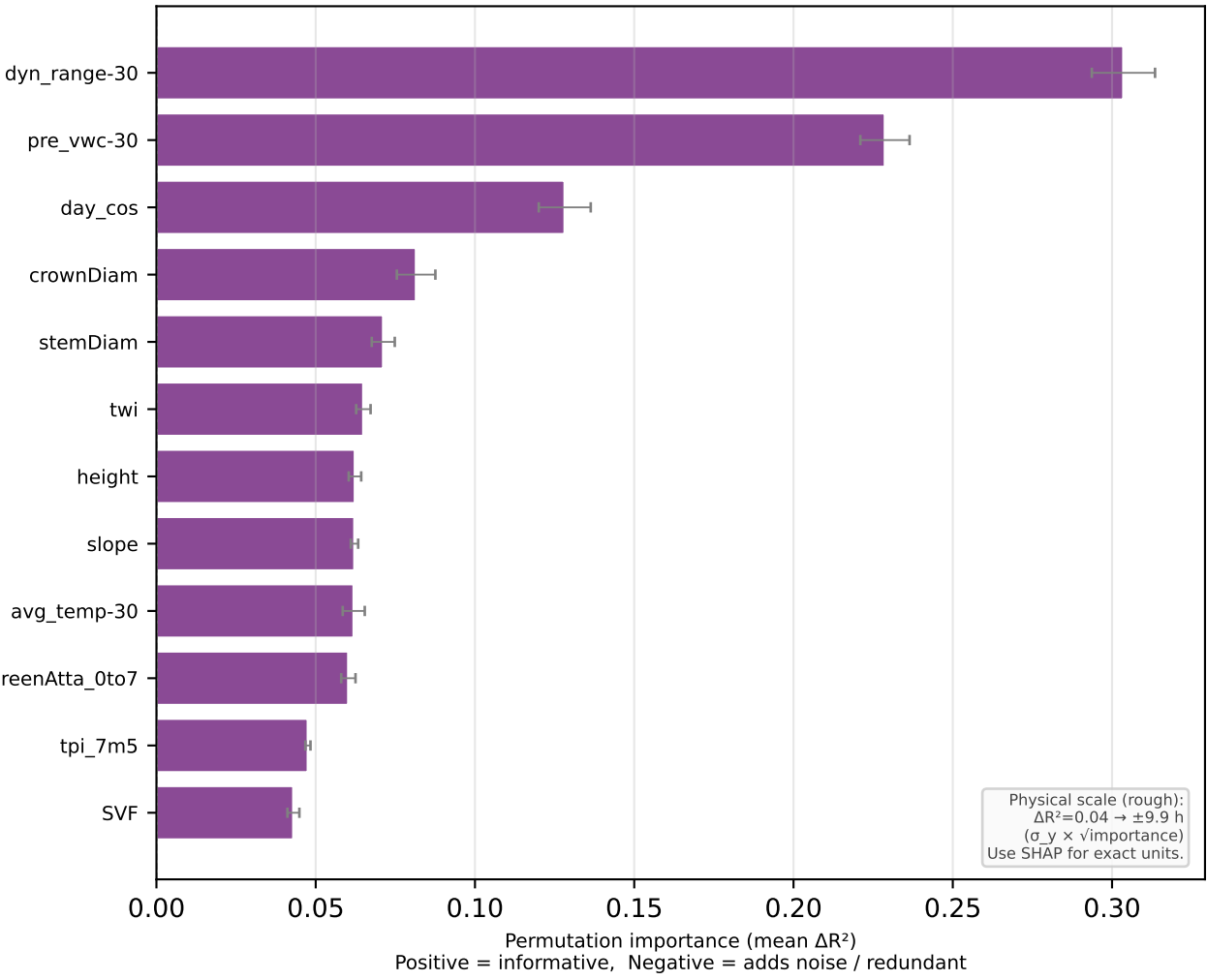


	hi(fv)	lo(fv)	p(fv,sv)	μ SHAP
dyn_range-10	23.6	6	+0.72	-0.286
pre_vwc-10	42.5	13	-0.86	-0.172
day_cos	0.467	-0.997	-0.39	-0.23
stemDiam	0.32 m	0.054 m	-0.78	+0.159
height	10 m	4 m	+0.03	+0.196
crownDiam	7 m	1 m	+0.16	-0.122
slope	5.78	0.463	+0.63	-0.0165
avg_temp-10	26.4 °C	10 °C	-0.24	+0.156
twi	6.53	2.63	-0.09	-0.0721
greenAtta_0to7	175	7.88	-0.13	+0.102
SVF	0.936	0.475	+0.52	+0.132
tpi_7m5	0.275	-0.0653	-0.25	+0.124

SHAP beeswarm — Depth -10 cm · mean = 62.90 h

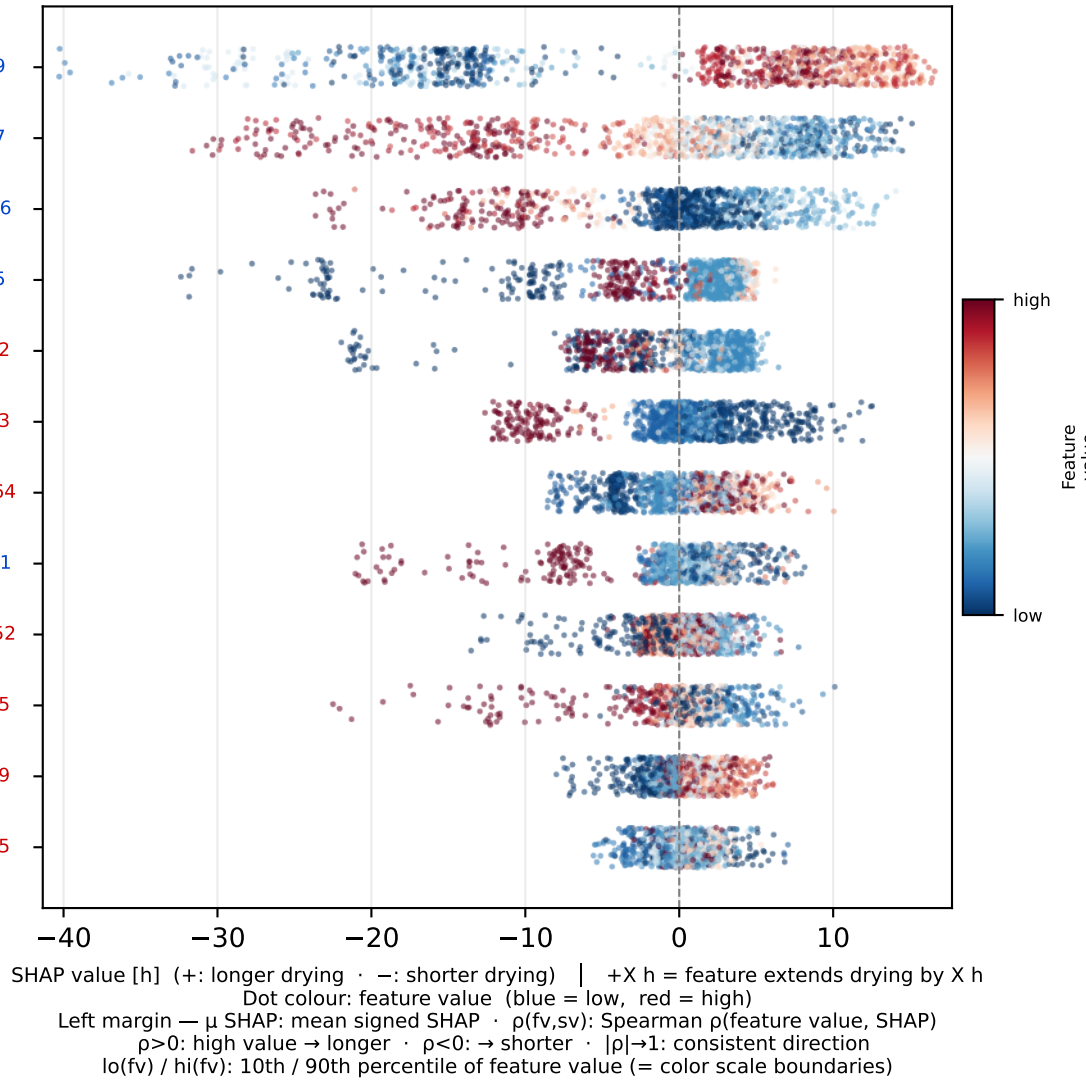


Depth -30 cm — permutation importance
CV R² = 0.526 ± 0.062 n = 1,075

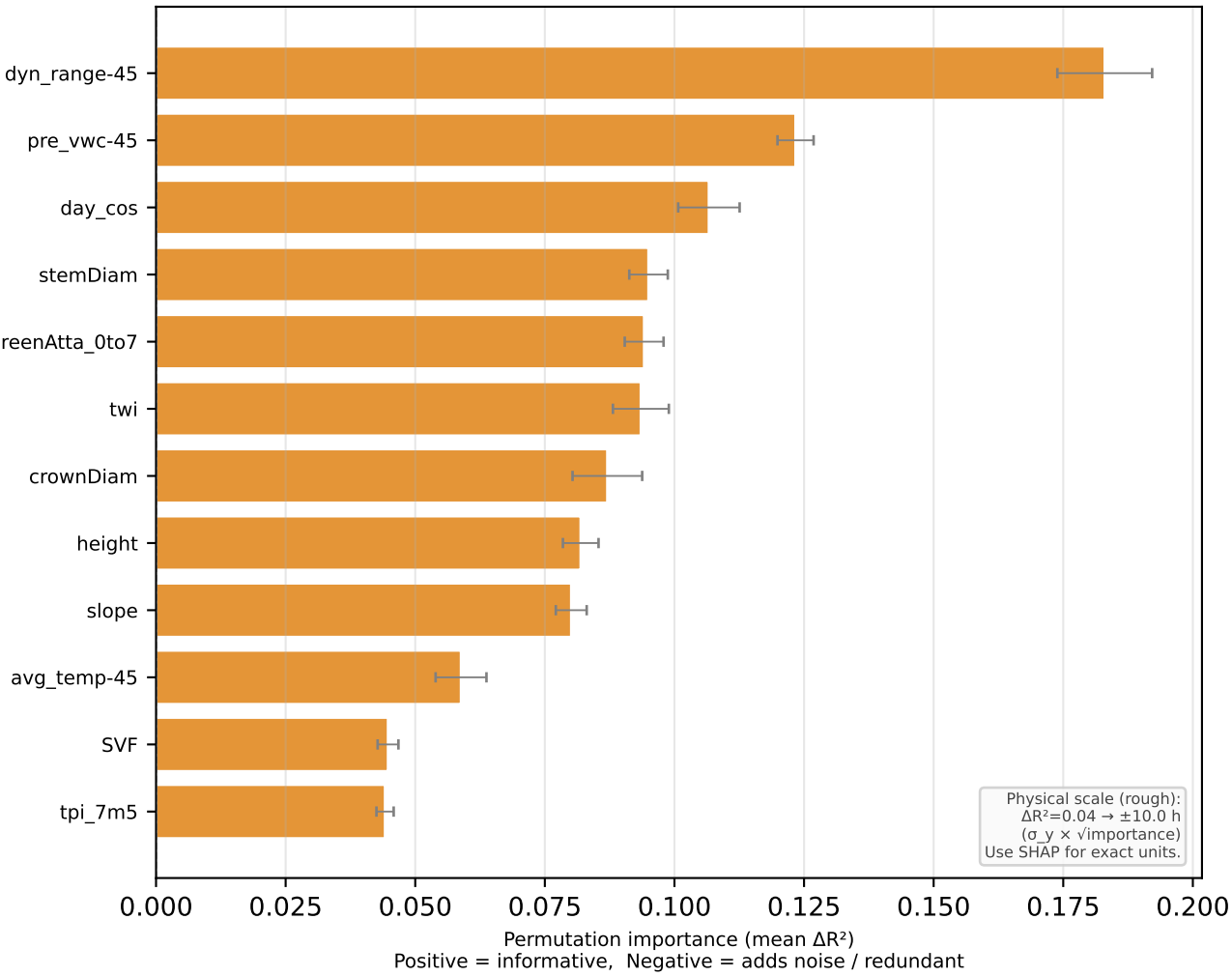


	hi(fv)	lo(fv)	p(fv,sv)	μ SHAP
dyn_range-30	21.1	4.99	+0.61	-0.539
pre_vwc-30	43.2	14.2	-0.86	-0.417
day_cos	0.498	-0.997	-0.25	-0.0906
crownDiam	6 m	1 m	+0.27	-0.175
height	9.72 m	4 m	+0.08	+0.112
stemDiam	0.32 m	0.054 m	-0.74	+0.183
slope	5.71	0.463	+0.80	+0.0764
twi	6.53	2.63	-0.29	-0.0471
greenAtta_0to7	175	7.76	+0.05	+0.0552
avg_temp-30	26.8 °C	10.4 °C	-0.60	+0.285
SVF	0.932	0.475	+0.66	+0.139
tpi_7m5	0.275	-0.062	+0.11	+0.165

SHAP beeswarm — Depth -30 cm · mean = 62.37 h

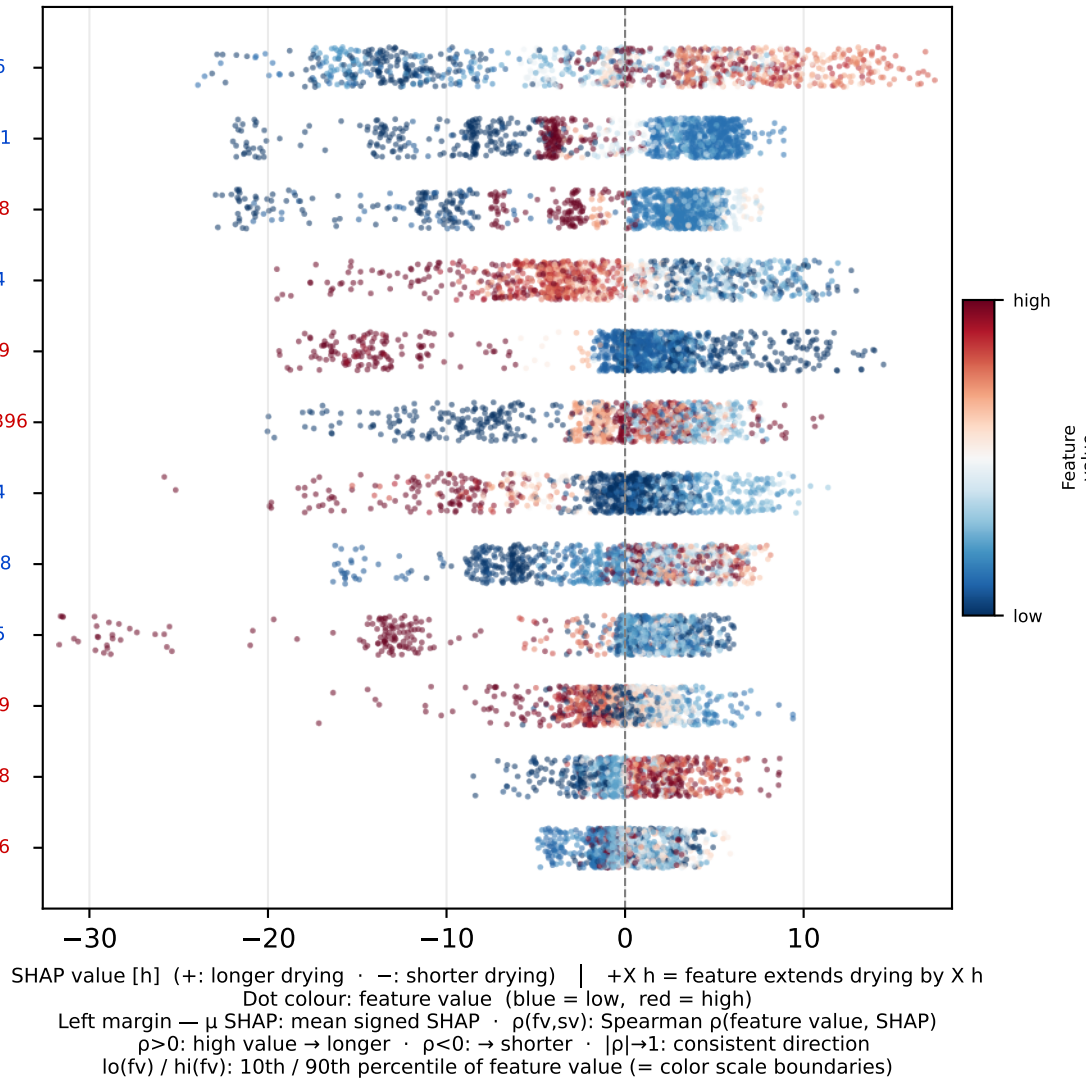


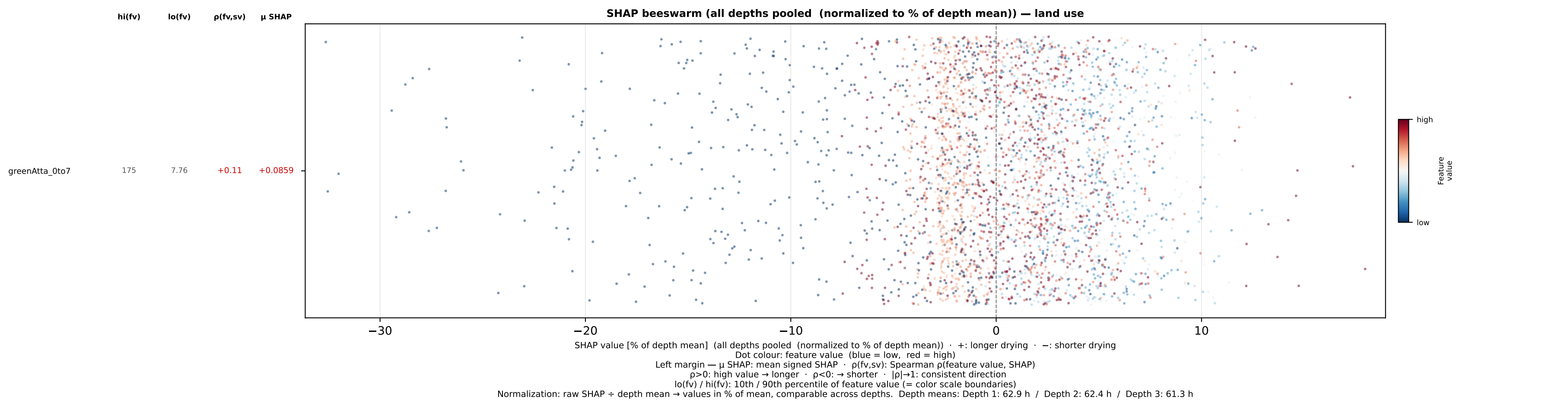
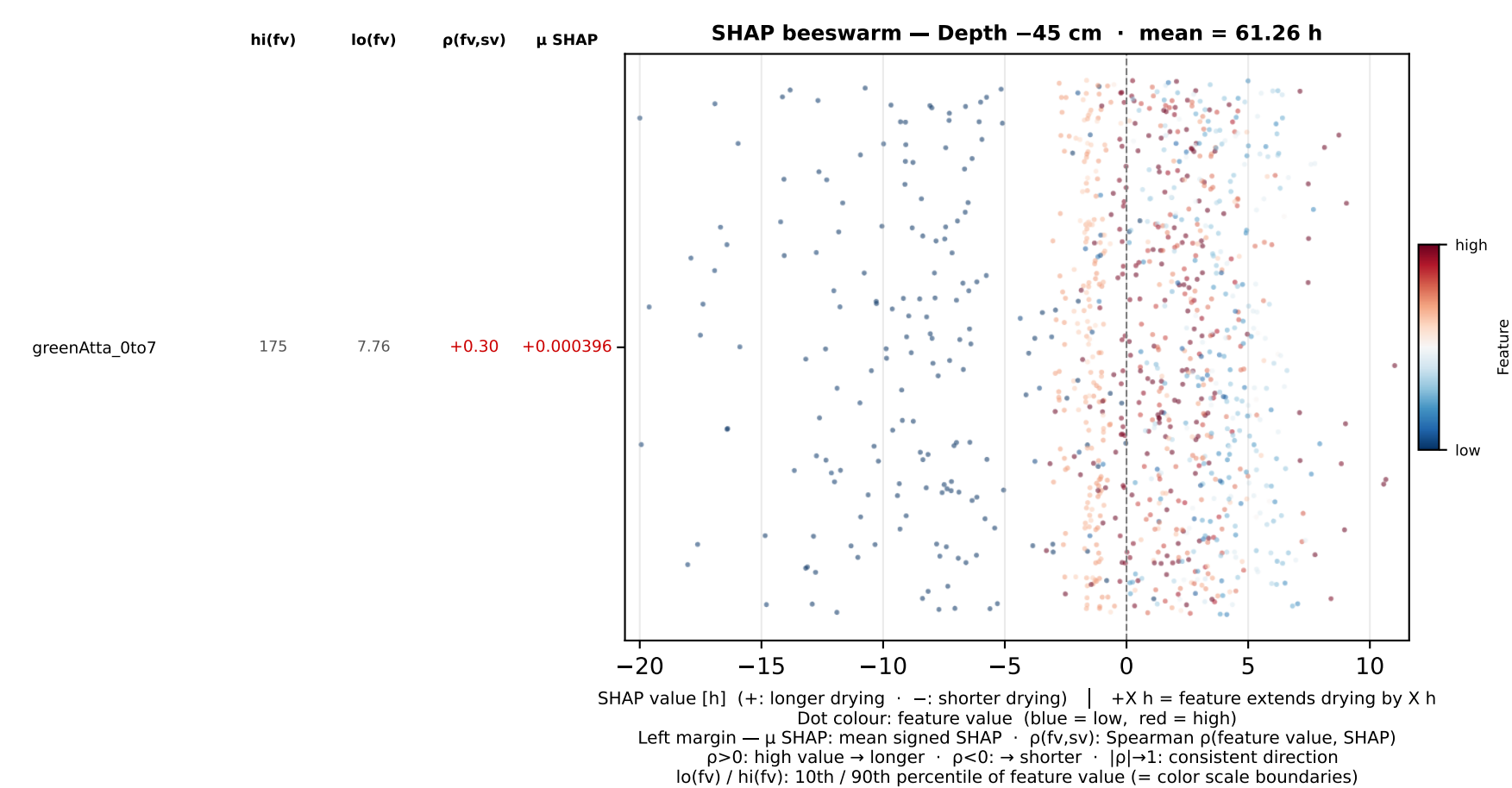
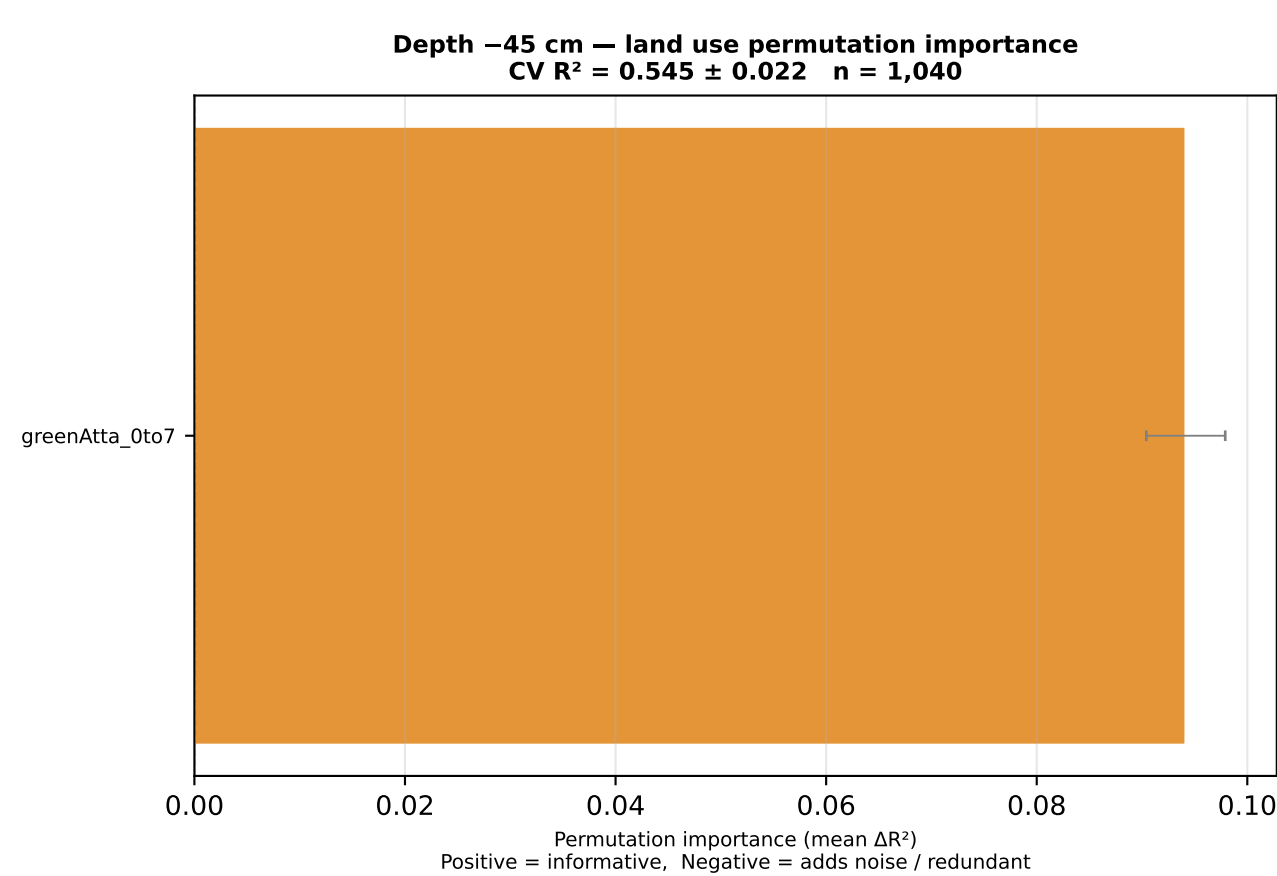
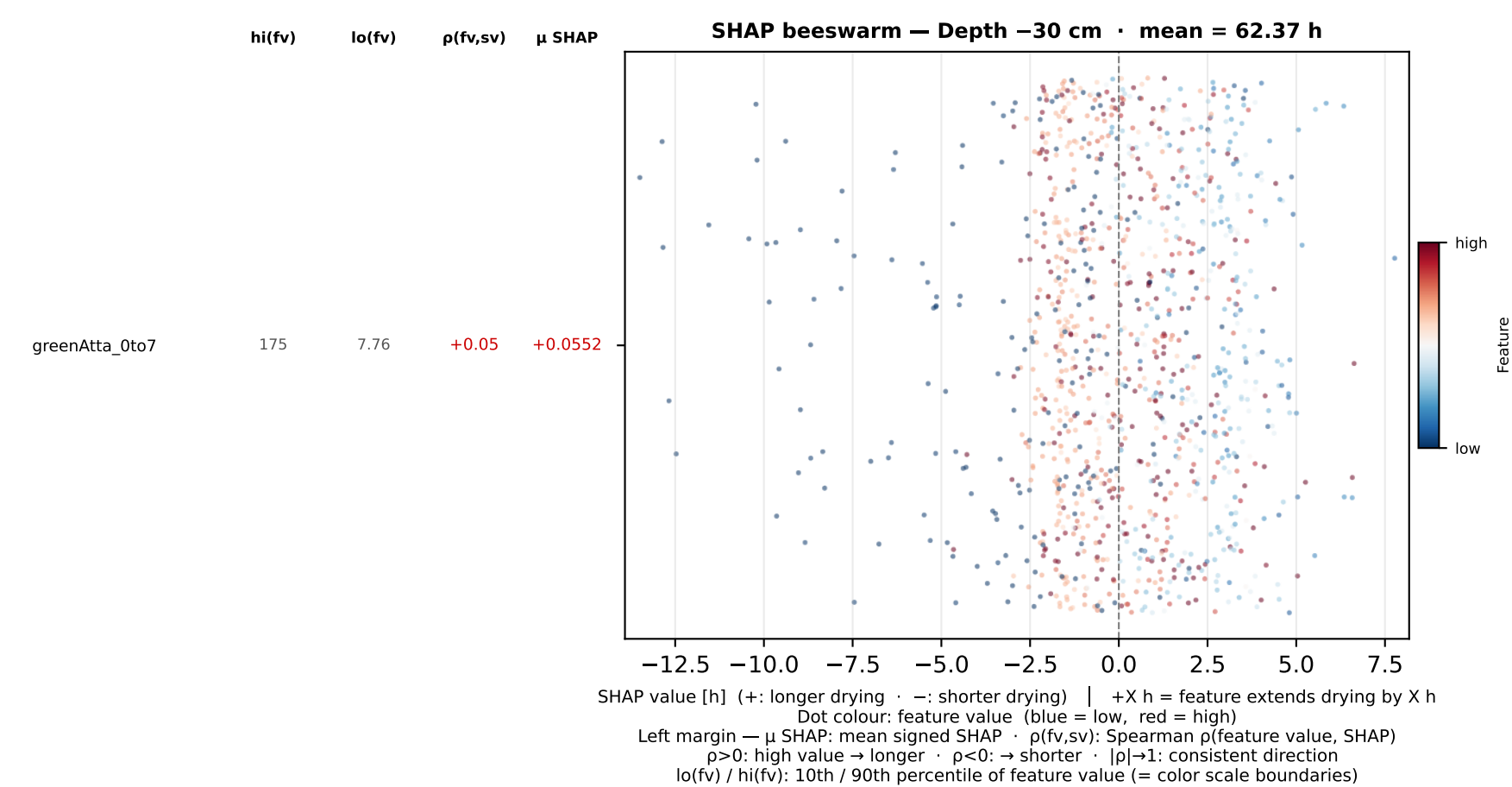
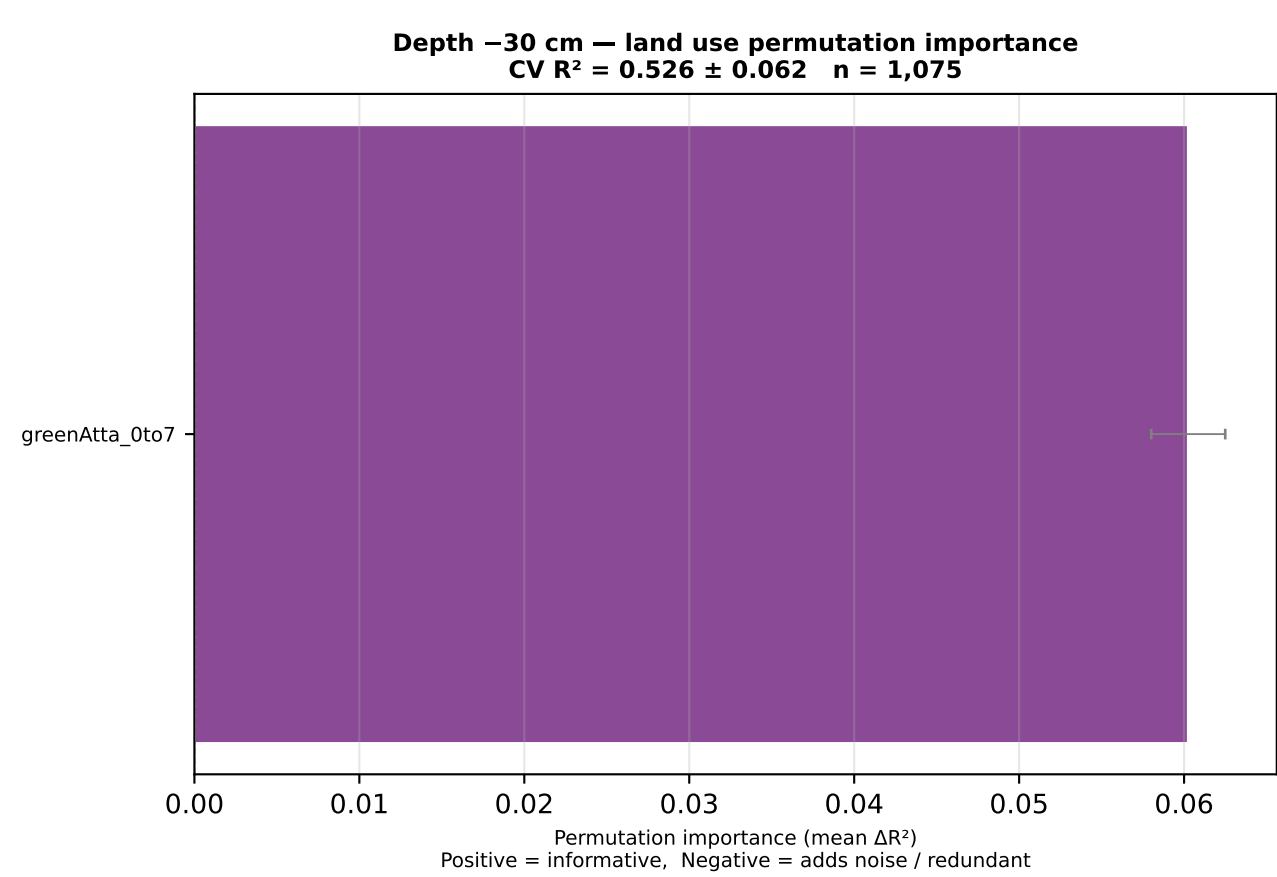
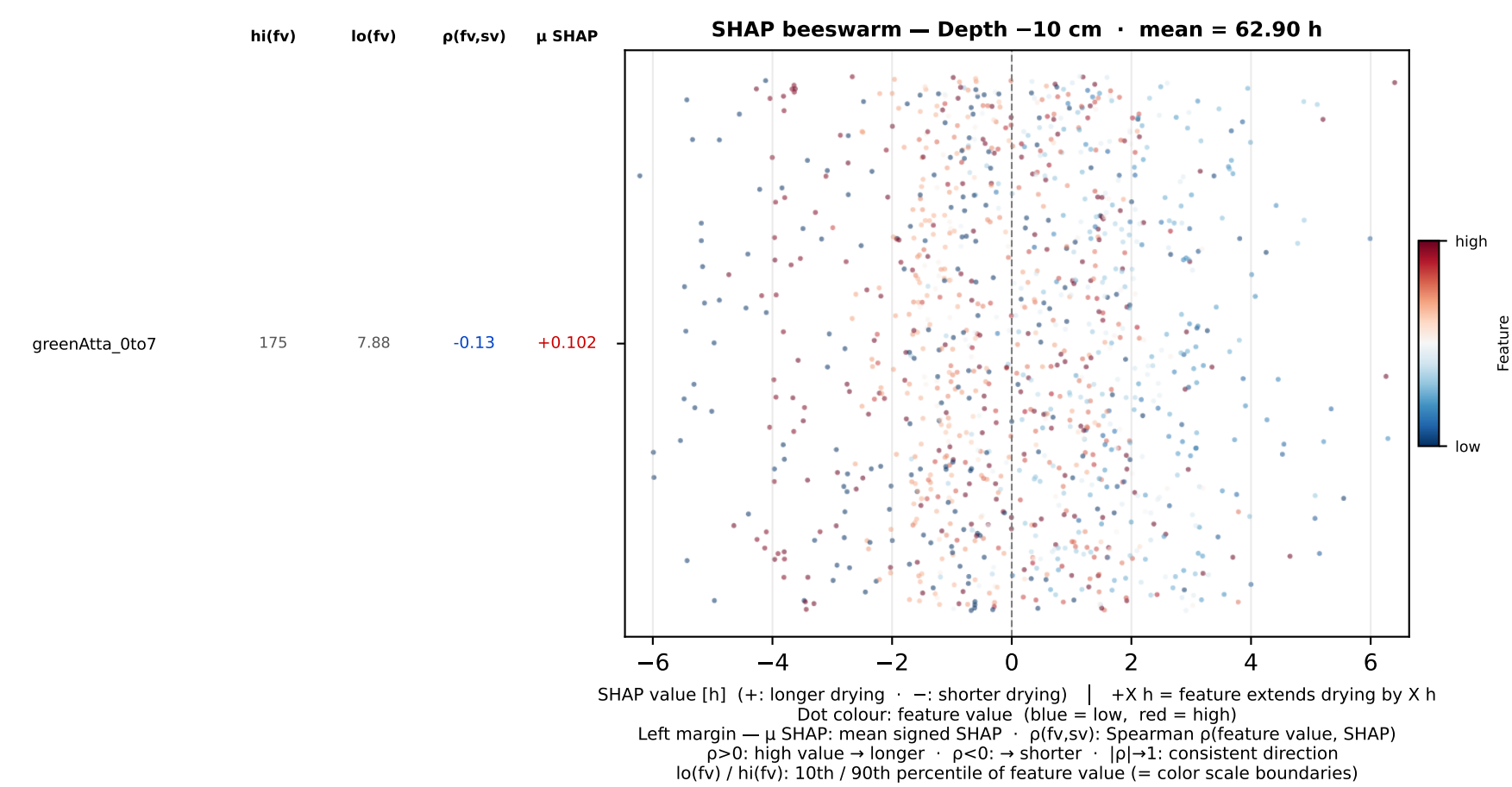
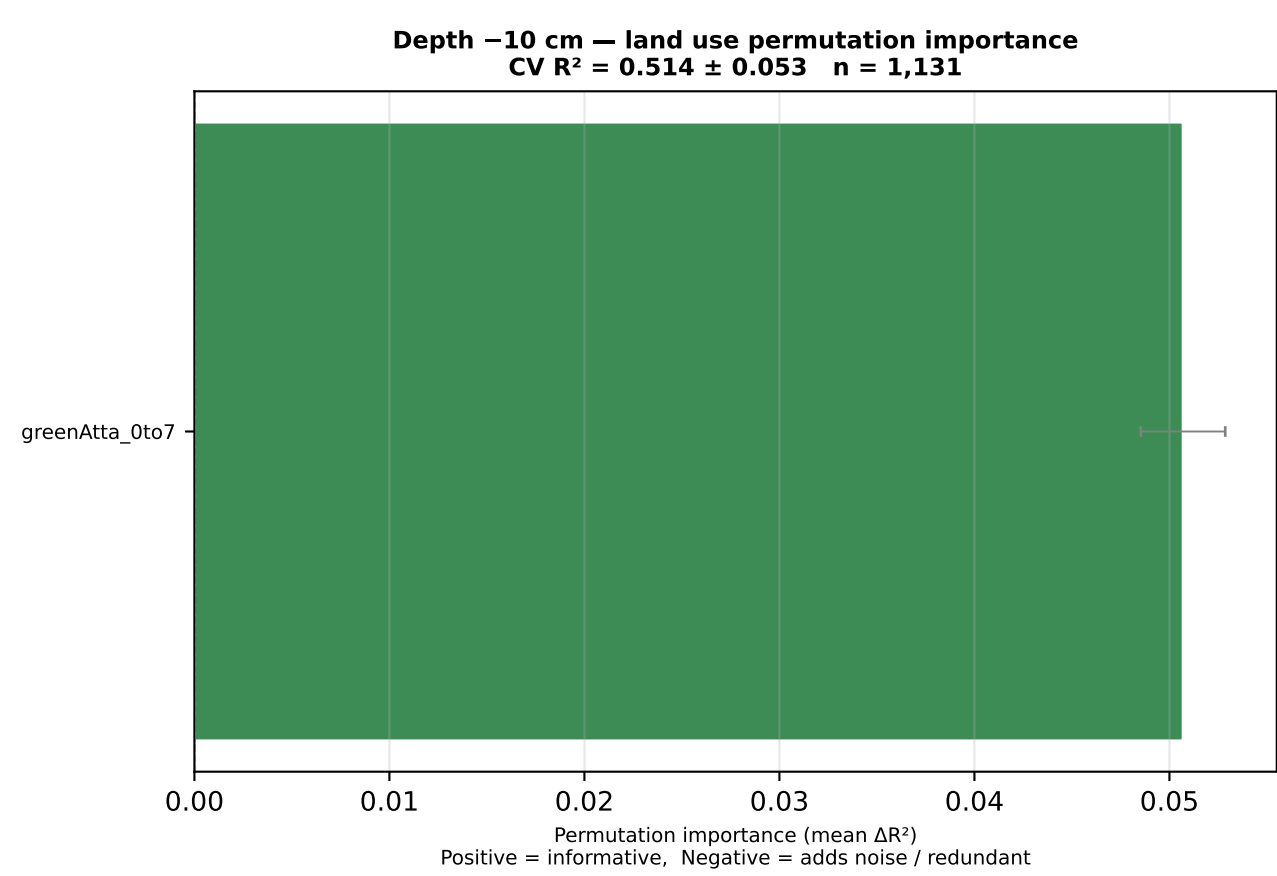
Depth -45 cm — permutation importance
CV R² = 0.545 ± 0.022 n = 1,040



	hi(fv)	lo(fv)	p(fv,sv)	μ SHAP
dyn_range-45	21.6	4.08	+0.71	-0.186
height	12 m	4 m	+0.24	-0.0151
crownDiam	8 m	1 m	+0.32	+0.018
pre_vwc-45	43.6	17	-0.86	-0.244
stemDiam	0.325 m	0.054 m	-0.63	+0.309
greenAtta_0to7	175	7.76	+0.30	+0.000396
day_cos	0.467	-0.997	-0.22	-0.204
slope	5.78	0.462	+0.72	-0.0968
twi	6.53	2.63	-0.43	-0.495
avg_temp-45	26.9 °C	10 °C	-0.56	+0.169
SVF	0.937	0.475	+0.78	+0.218
tpi_7m5	0.289	-0.0653	+0.12	+0.226

SHAP beeswarm — Depth -45 cm · mean = 61.26 h

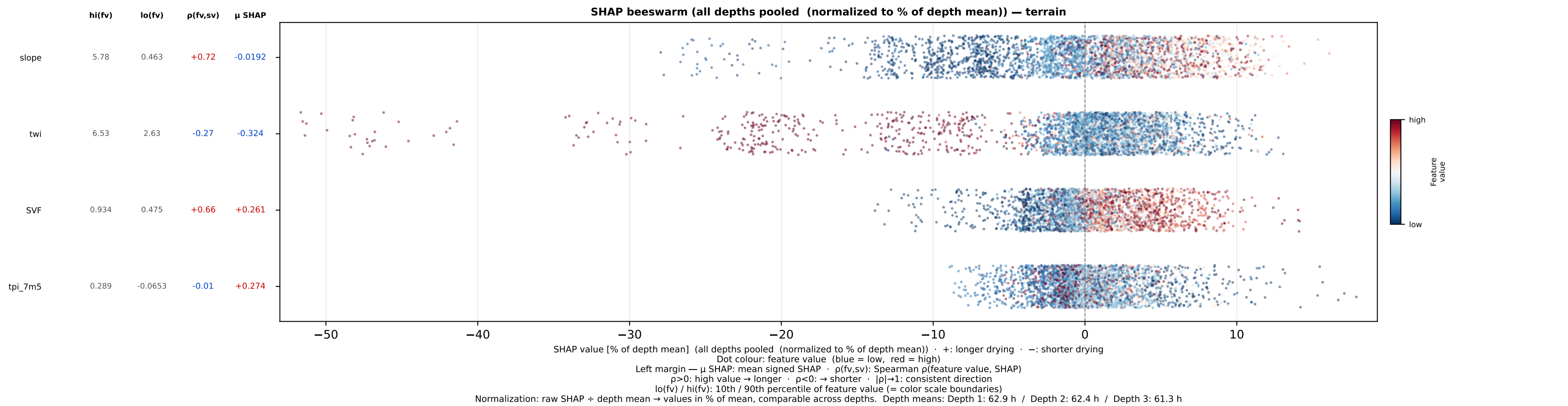
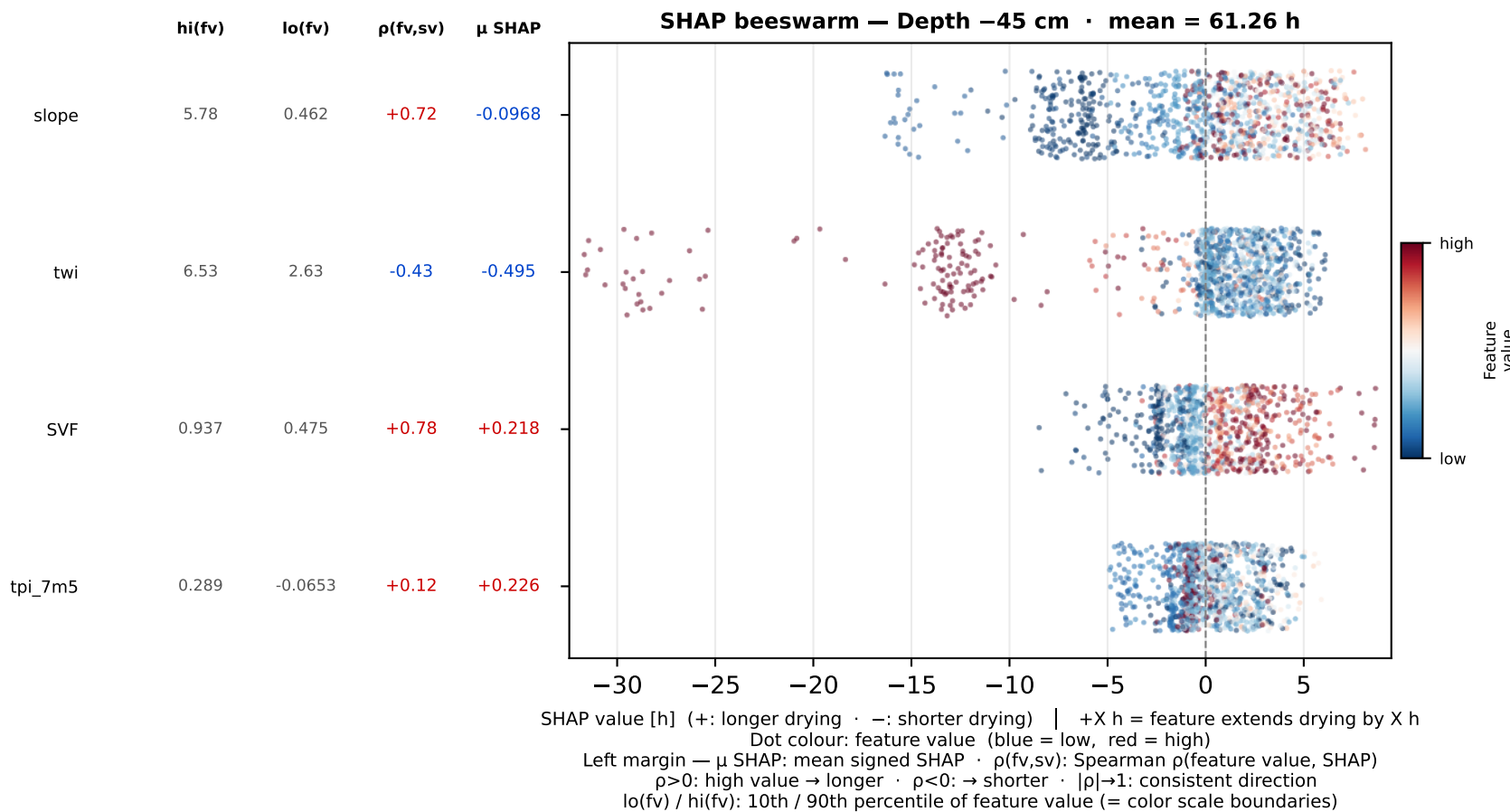
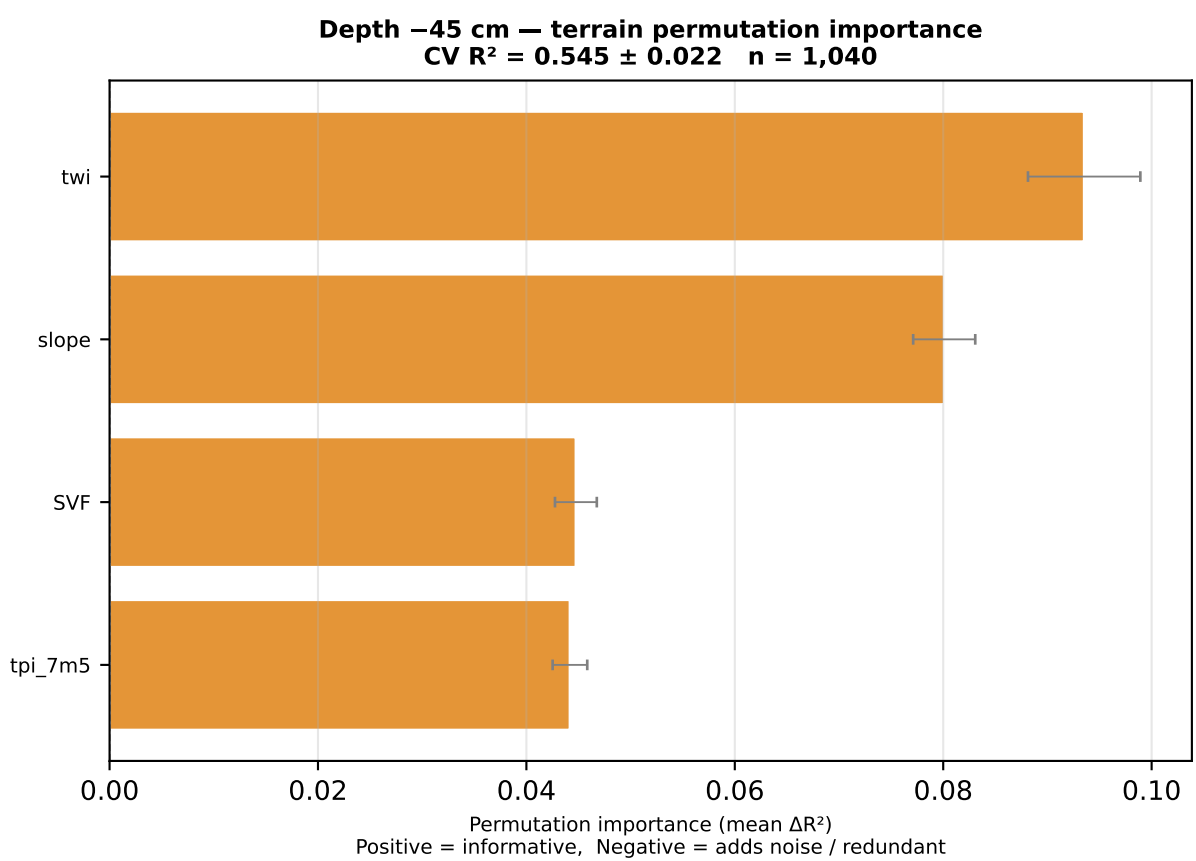
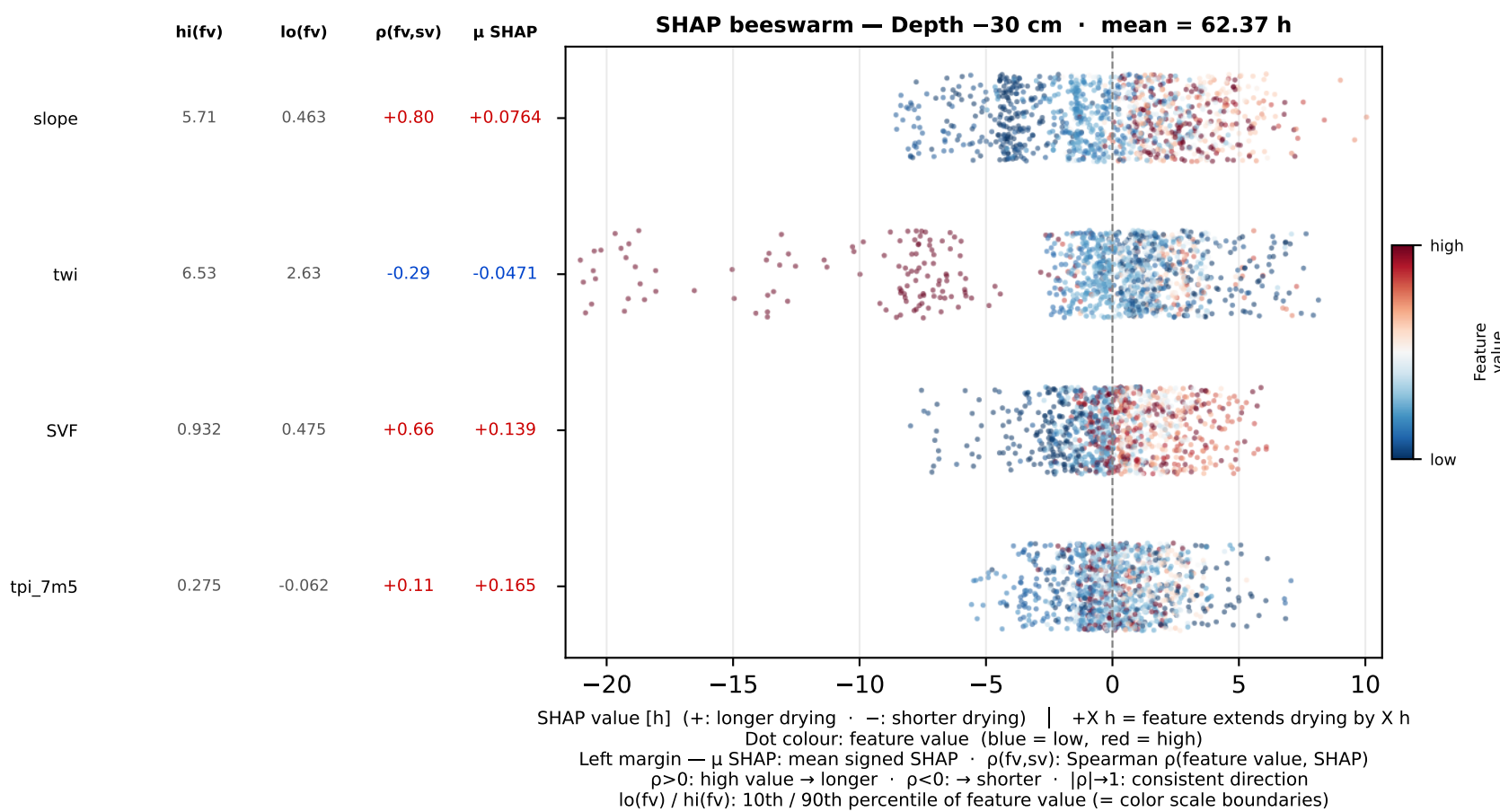
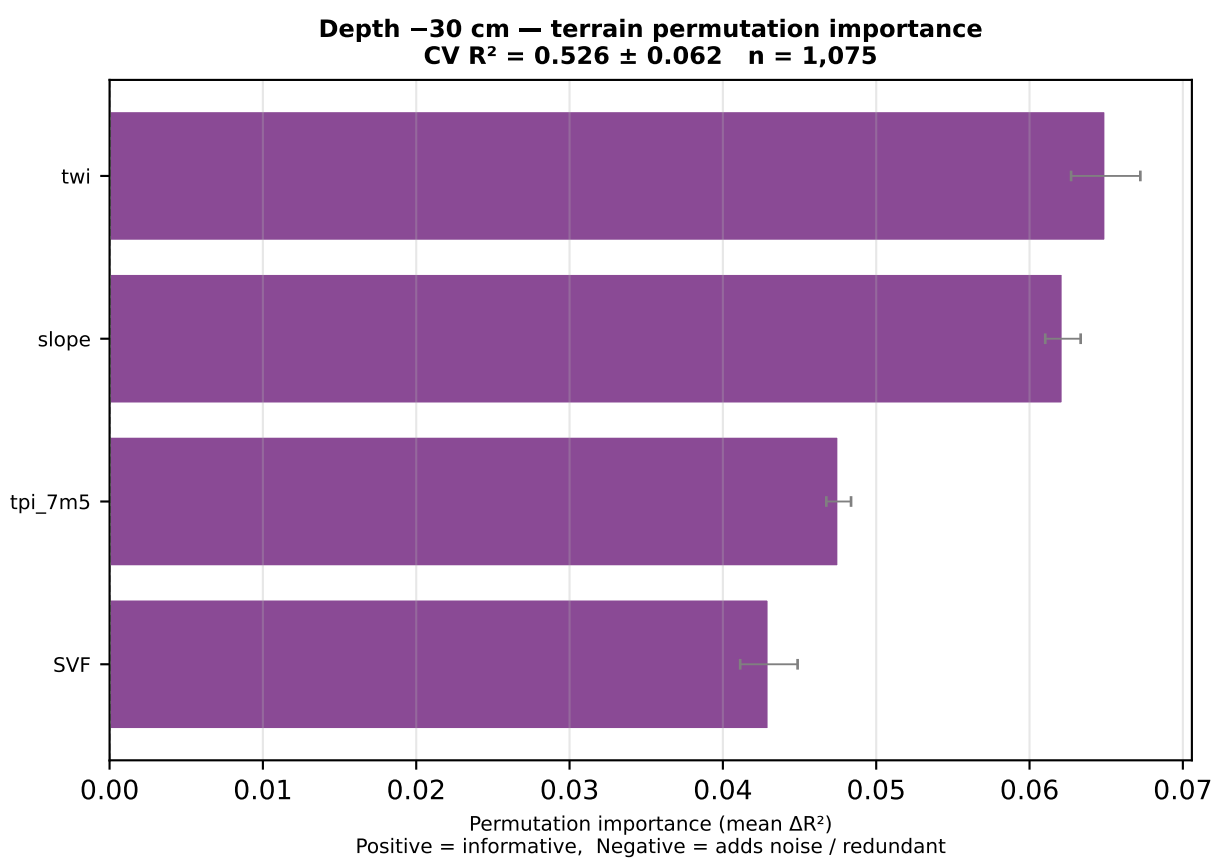
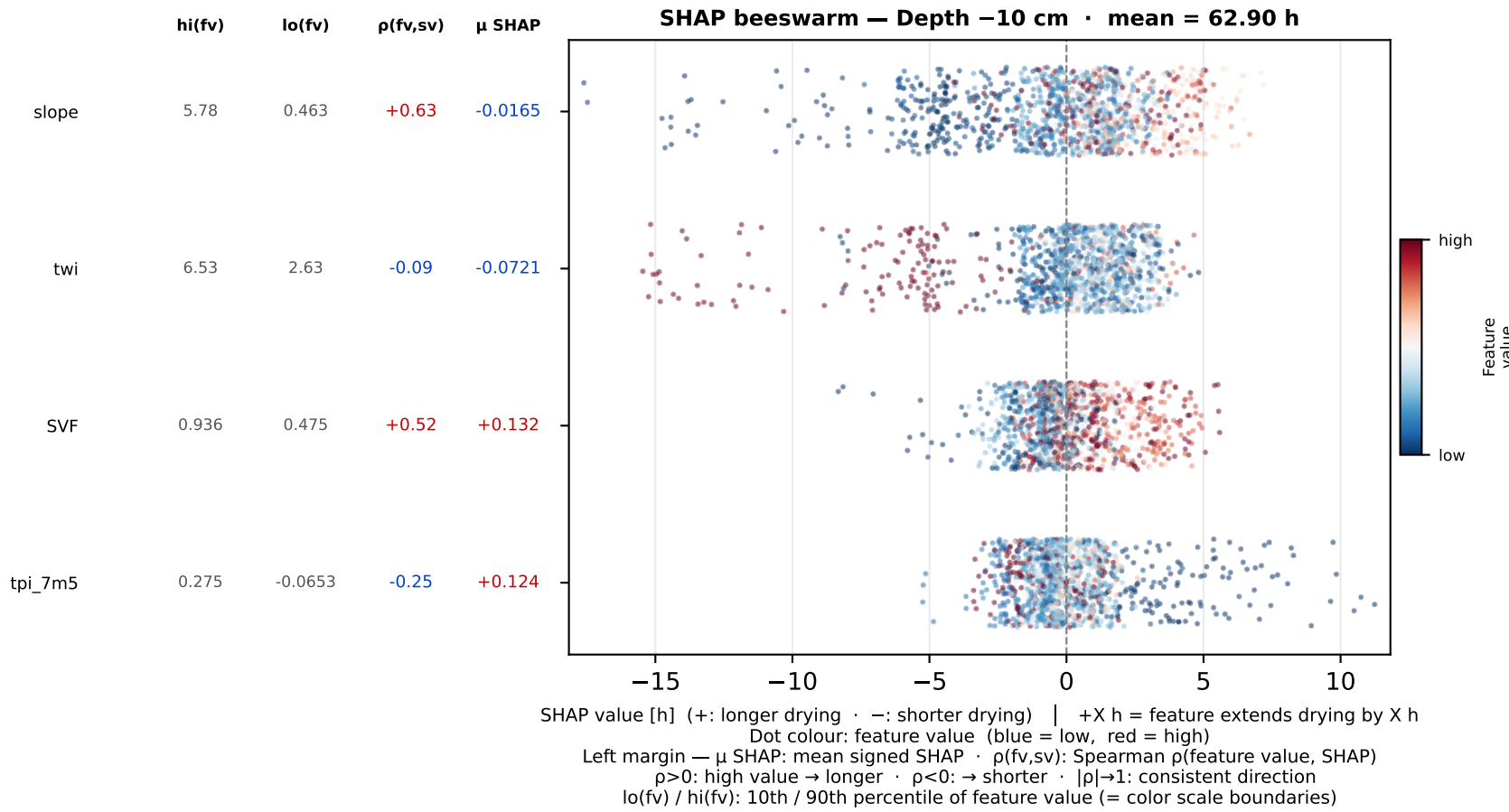
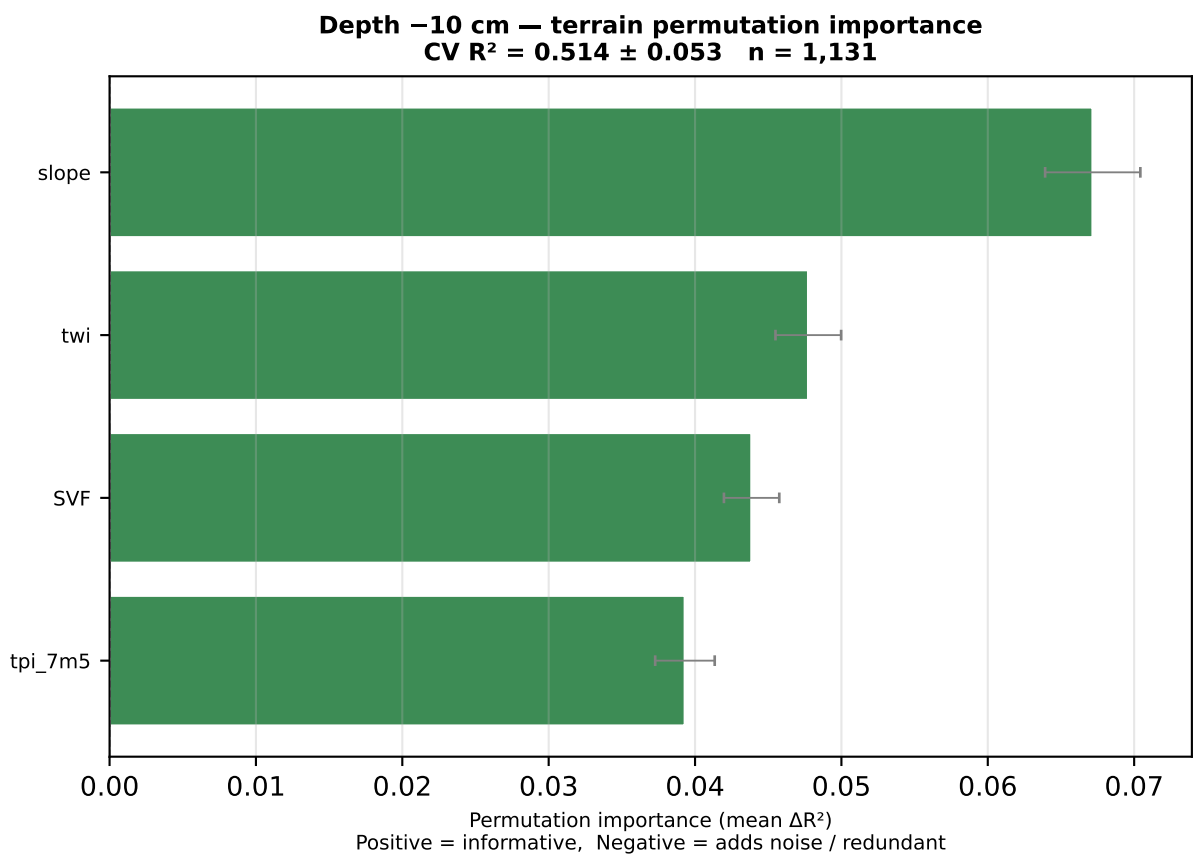




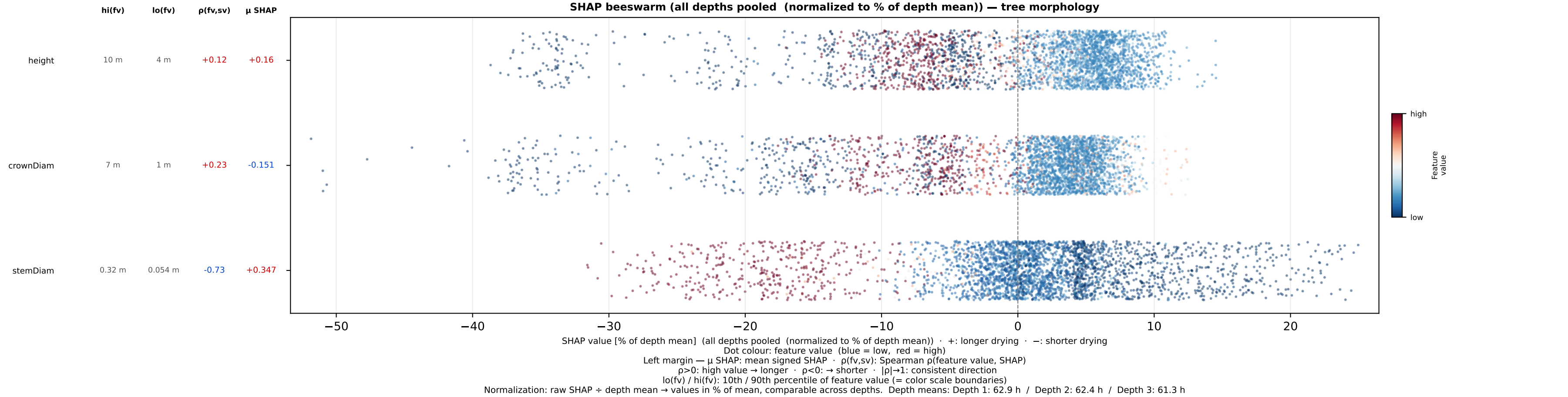
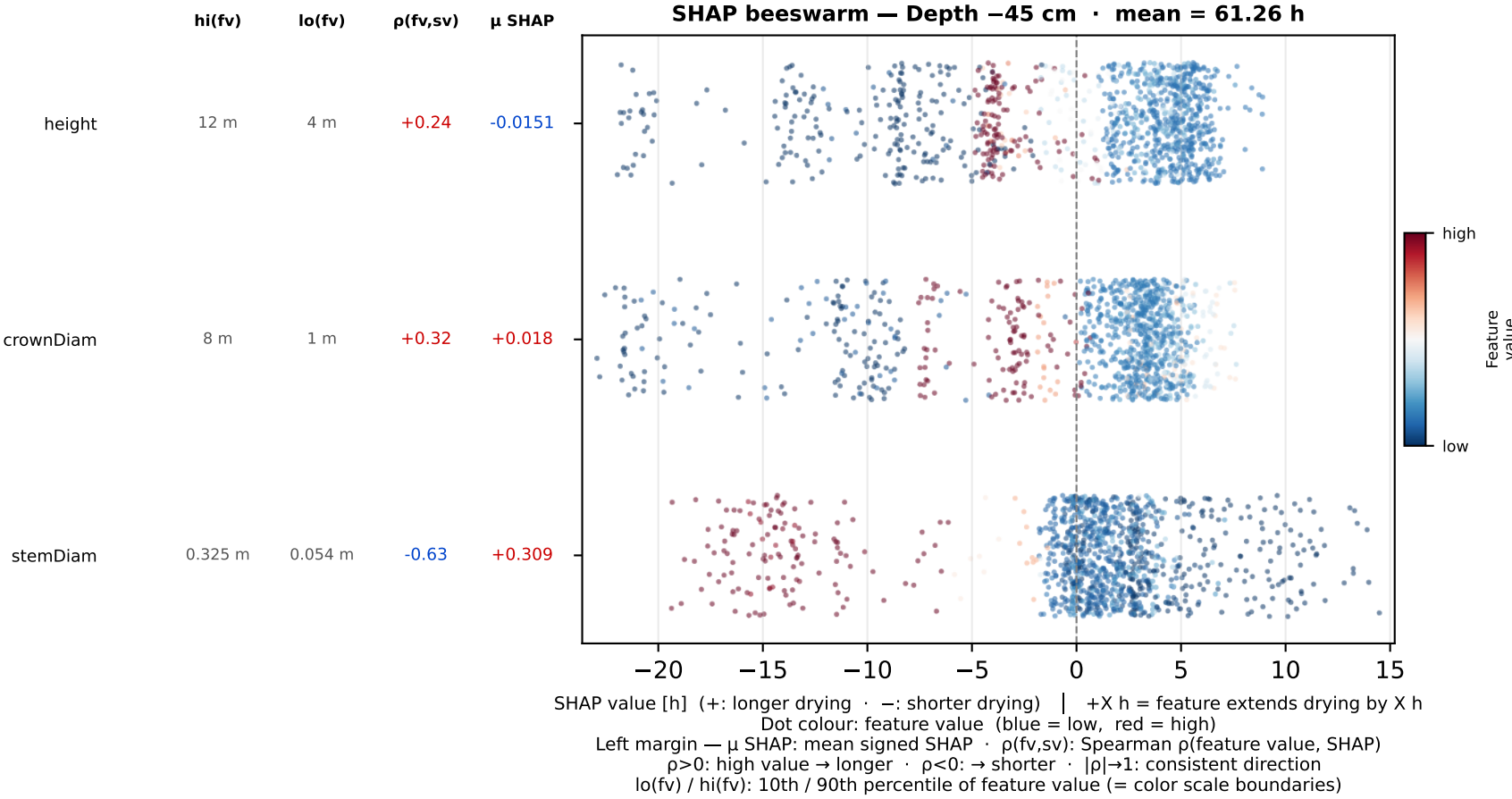
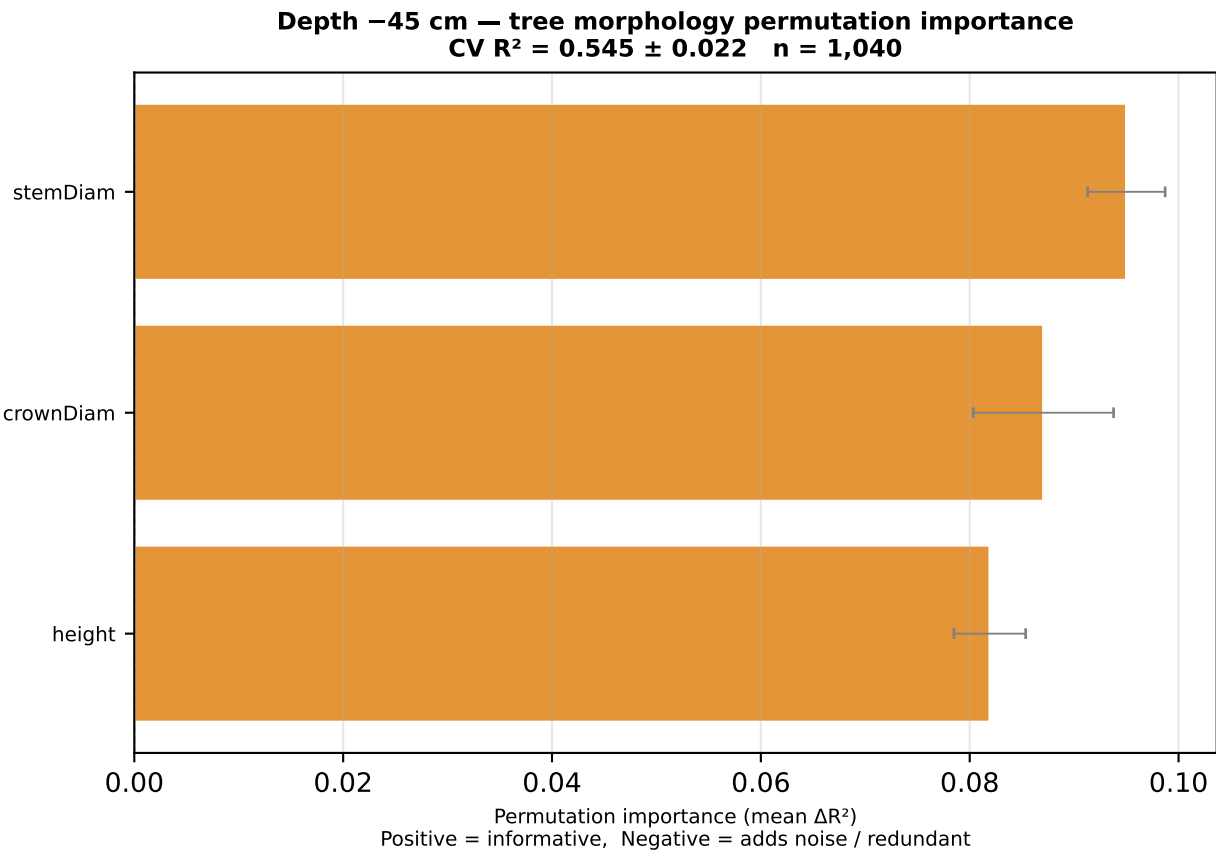
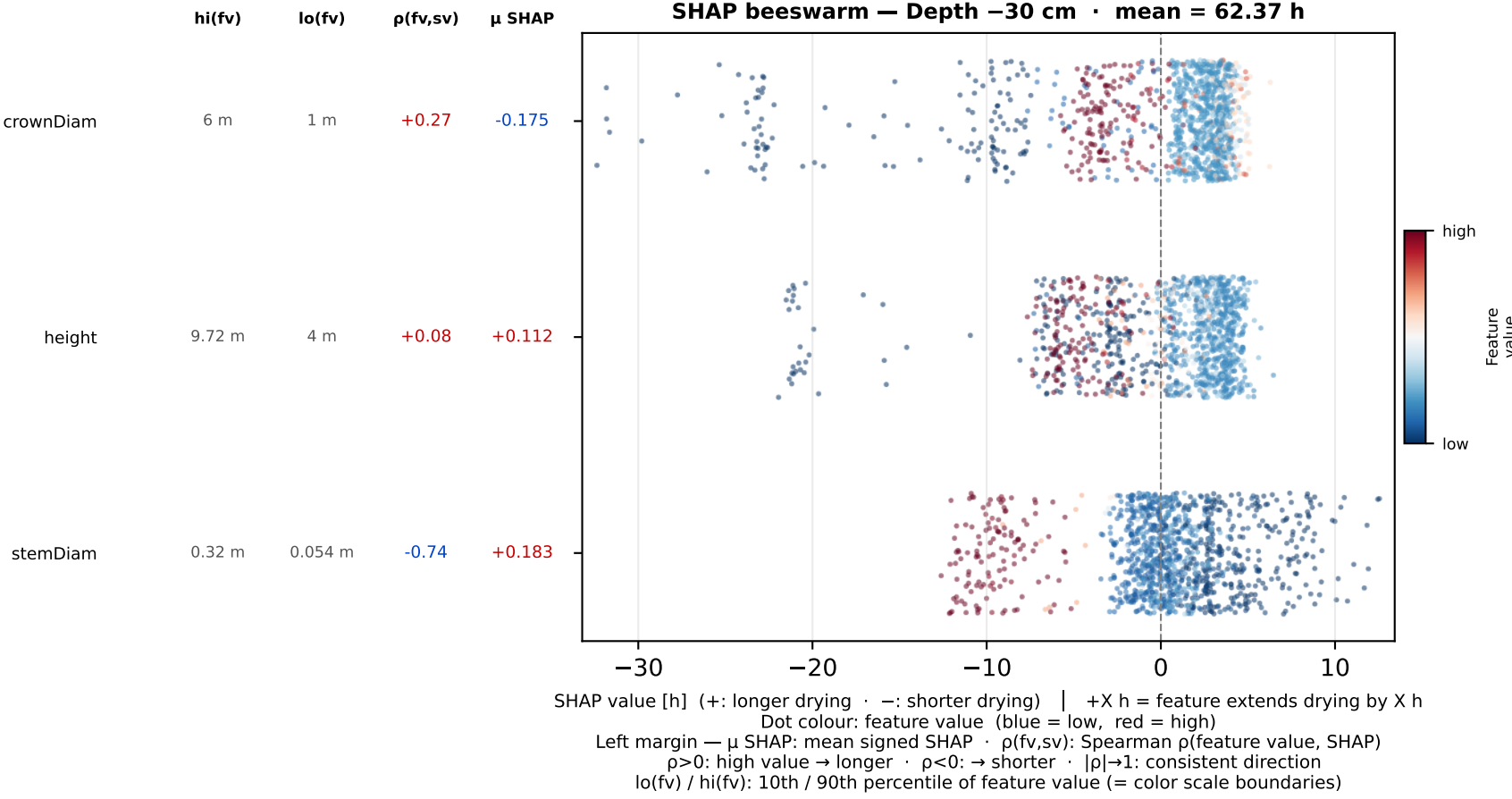
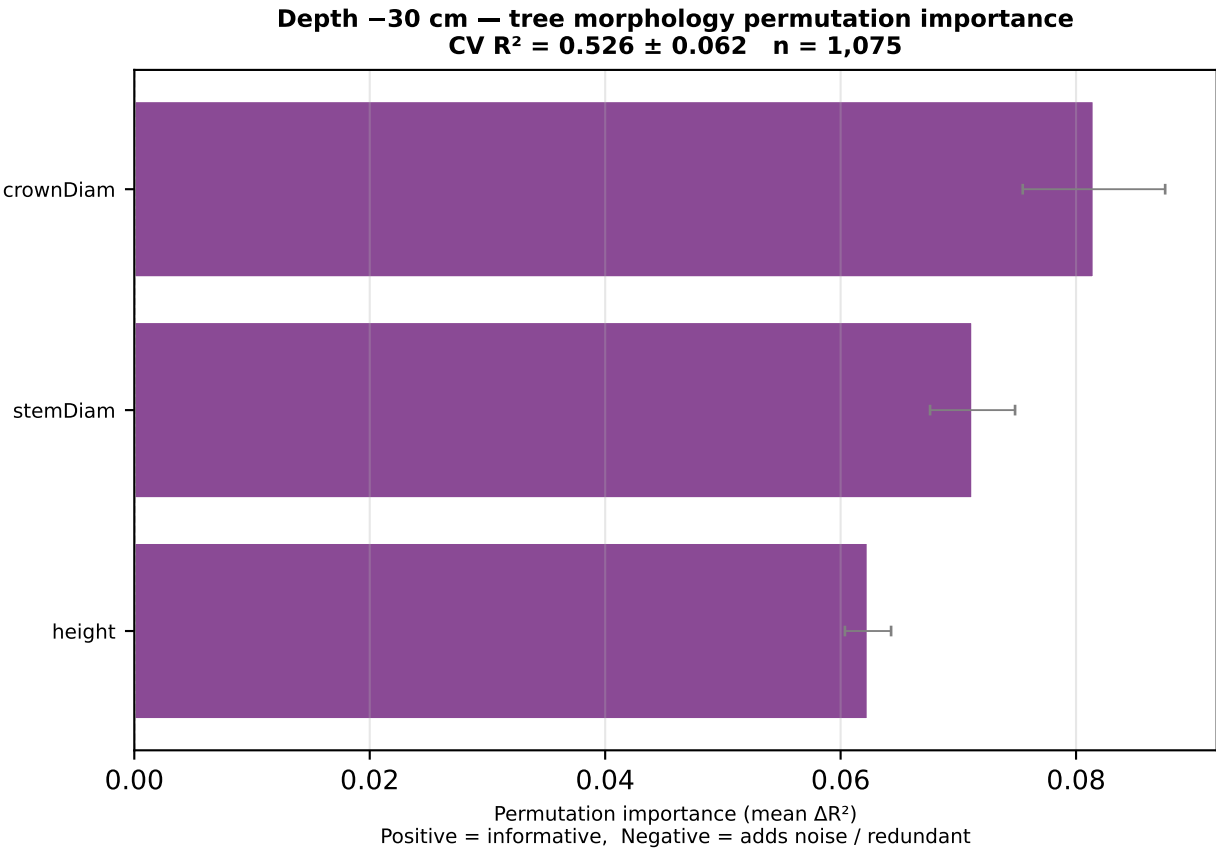
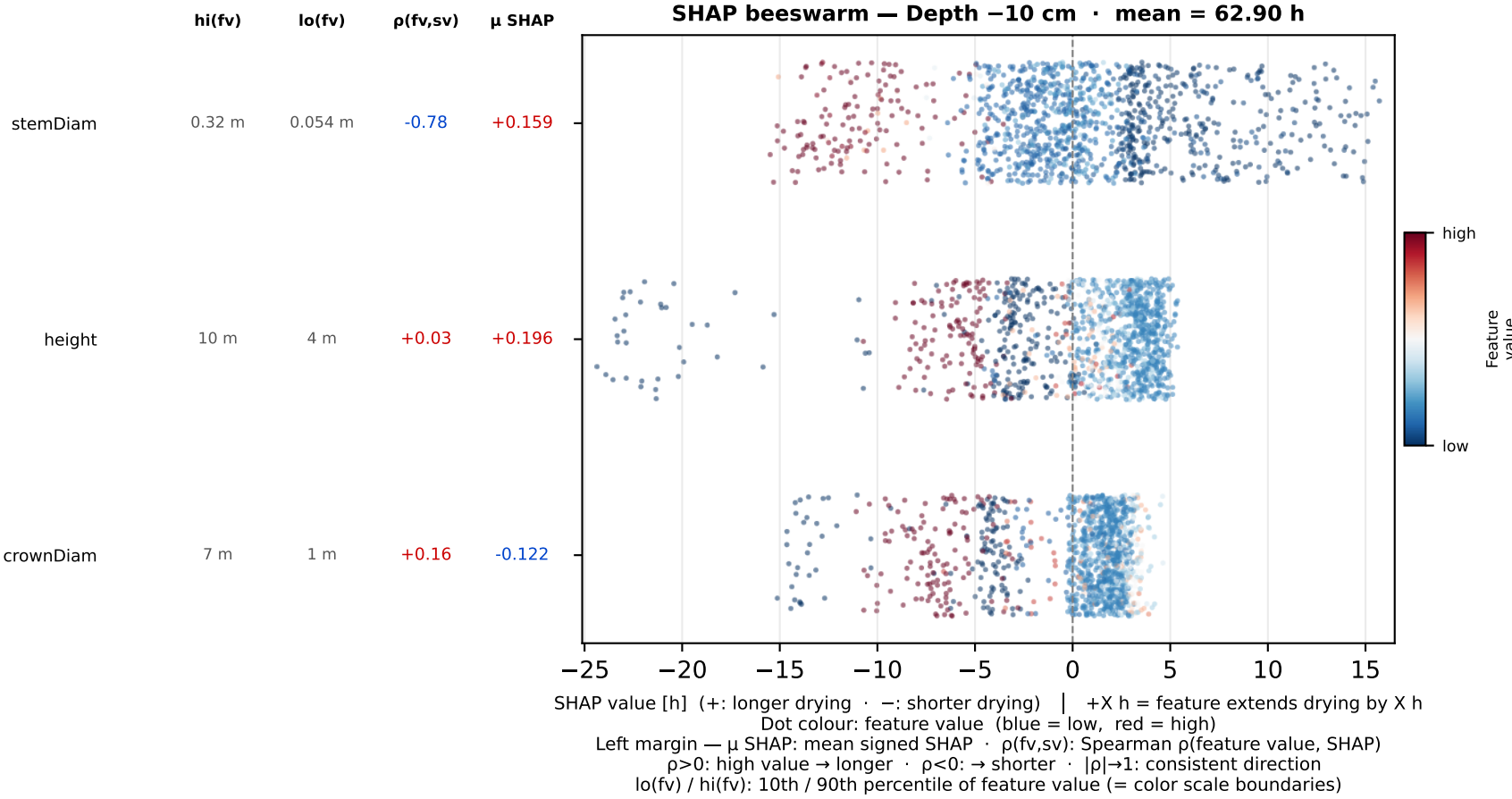
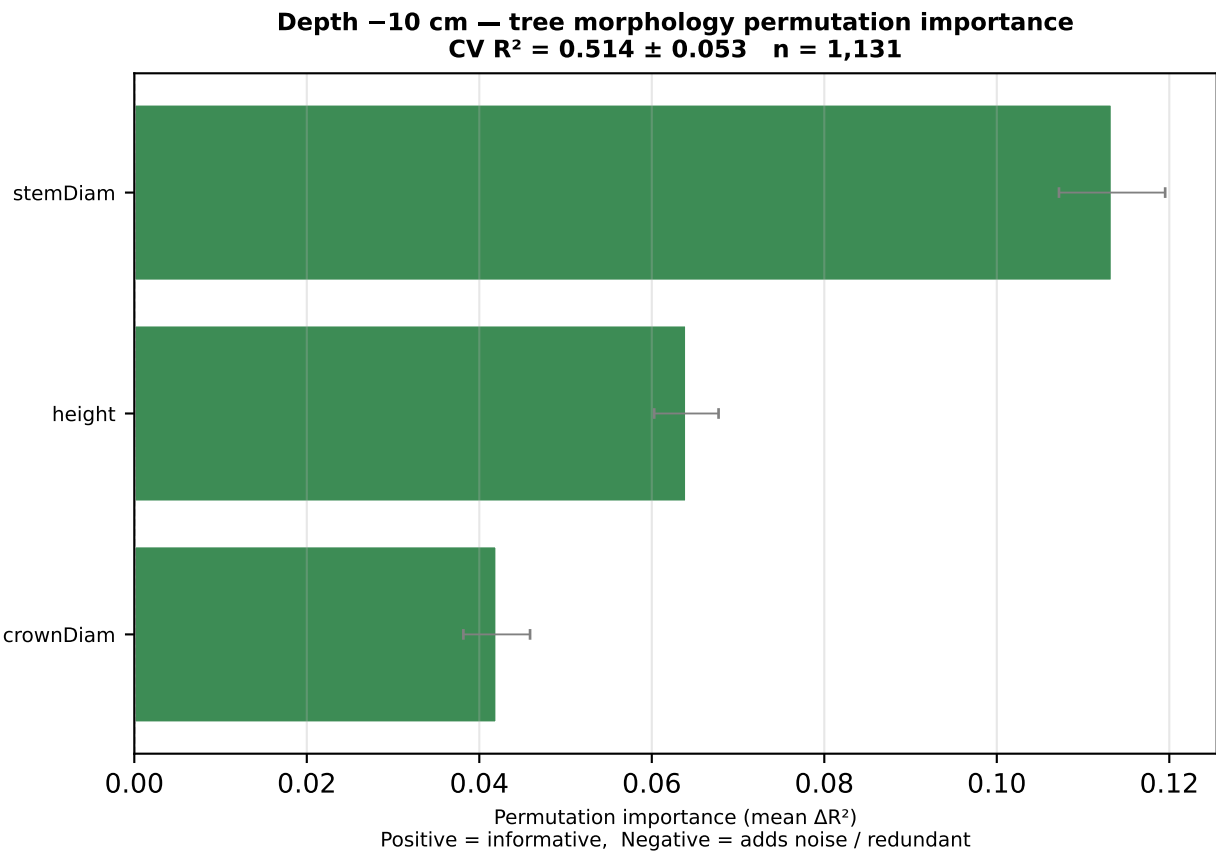
Terrain Feature Analysis — Irrigation

Target: drying time [h] per depth (5,242 events)

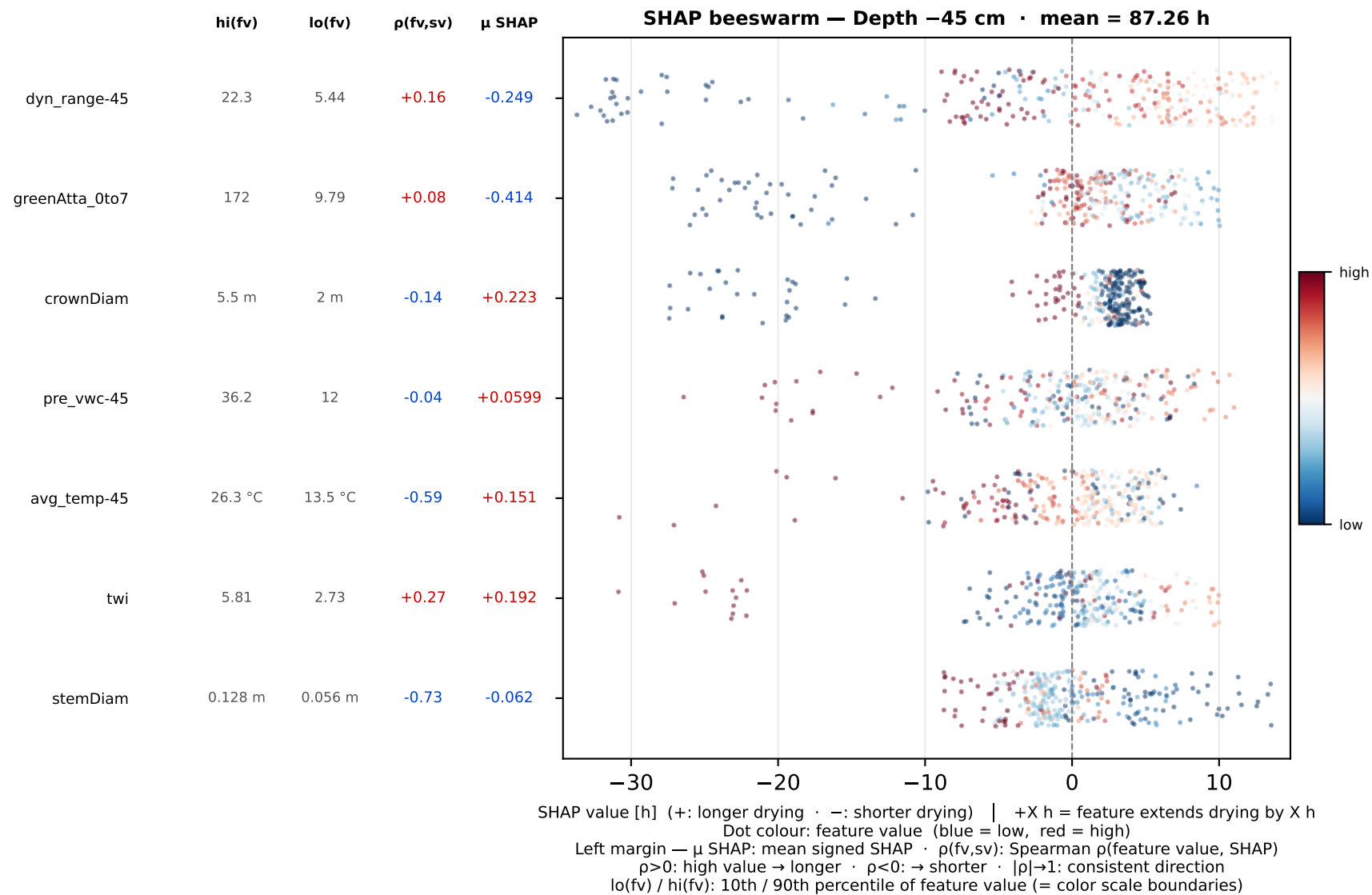
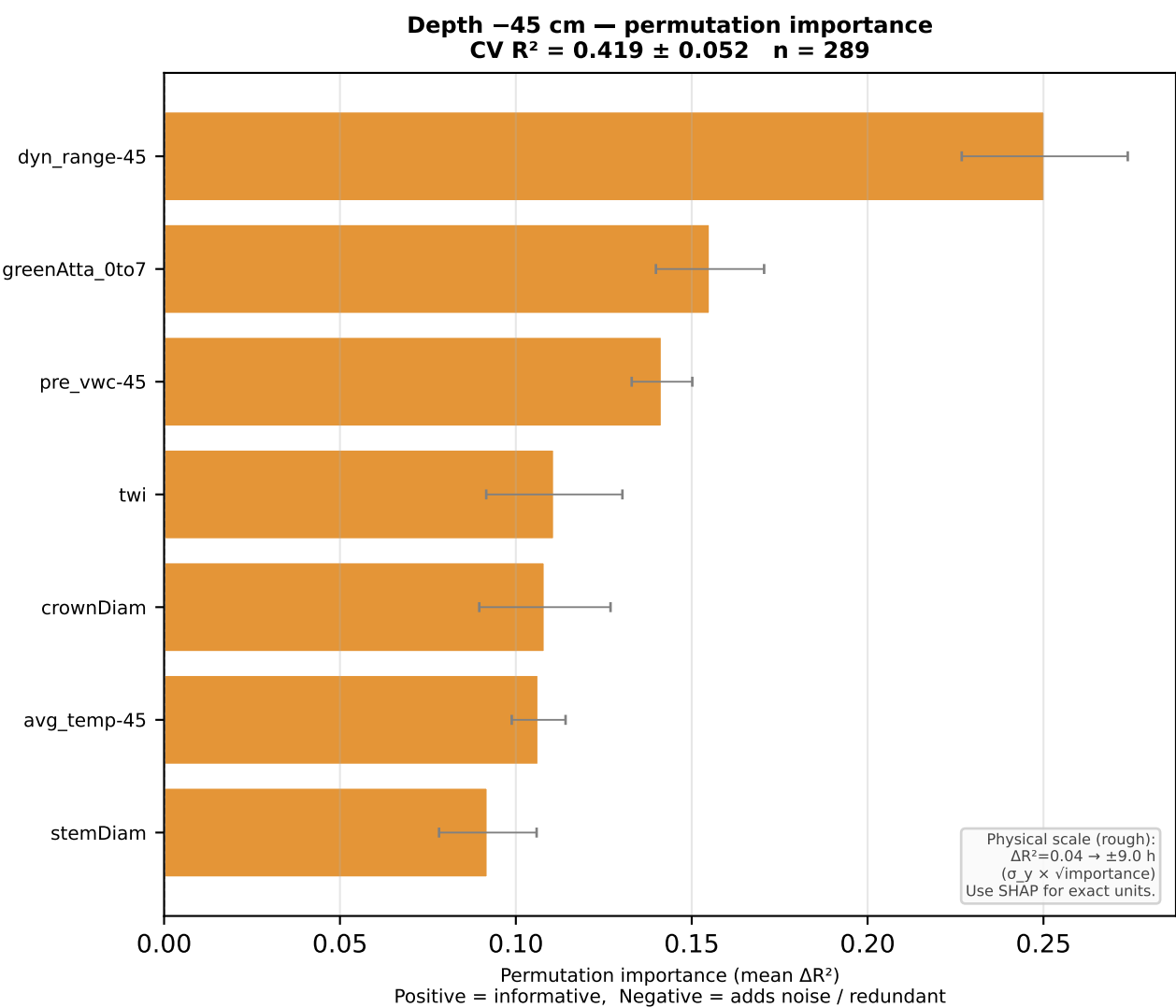
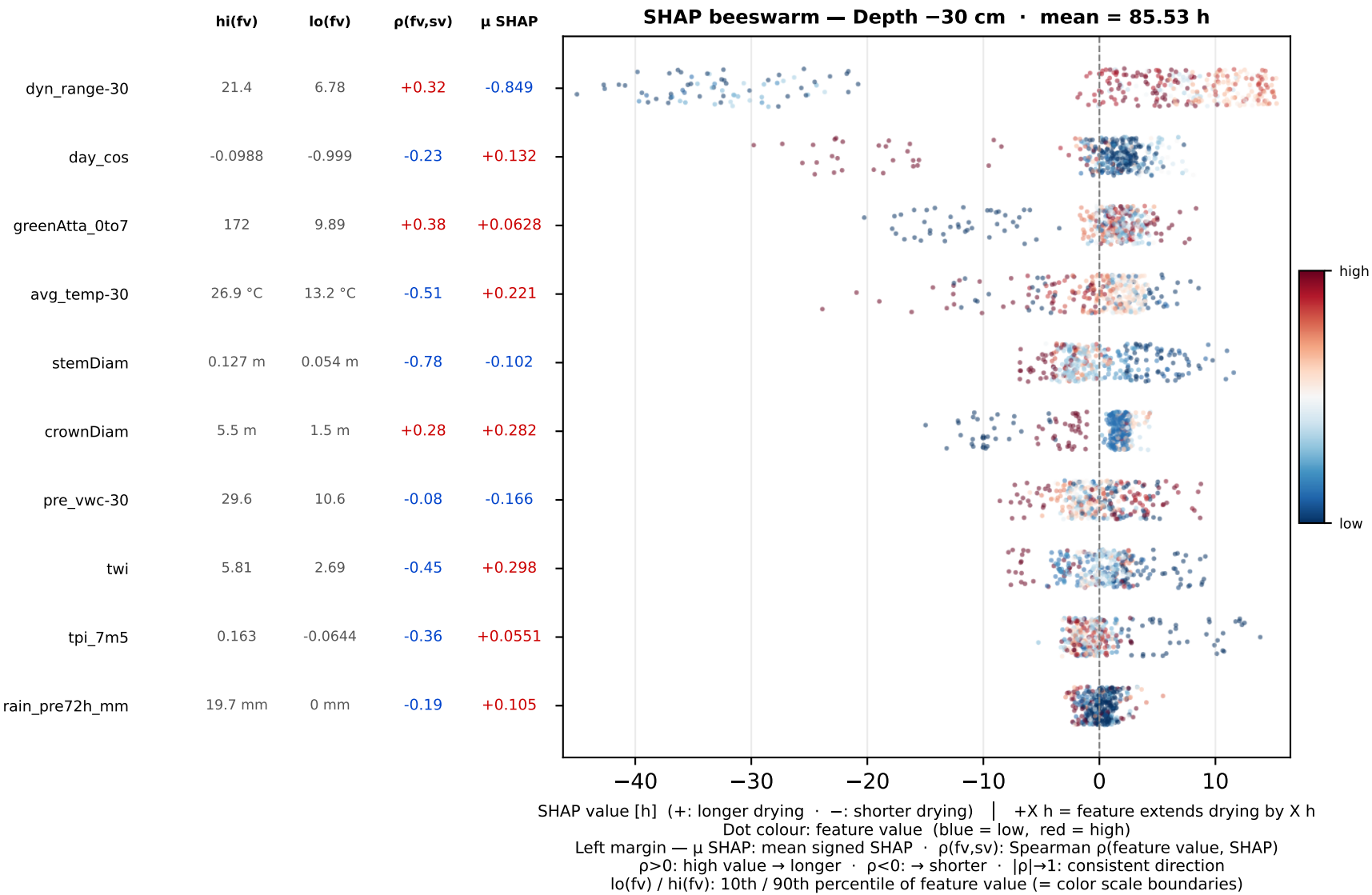
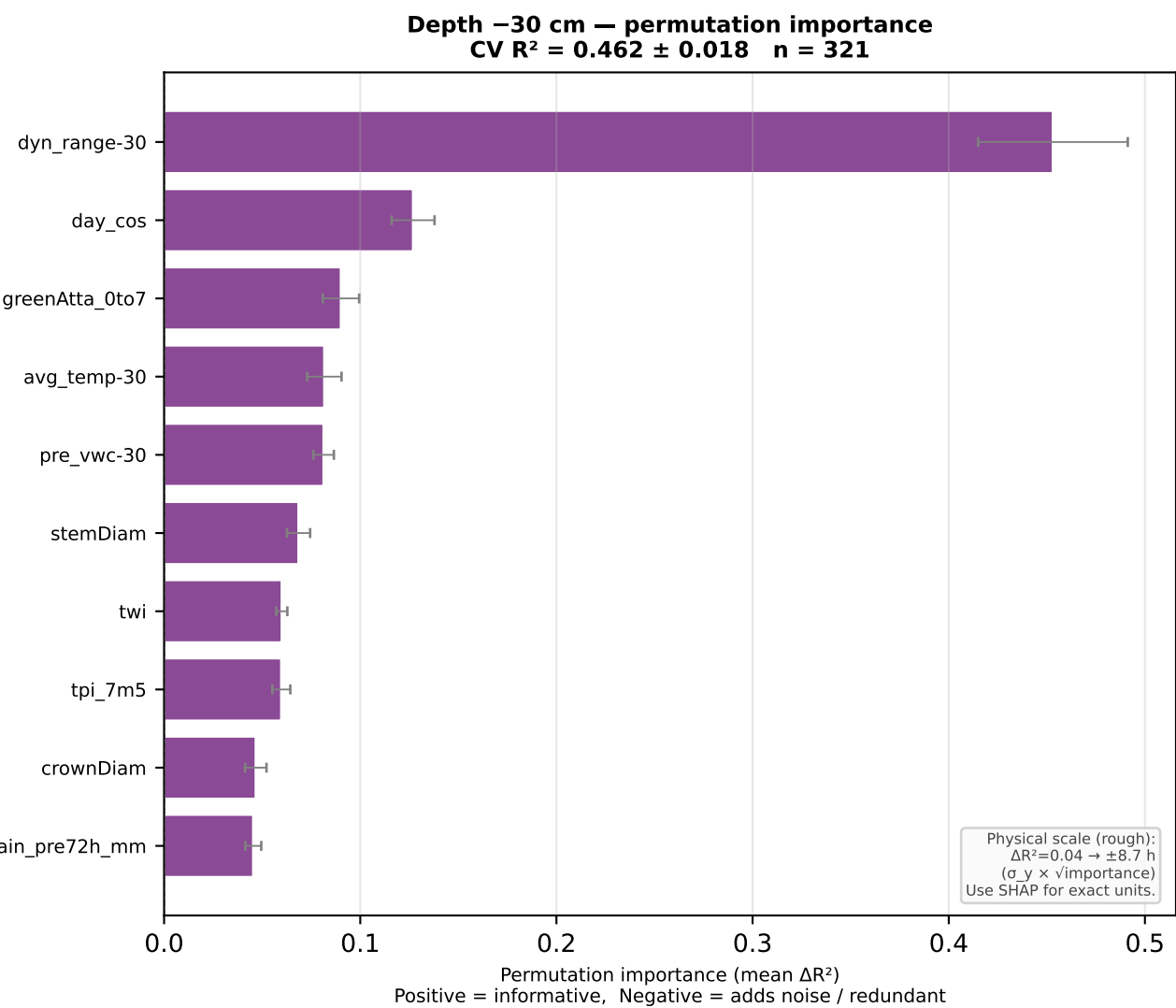
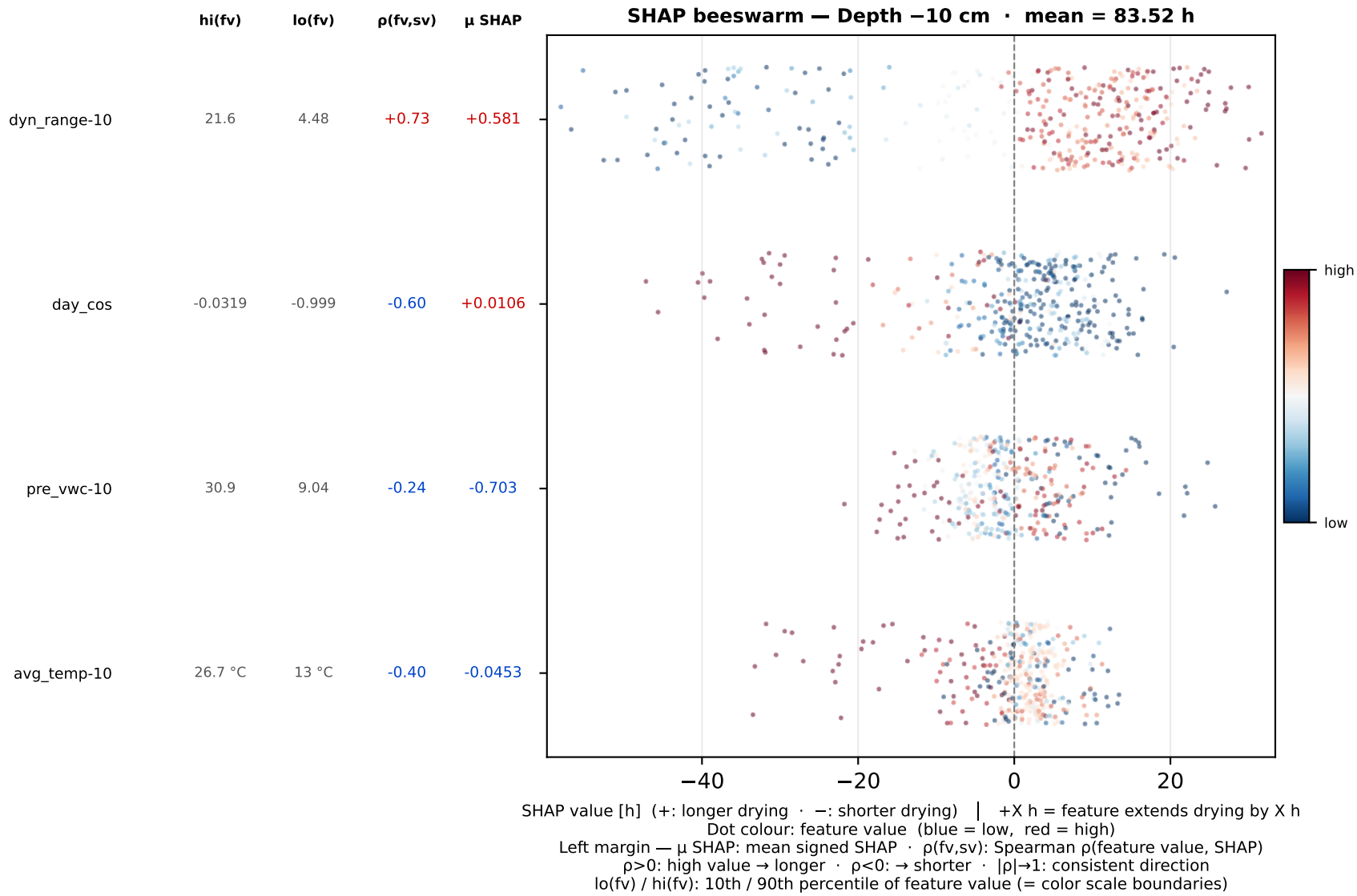
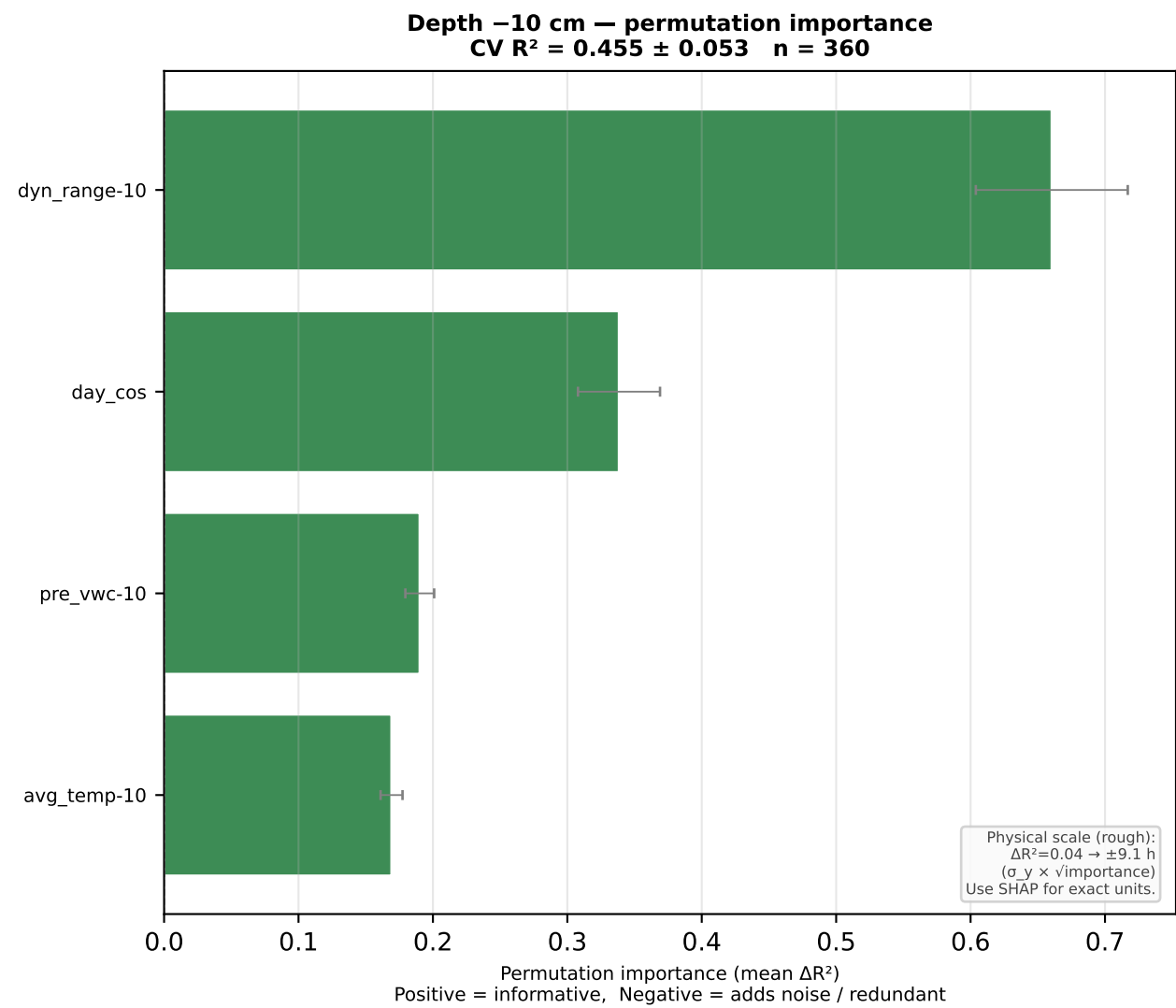
Features: TWI · TPI (2.5/5/7.5 m + crown-diam) · slope · amsl · flow accumulation · SVF · depth to groundwater

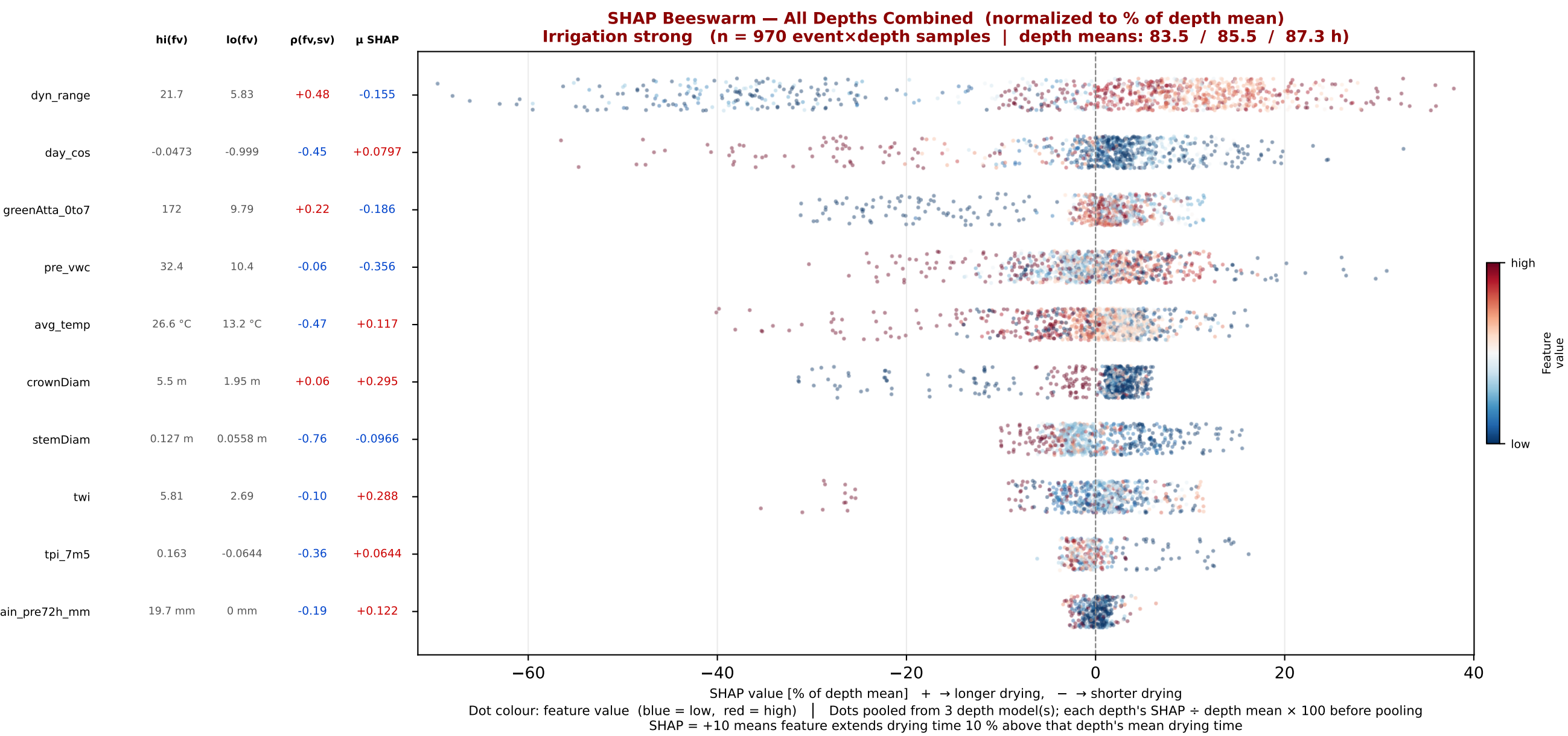


Tree Morphology Feature Analysis — Irrigation
Target: drying time [h] per depth (5,242 events)
Features: crown diameter · height · stem diameter · vitality

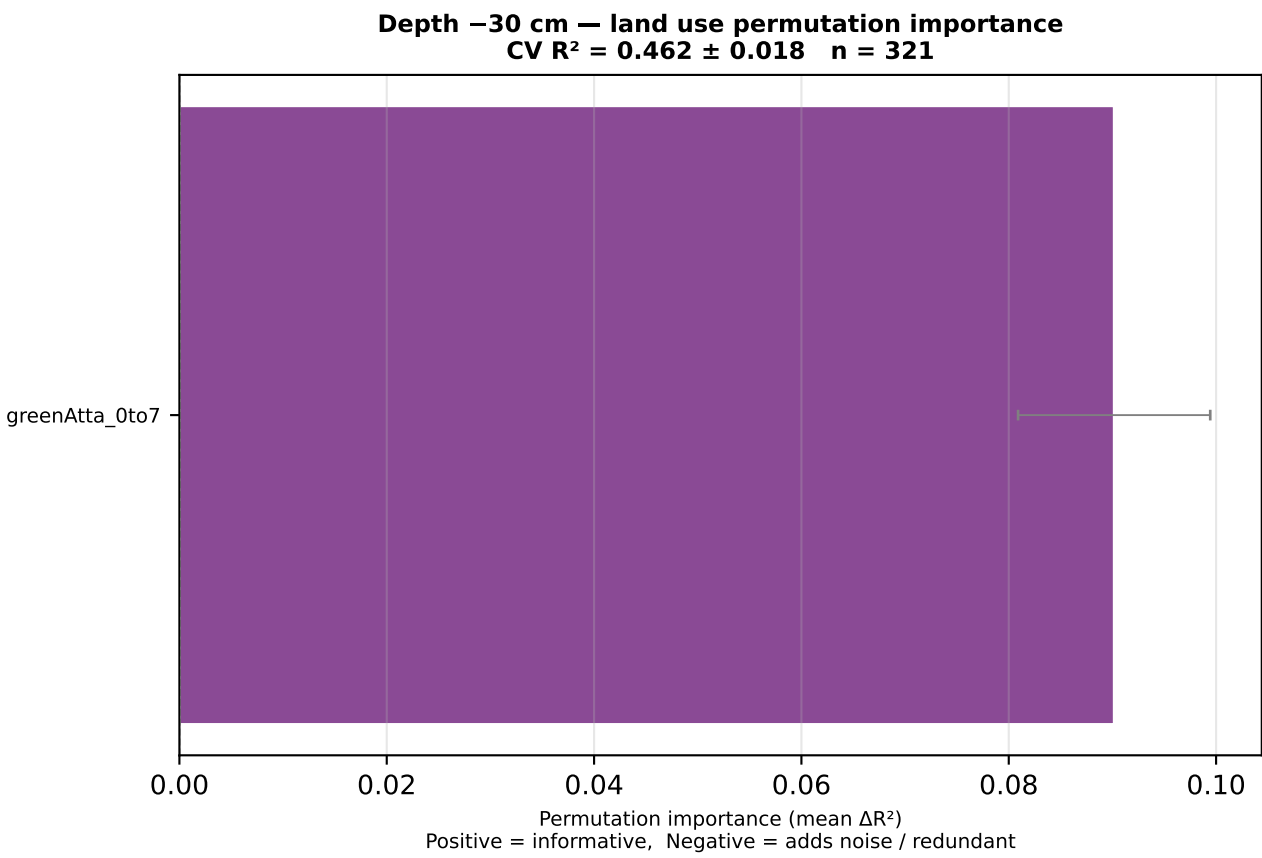
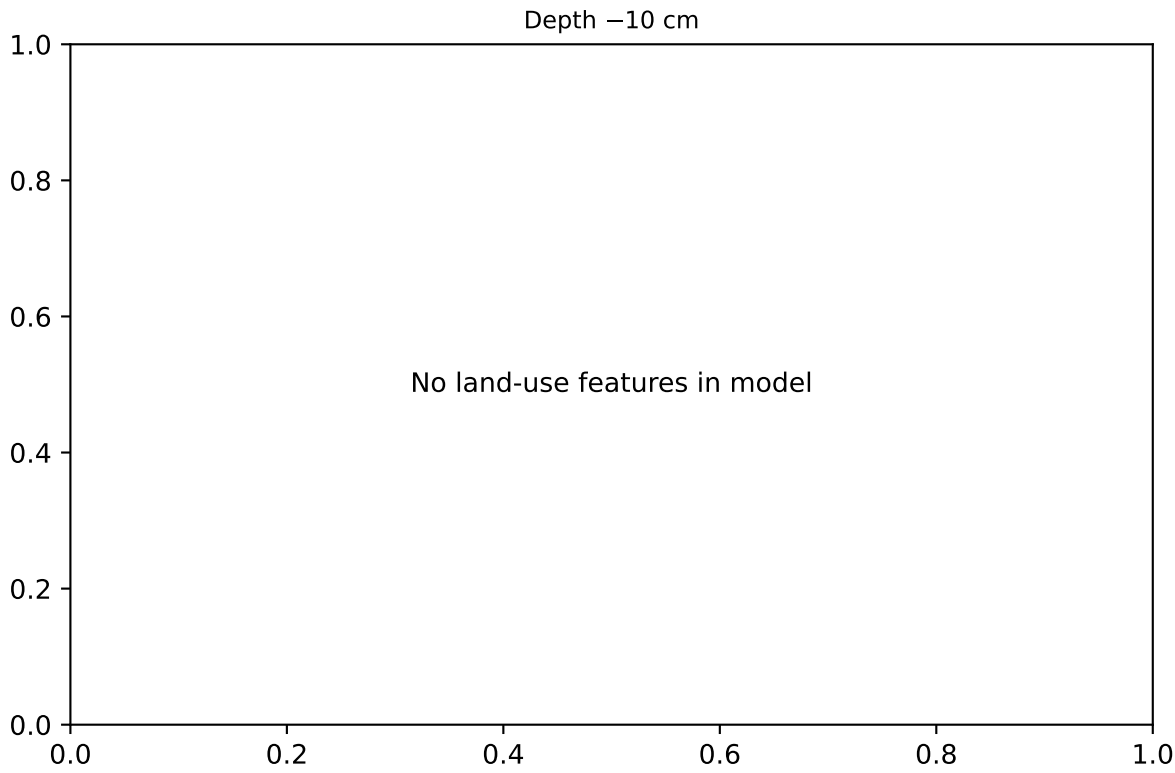


Random Forest Feature Importance — Irrigation strong
Target: Drying time (h) at each depth (1,161 events)

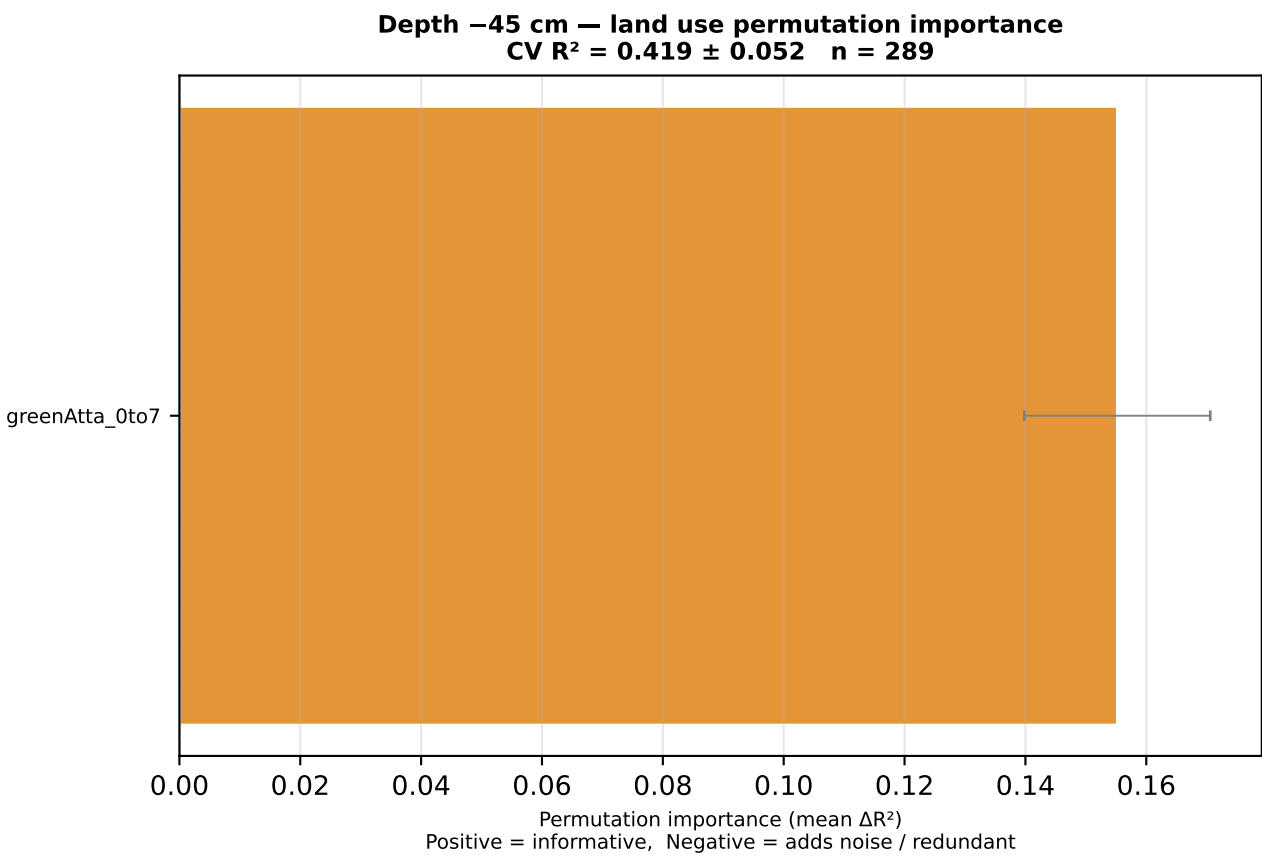
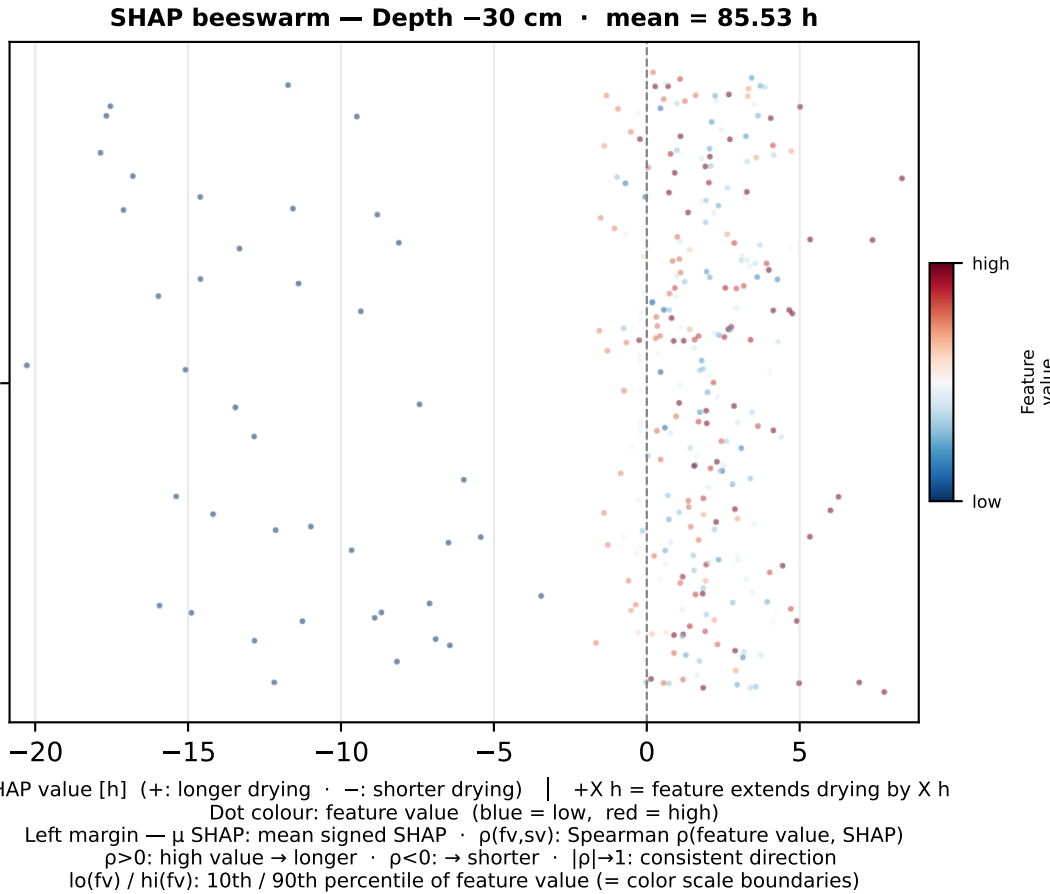




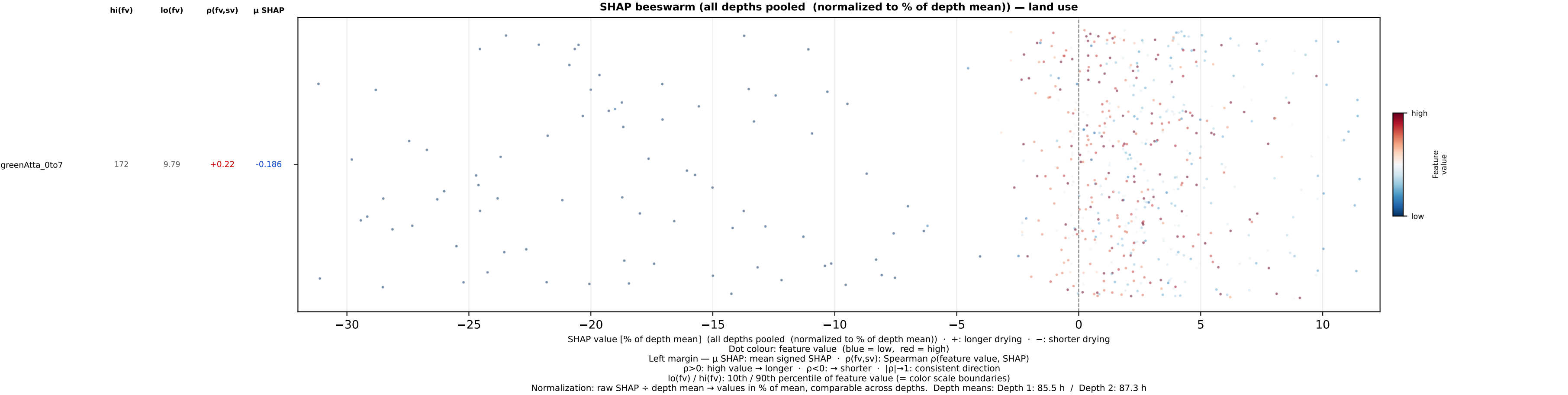
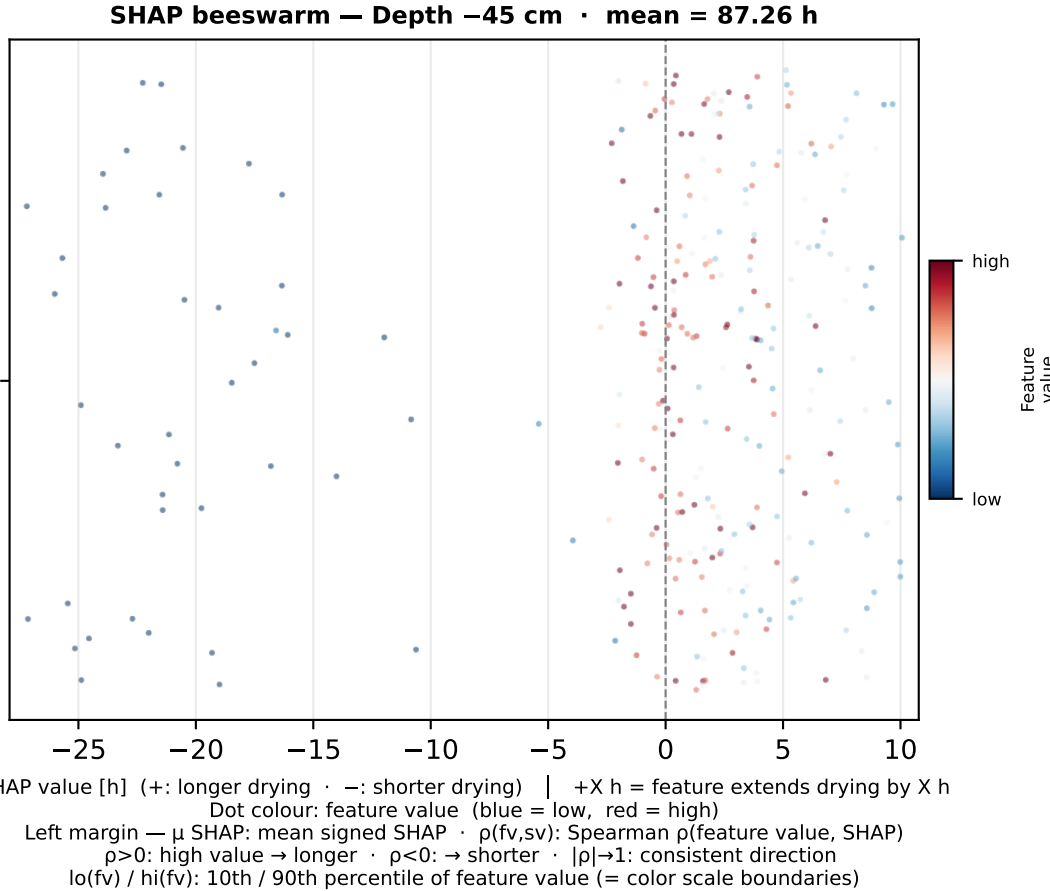
Land Use Feature Analysis — Irrigation strong
Target: drying time [h] per depth (1,161 events)
Features: green (combined) / greenAttached / greenDetached / buildings / sealedSurface (2.5 / 5 / 7.5 m rings, 0→5 m + 0→7.5 m aggregates, crown-diameter)



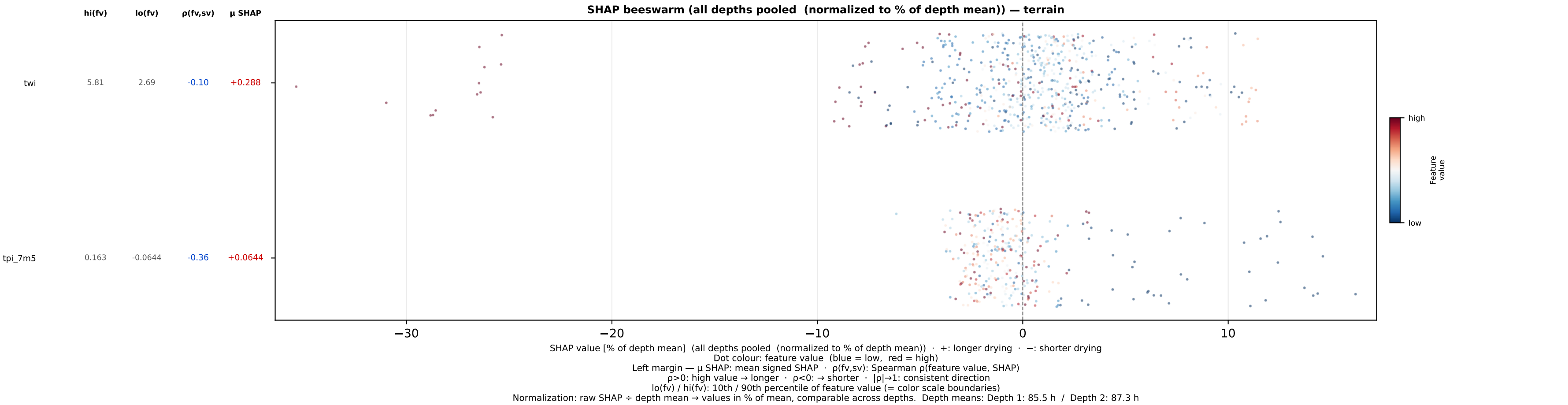
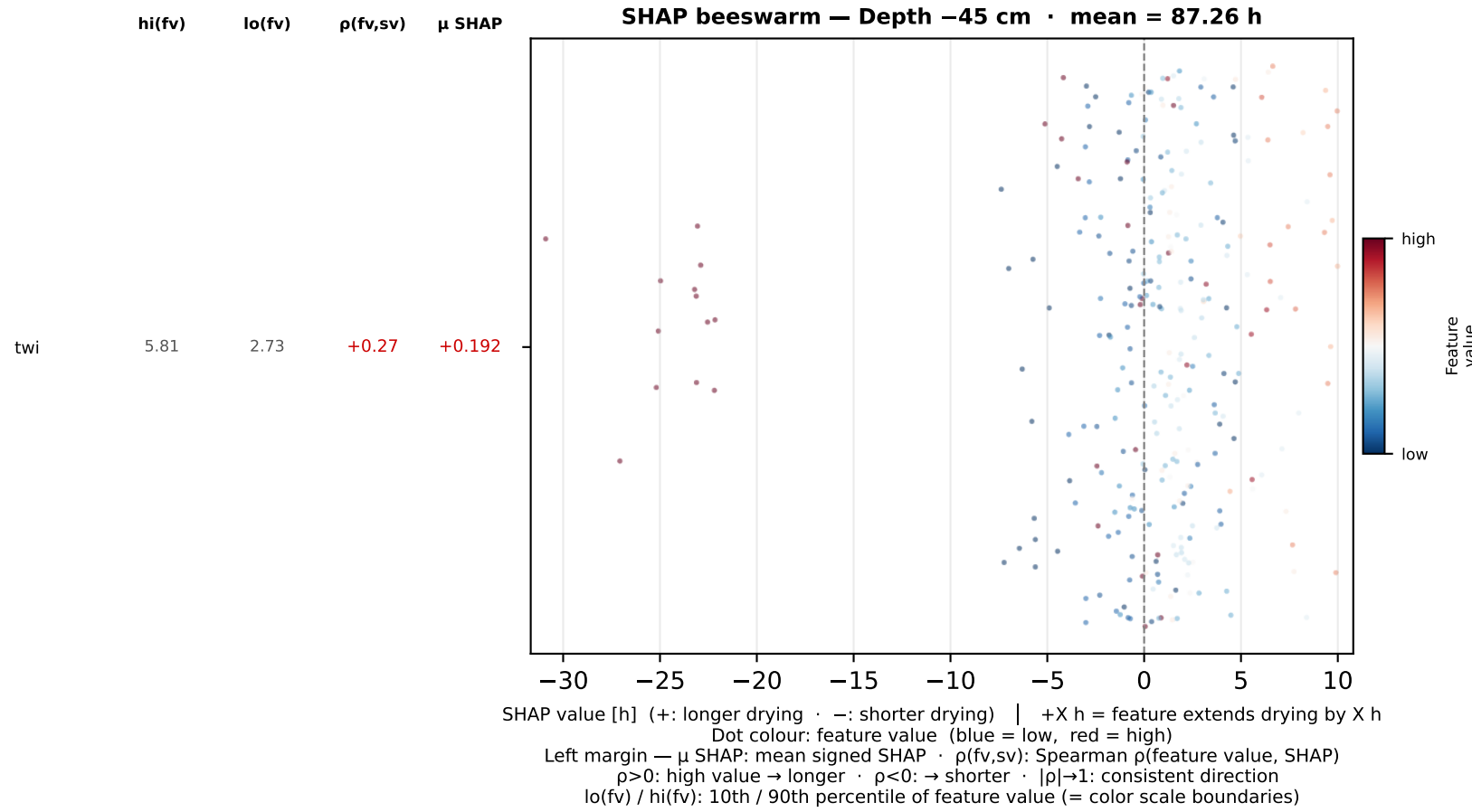
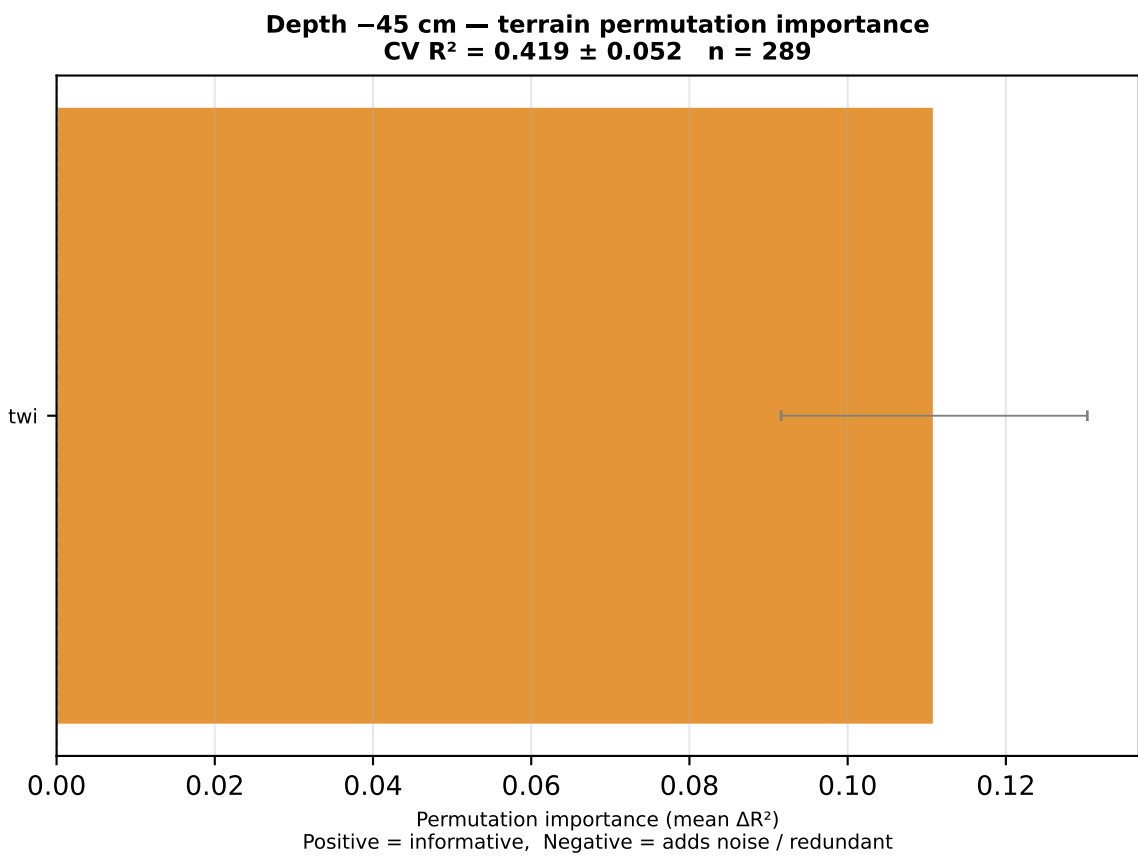
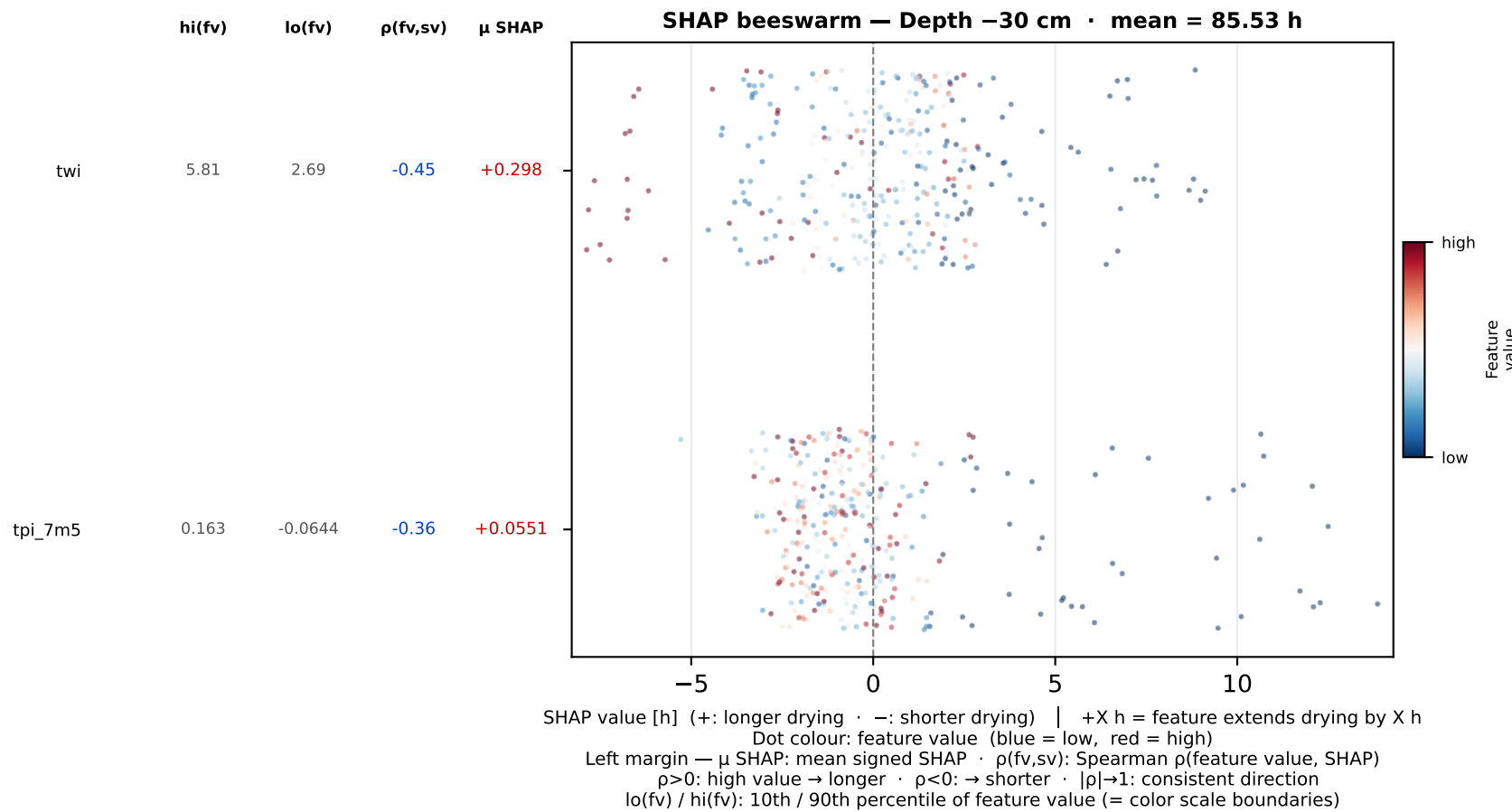
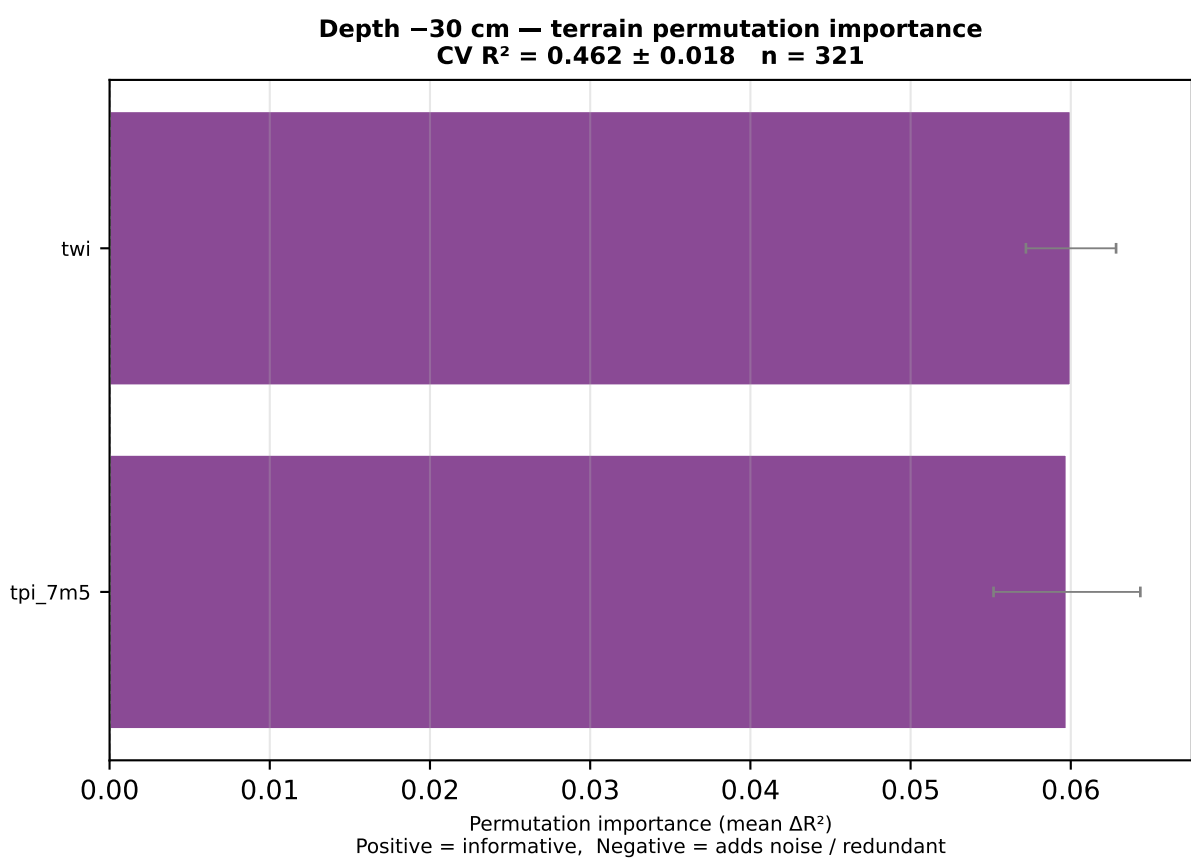
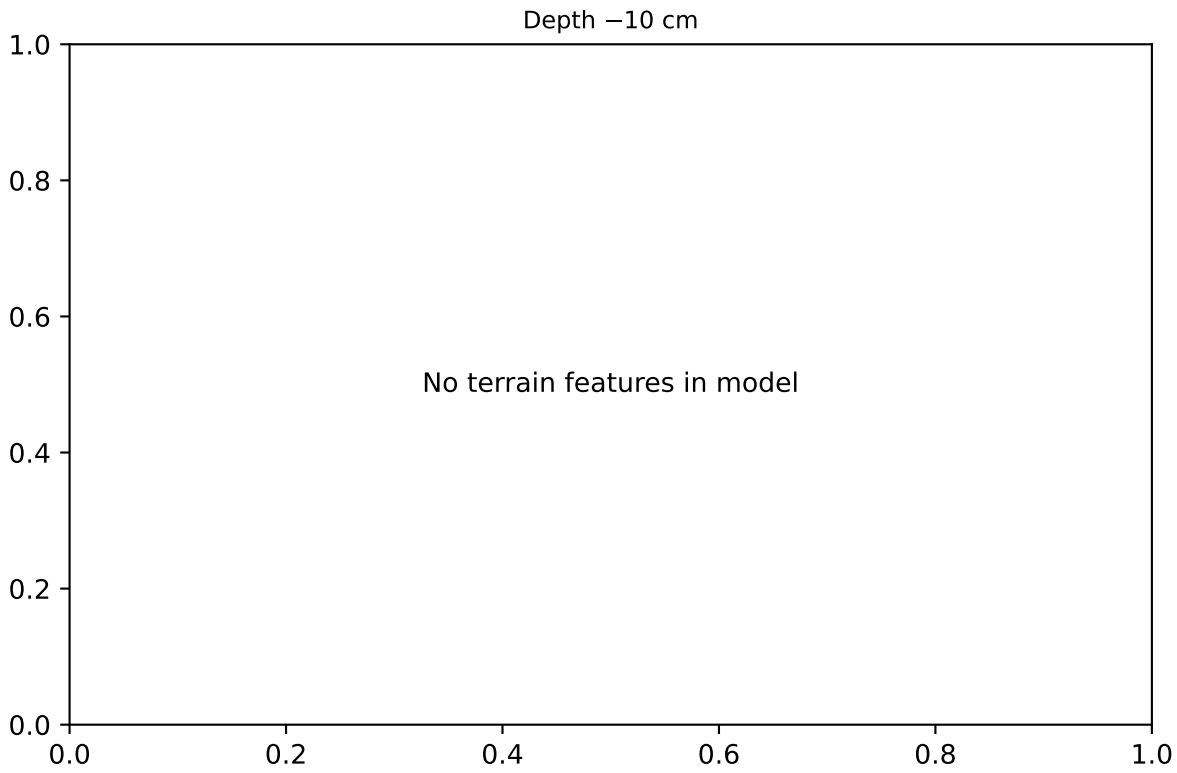
hi(fv)	lo(fv)	$\rho(fv,sv)$	μ SHAP
172	9.89	+0.38	+0.0628



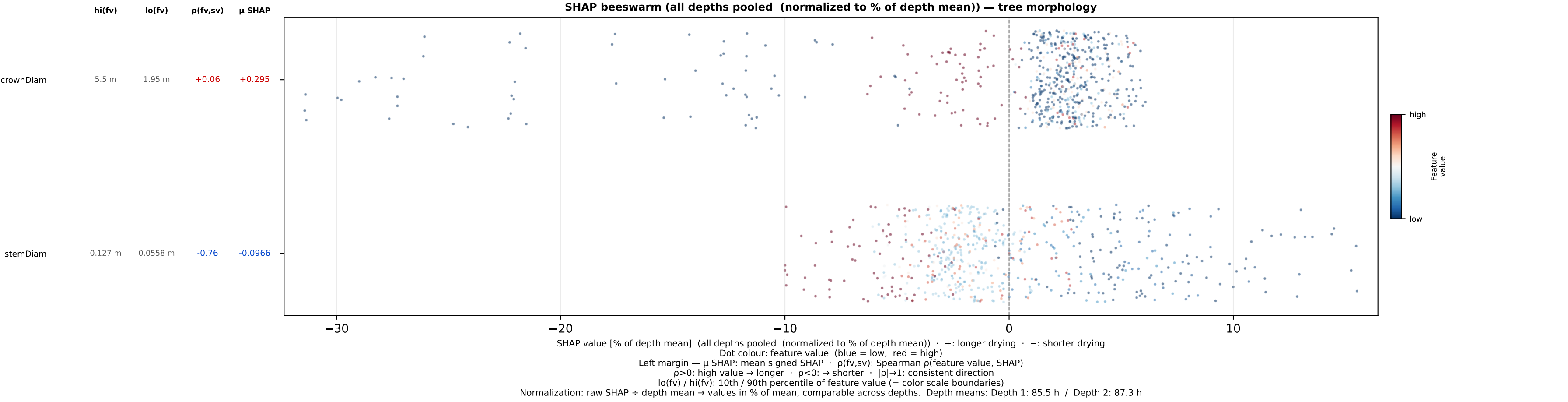
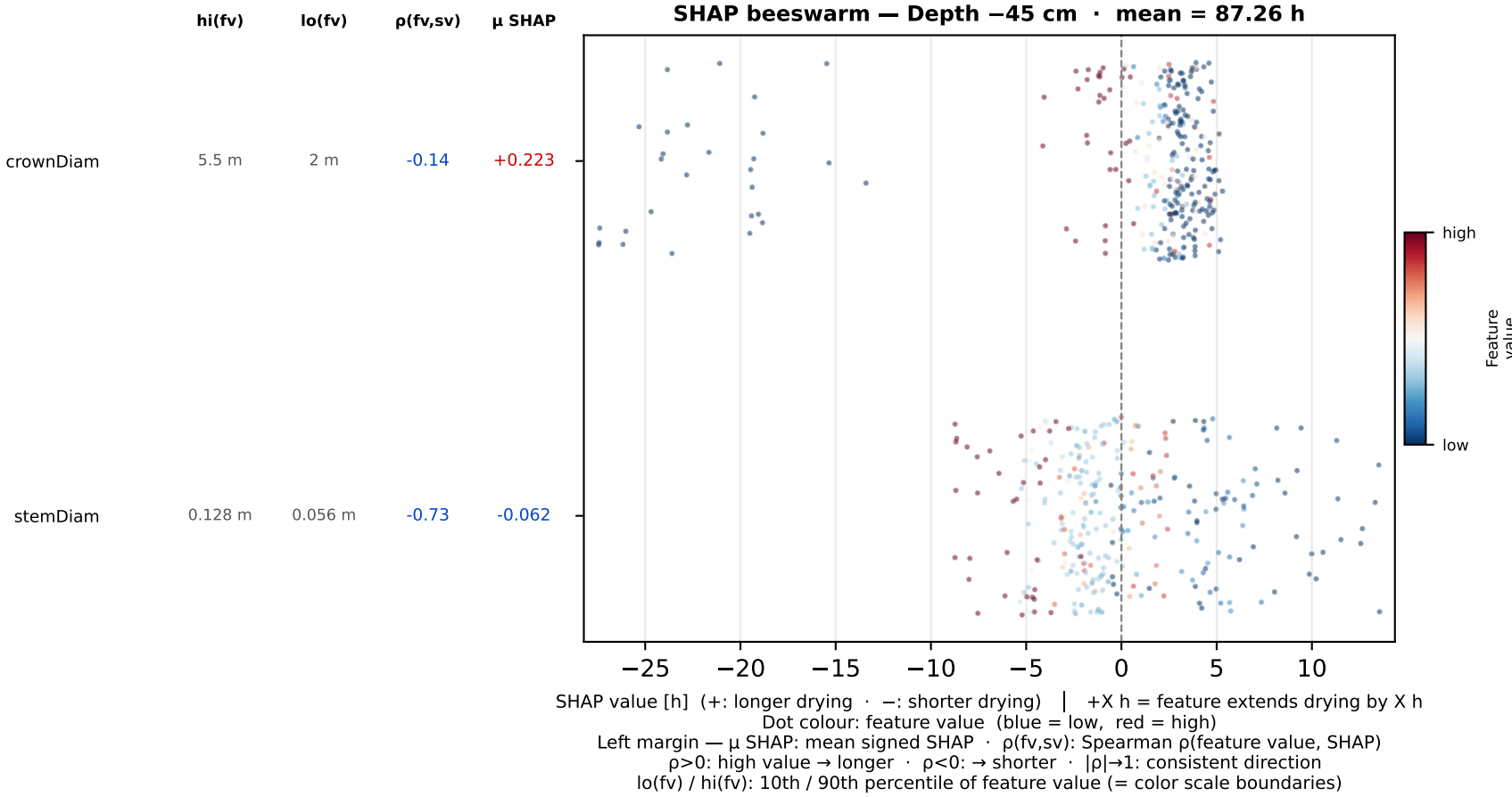
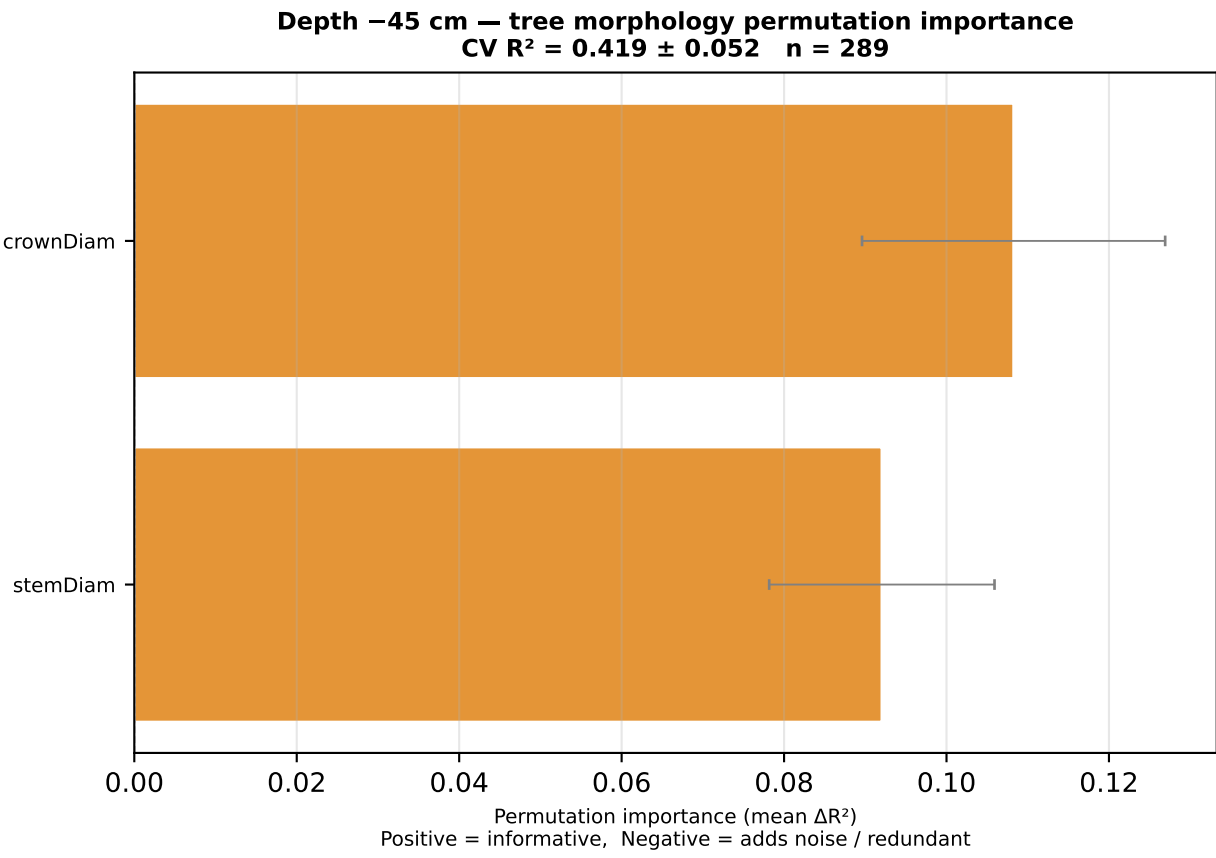
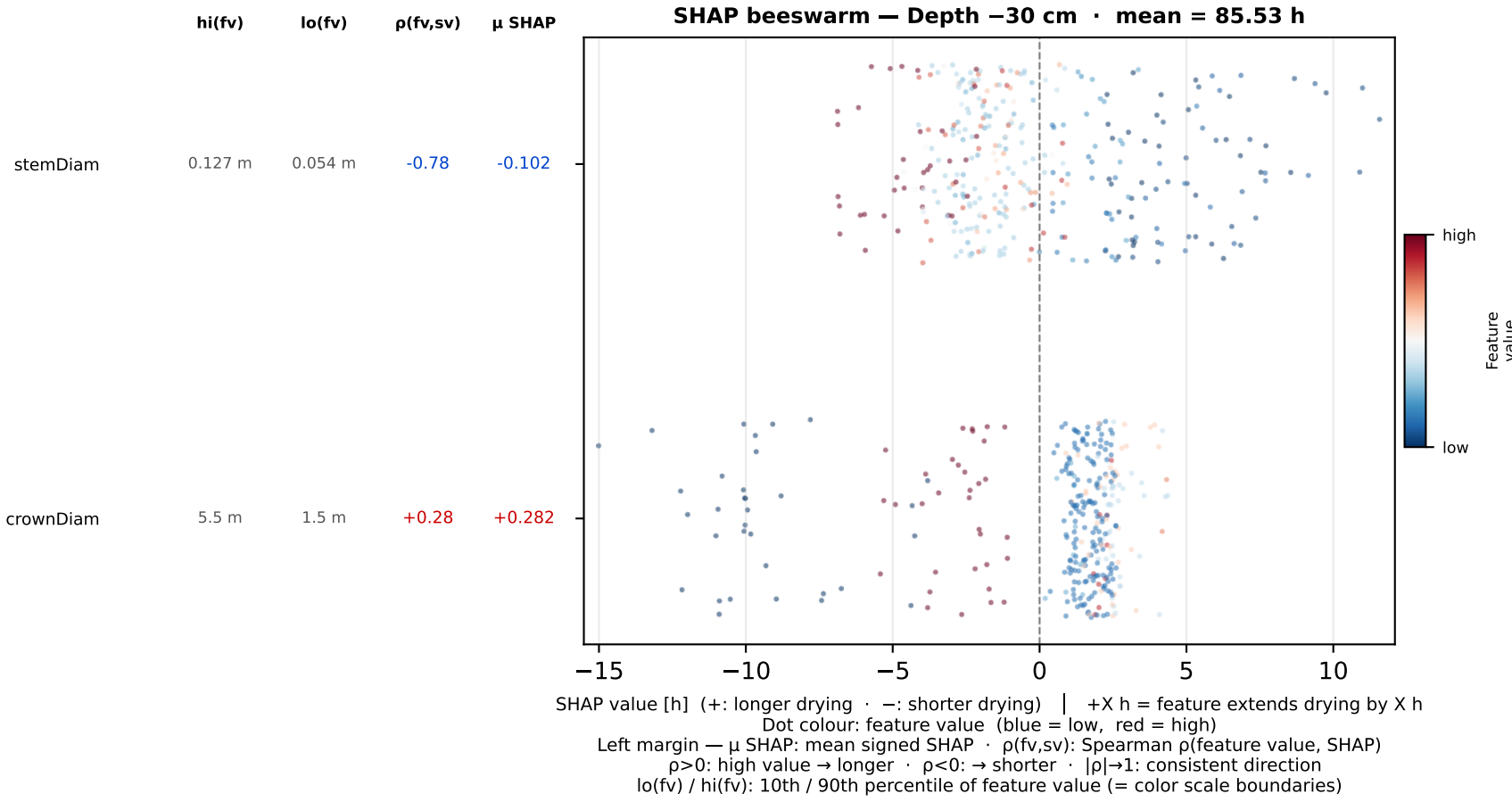
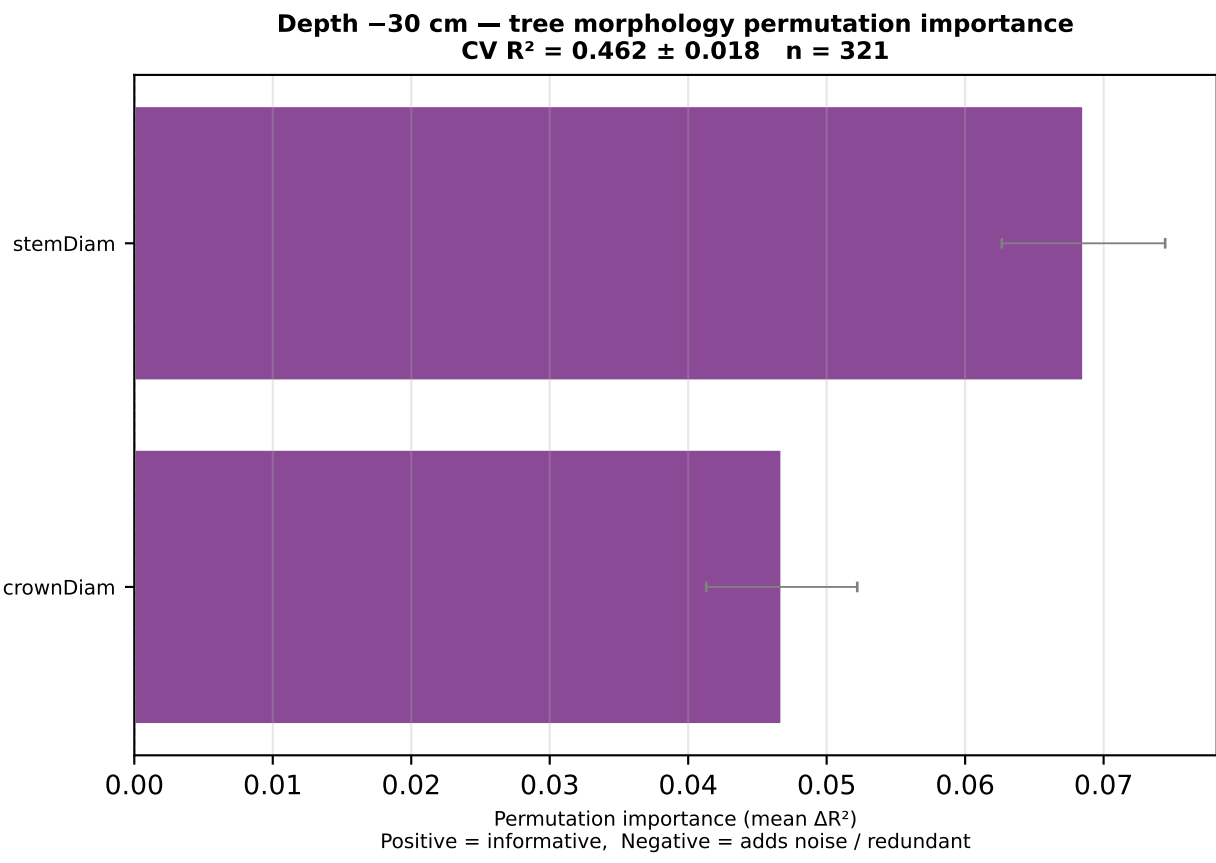
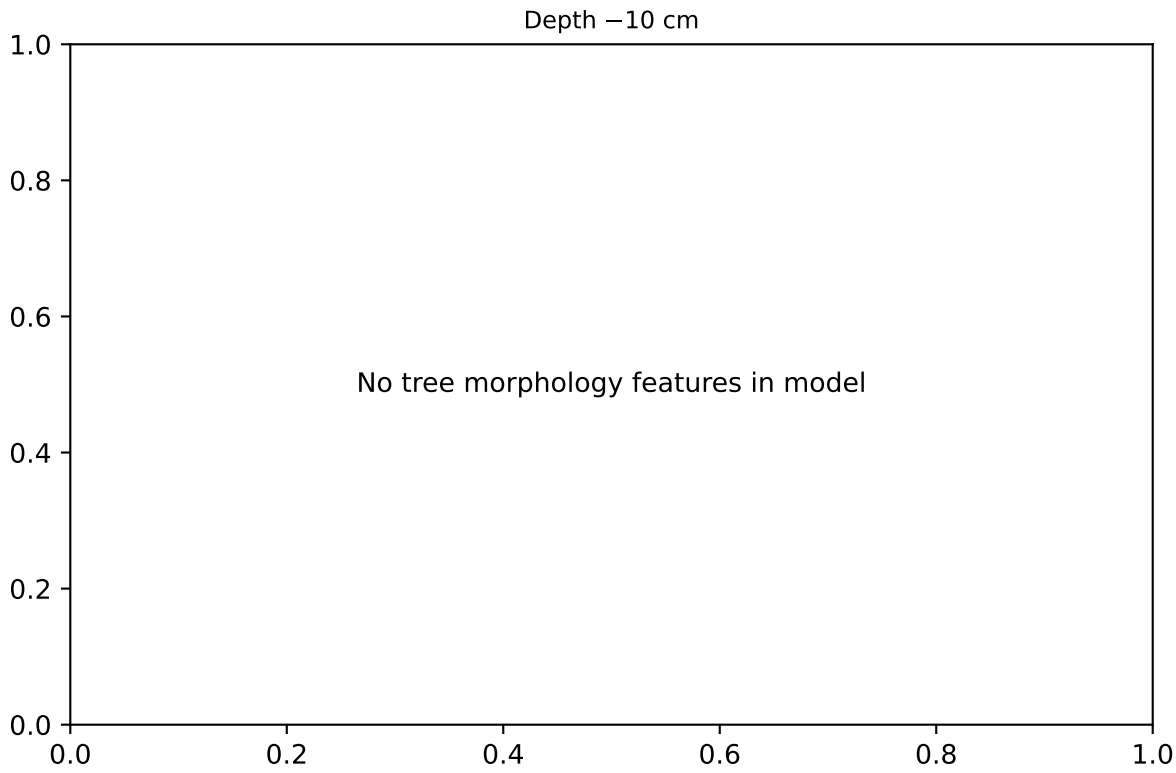
hi(fv)	lo(fv)	$\rho(fv,sv)$	μ SHAP
172	9.79	+0.08	-0.414

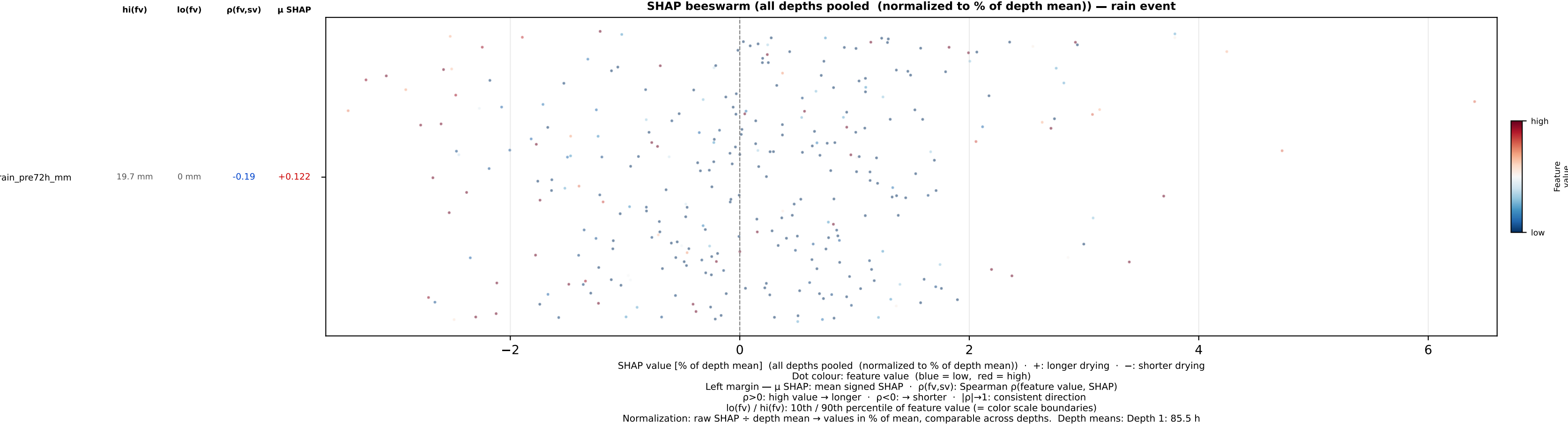
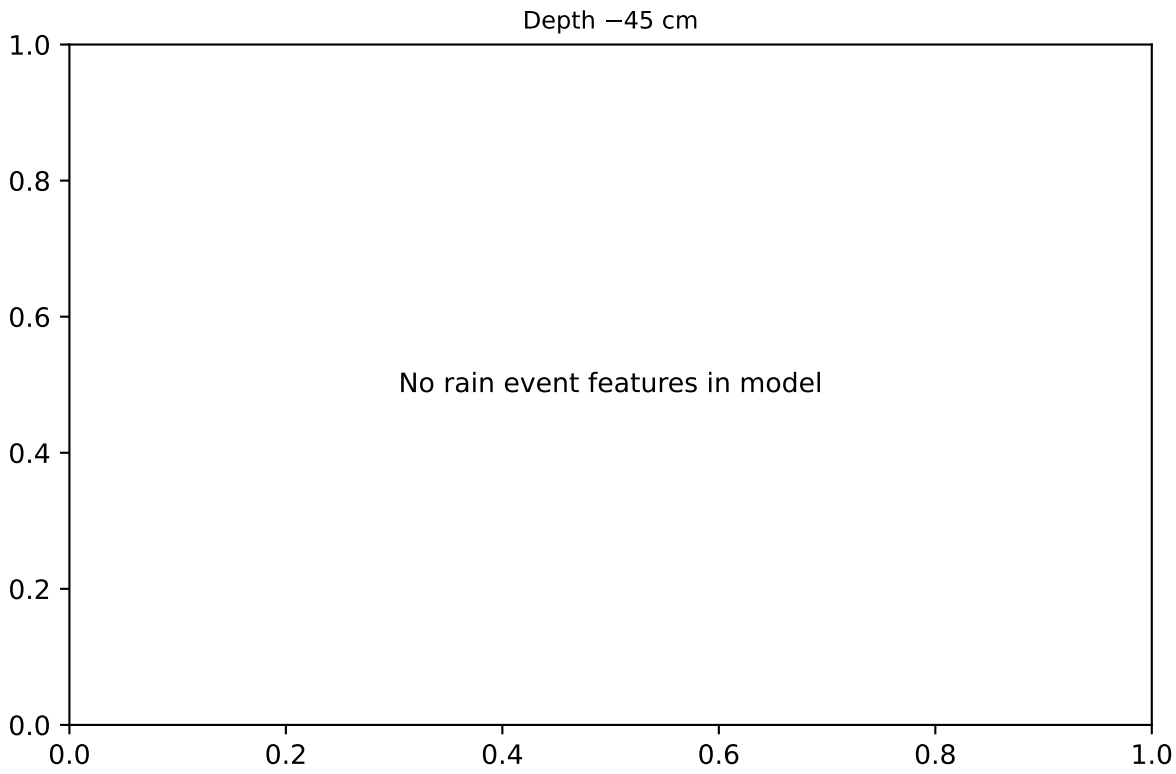
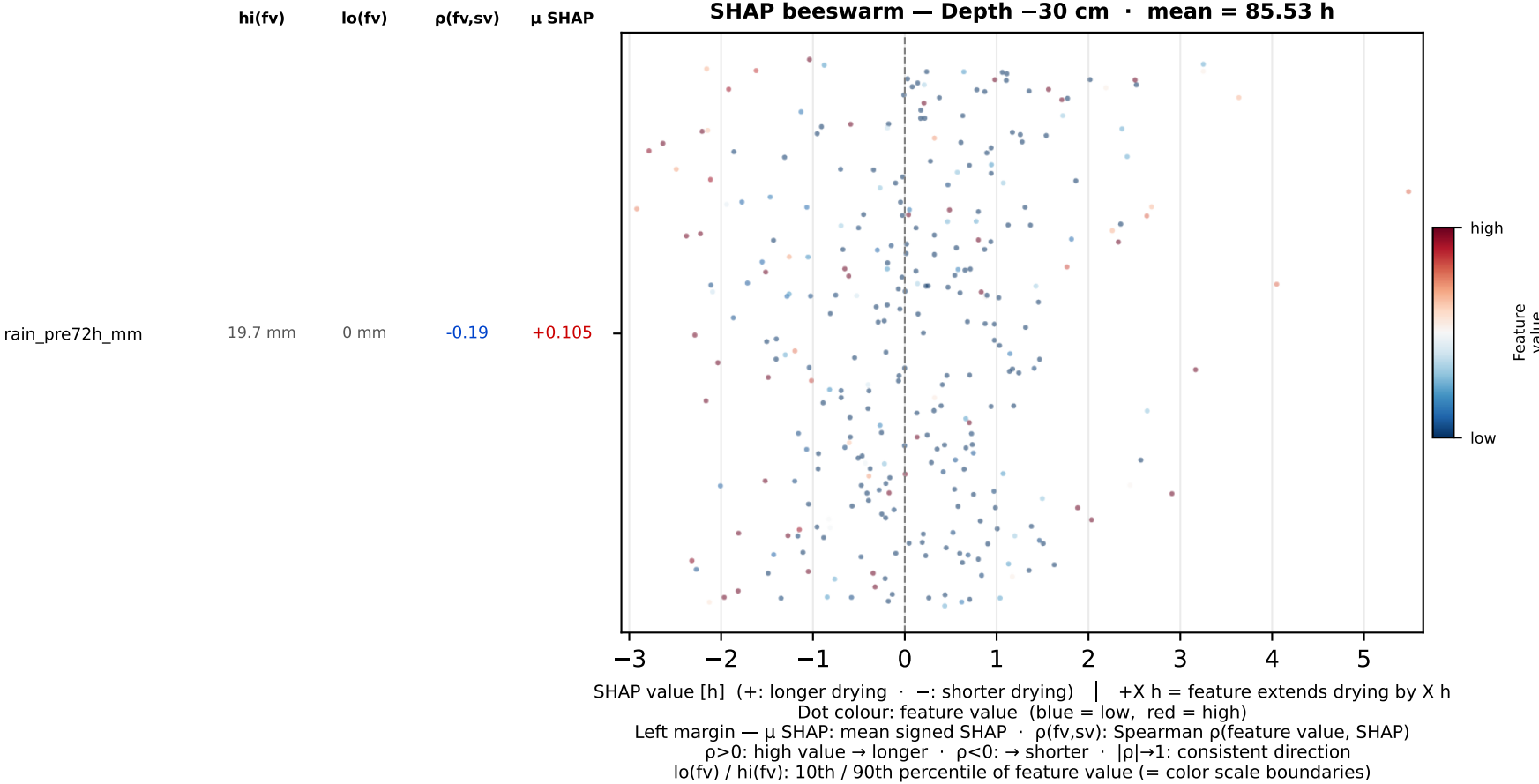
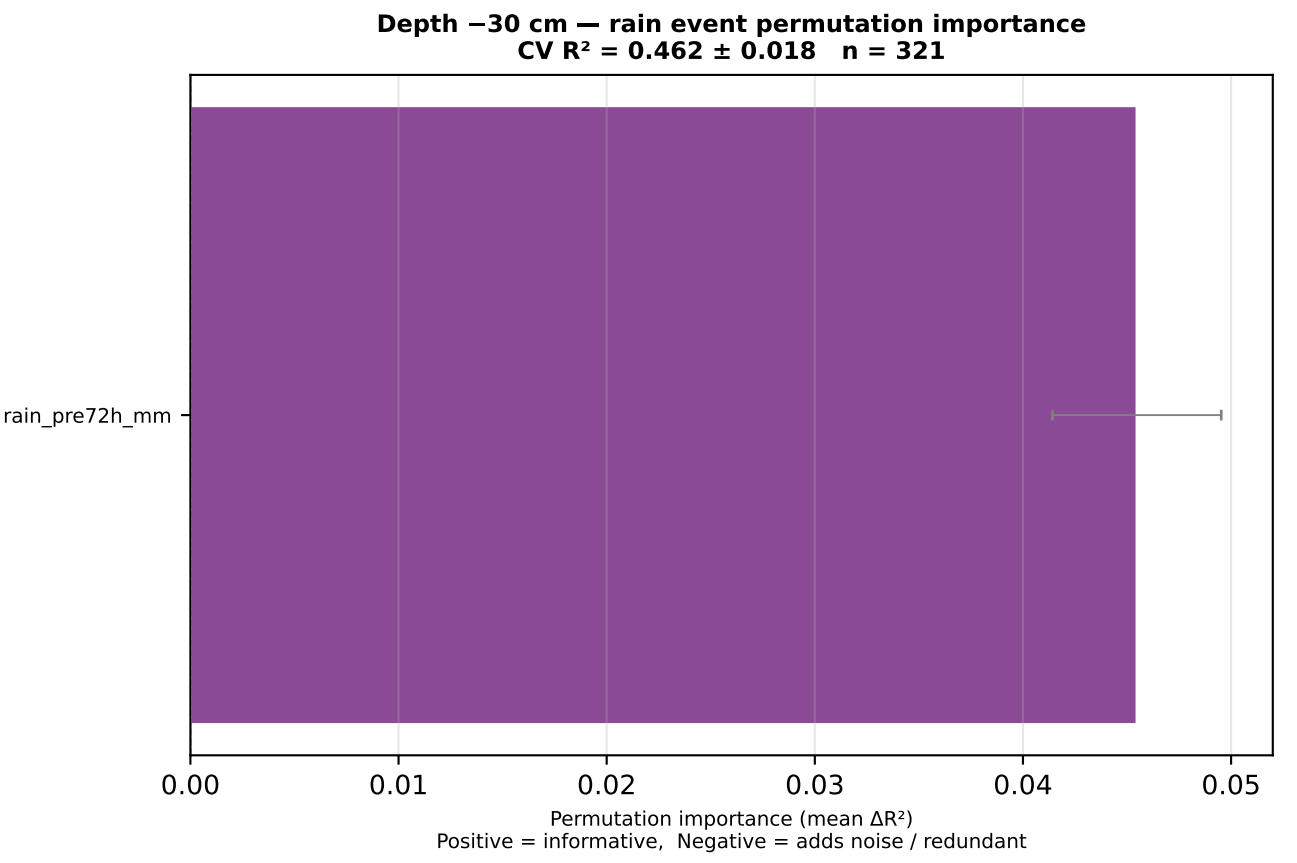
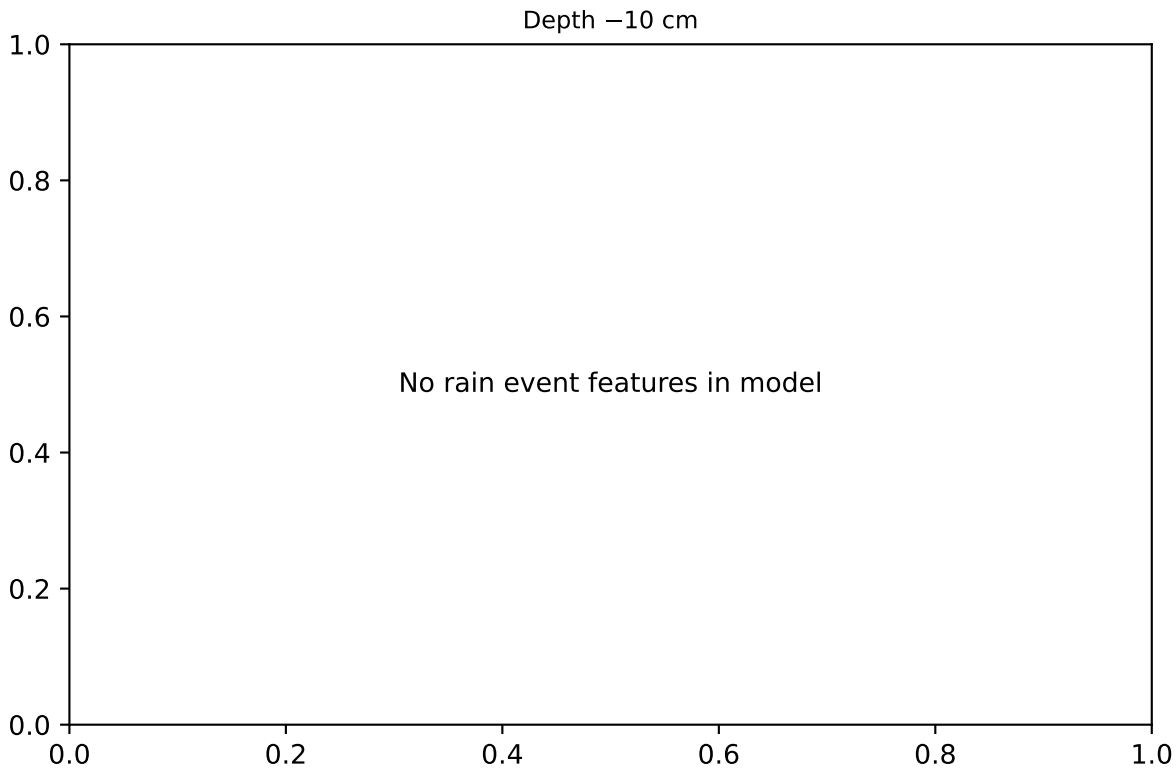


Terrain Feature Analysis — Irrigation strong
Target: drying time [h] per depth (1,161 events)
Features: TWI · TPI (2.5/5/7.5 m + crown-diam) · slope · amsl · flow accumulation · SVF · depth to groundwater



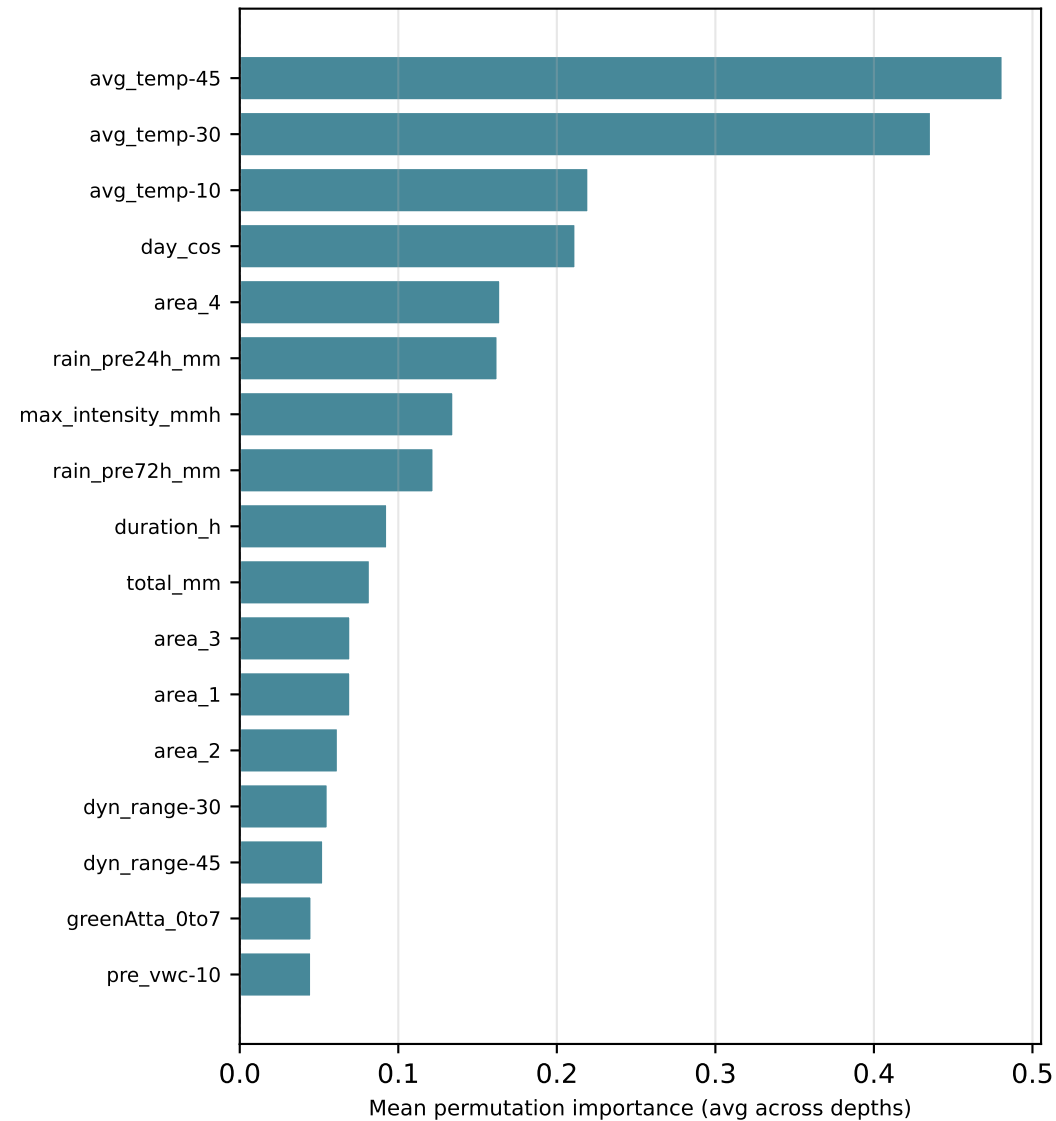
Tree Morphology Feature Analysis — Irrigation strong
Target: drying time [h] per depth (1,161 events)
Features: crown diameter · height · stem diameter · vitality



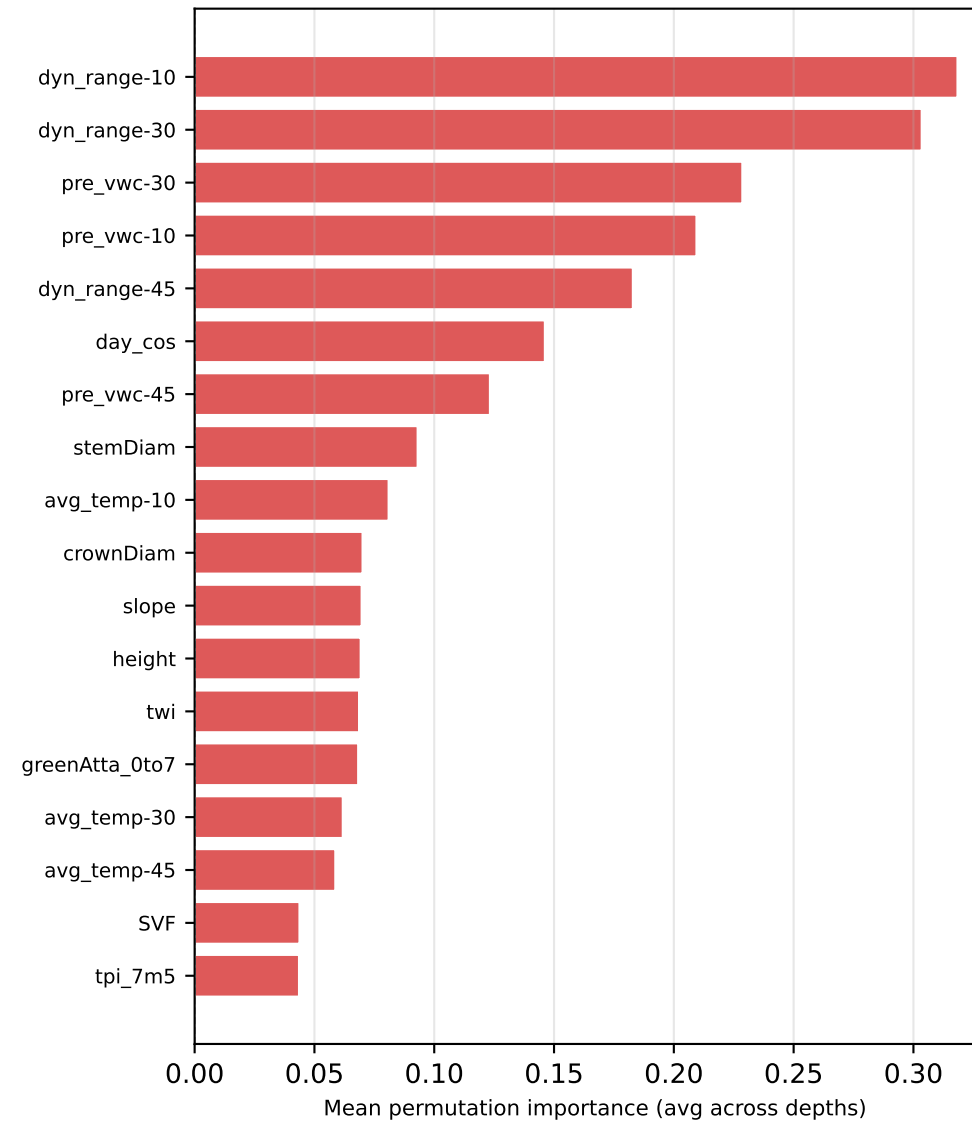


Feature Importance Summary — mean permutation importance across depths

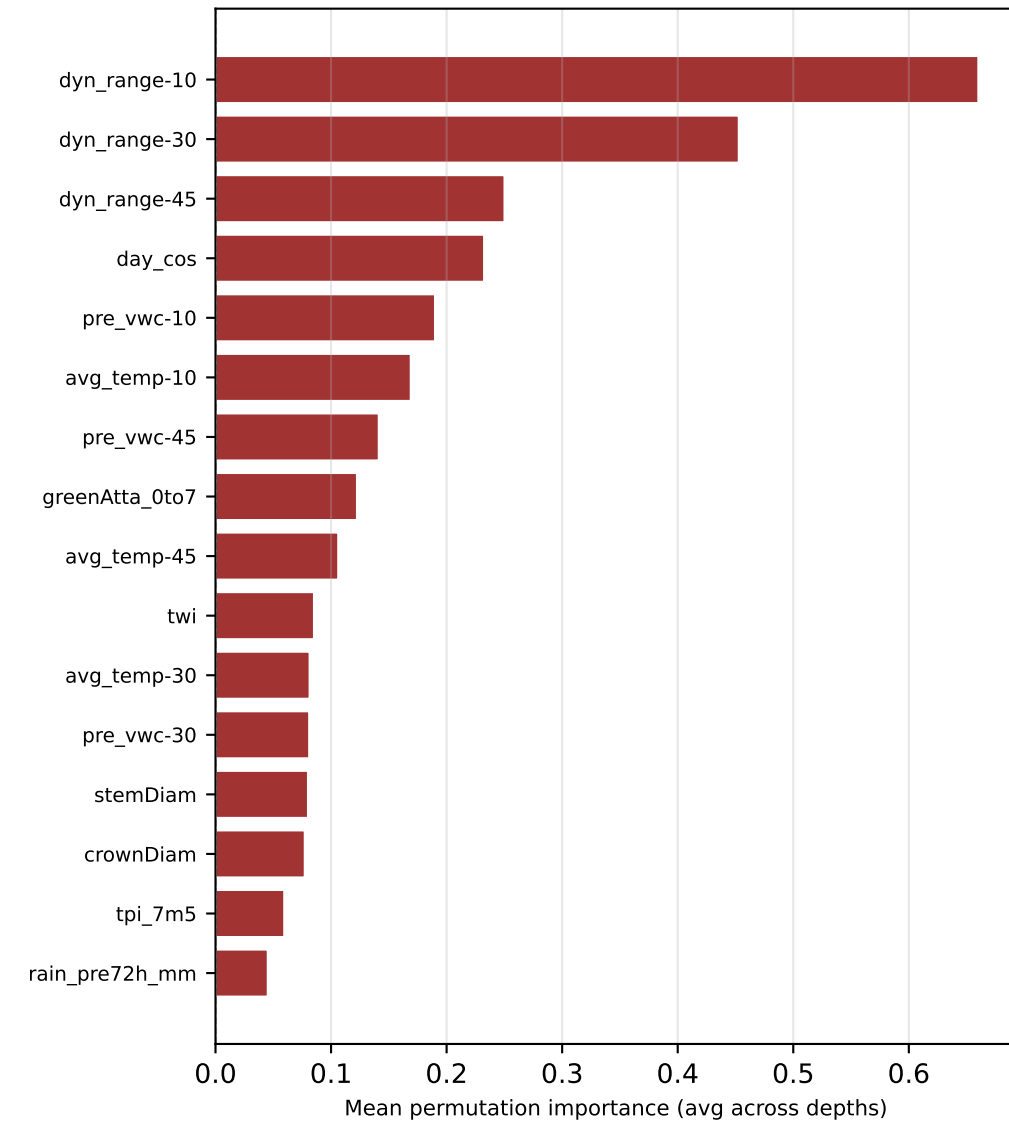
BSC 00XX + Huff Q3/Q4 (back-loaded rain)
avg $R^2 = 0.437$



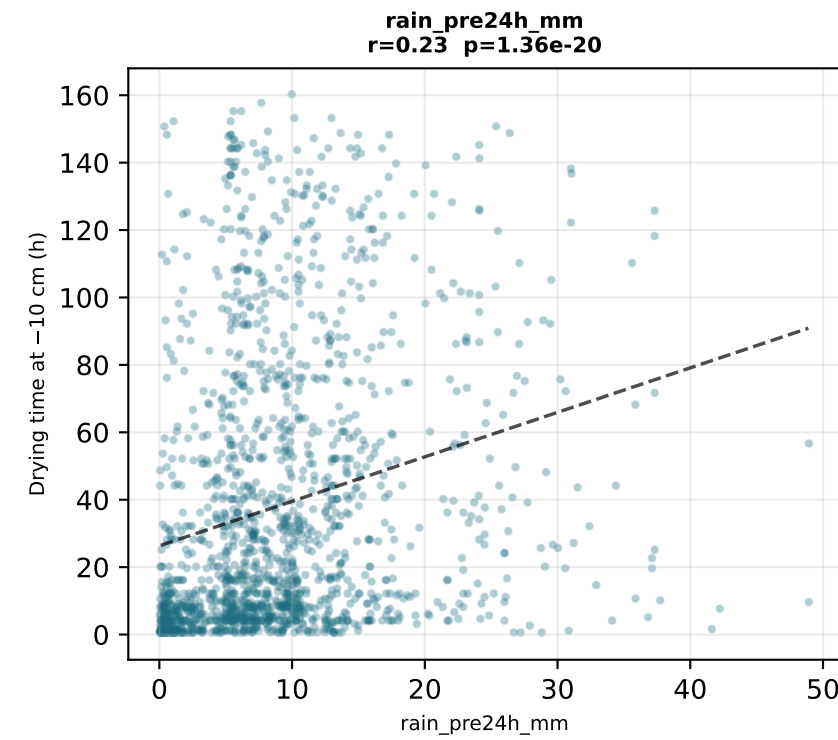
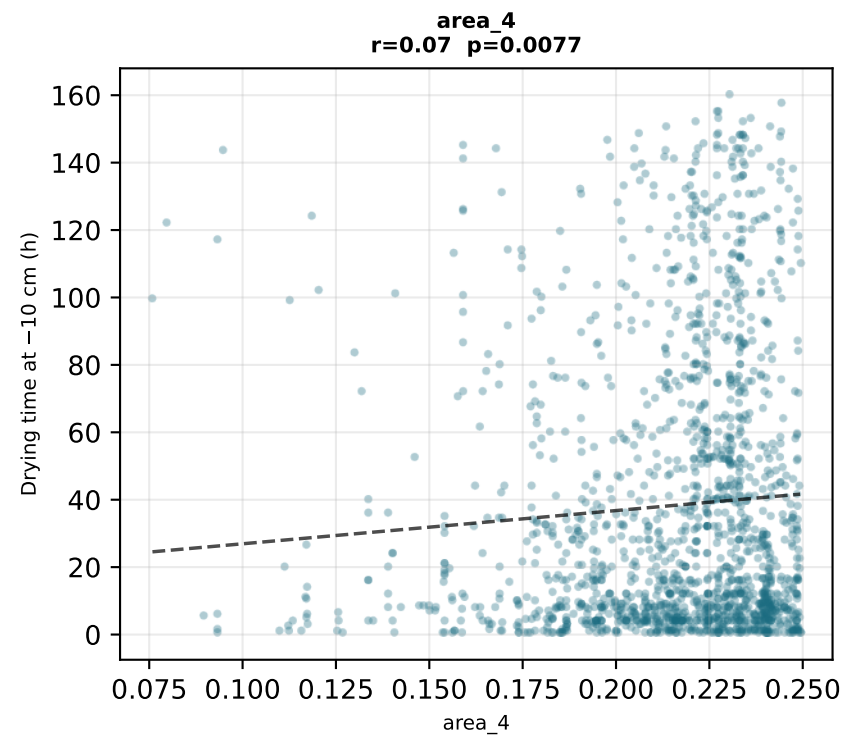
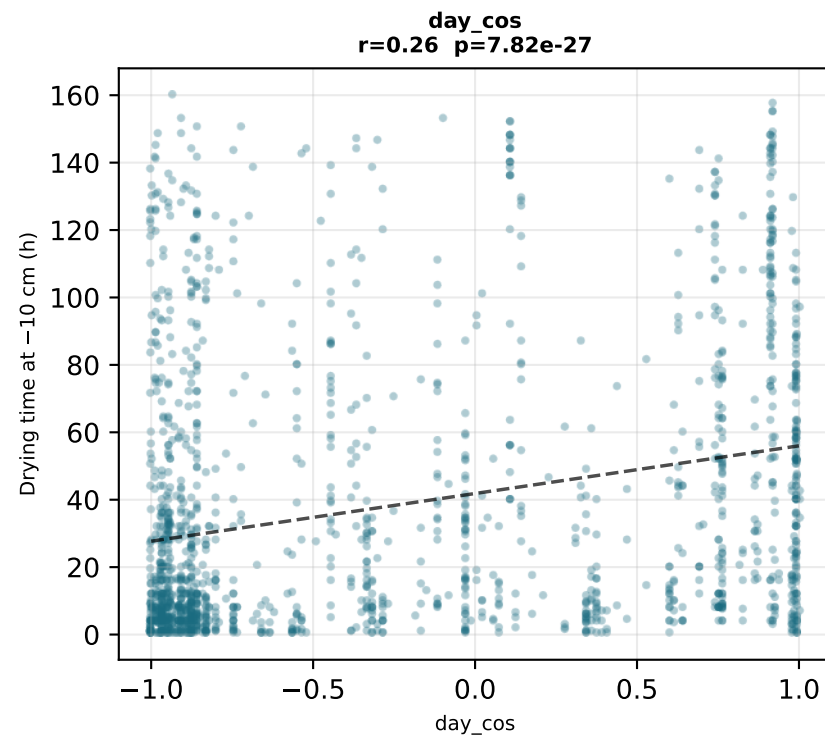
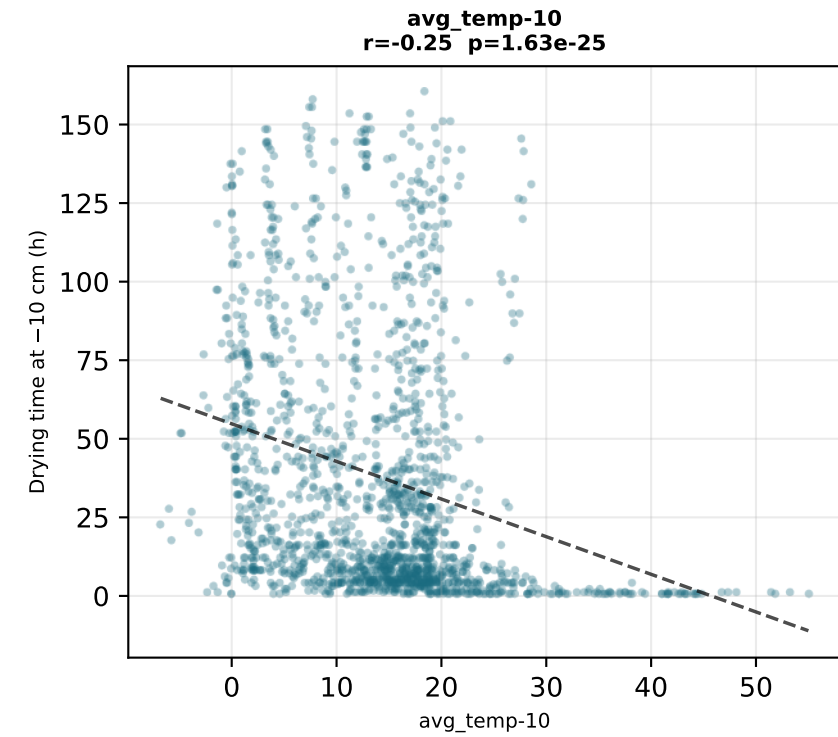
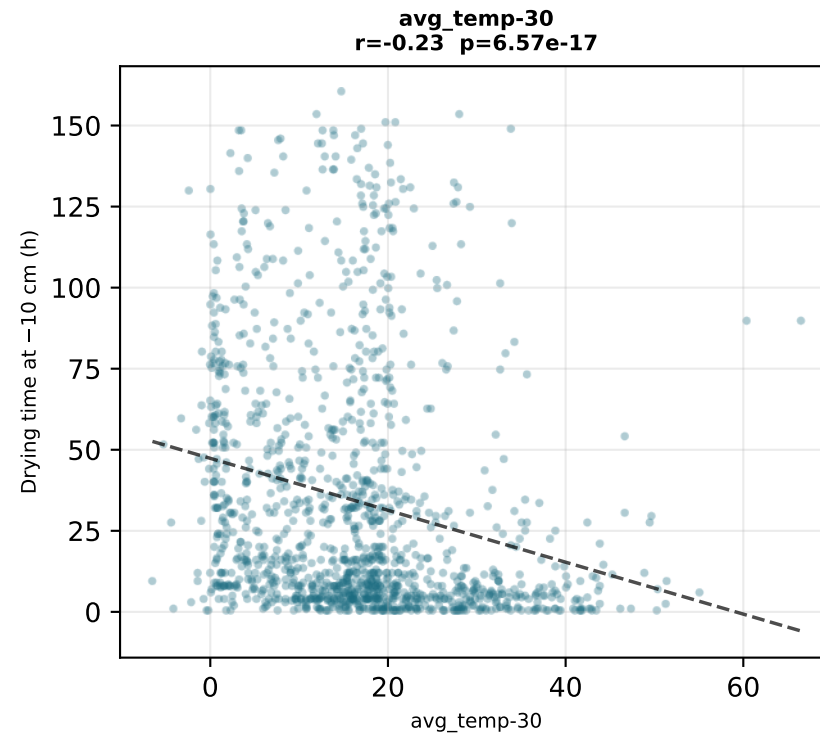
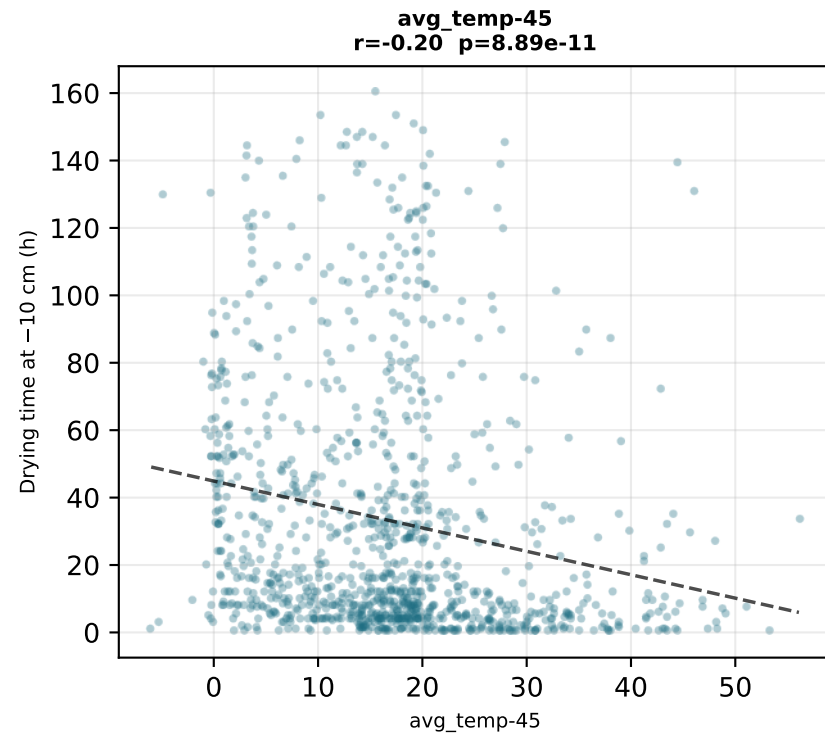
Irrigation
avg $R^2 = 0.529$



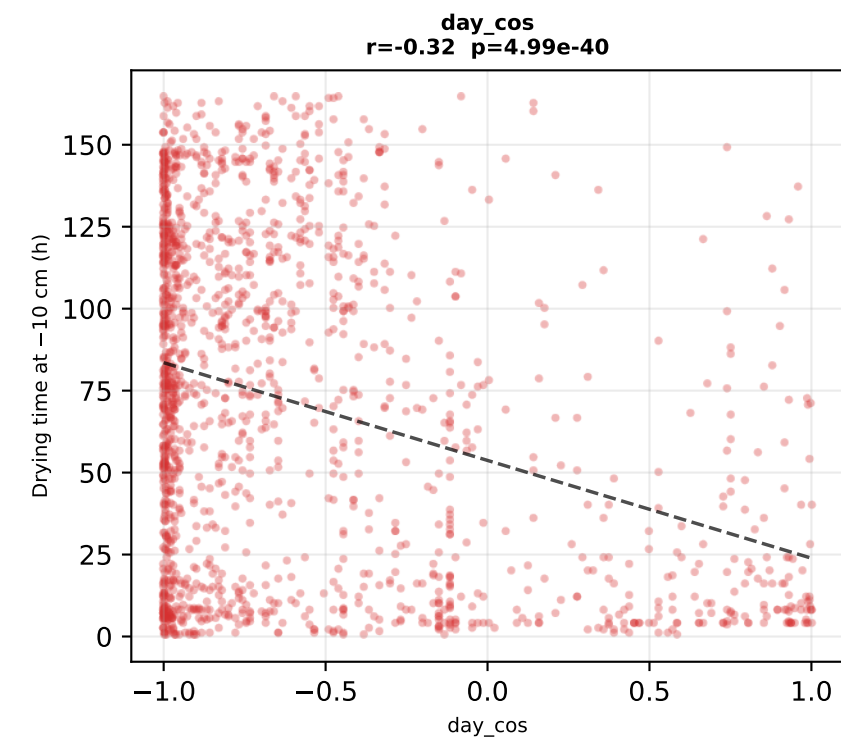
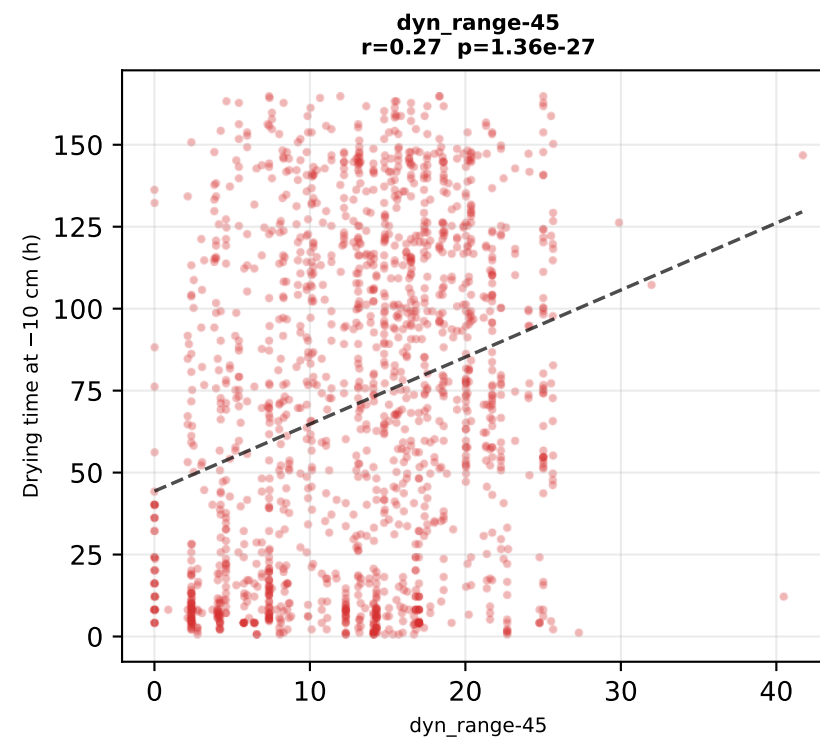
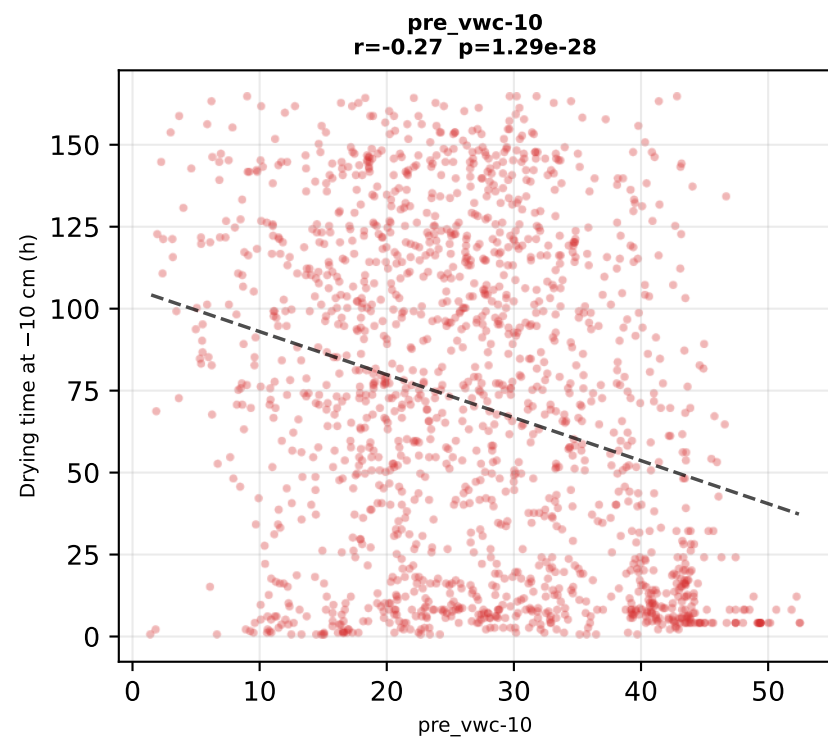
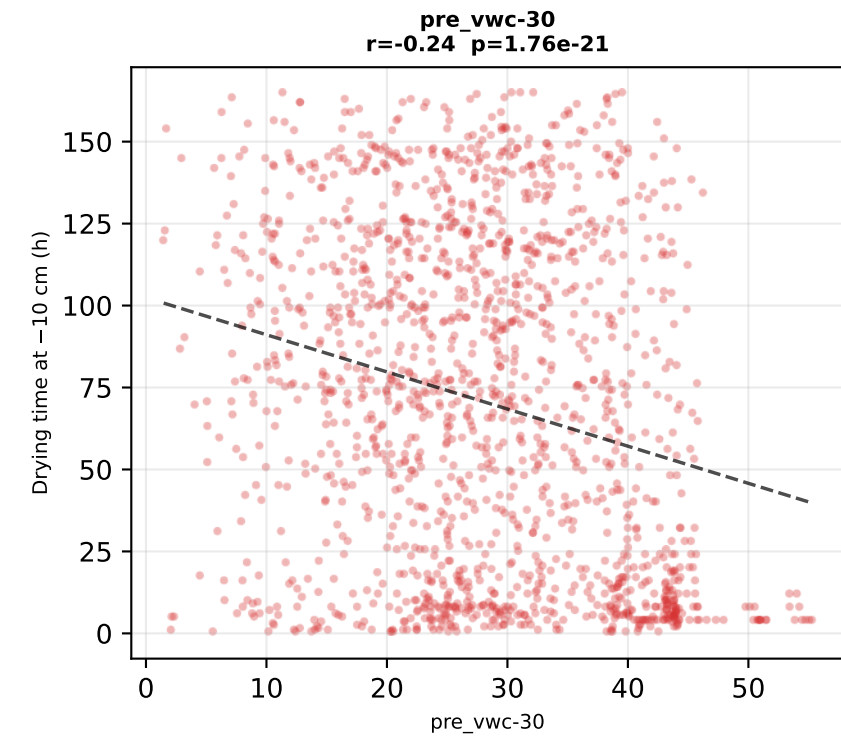
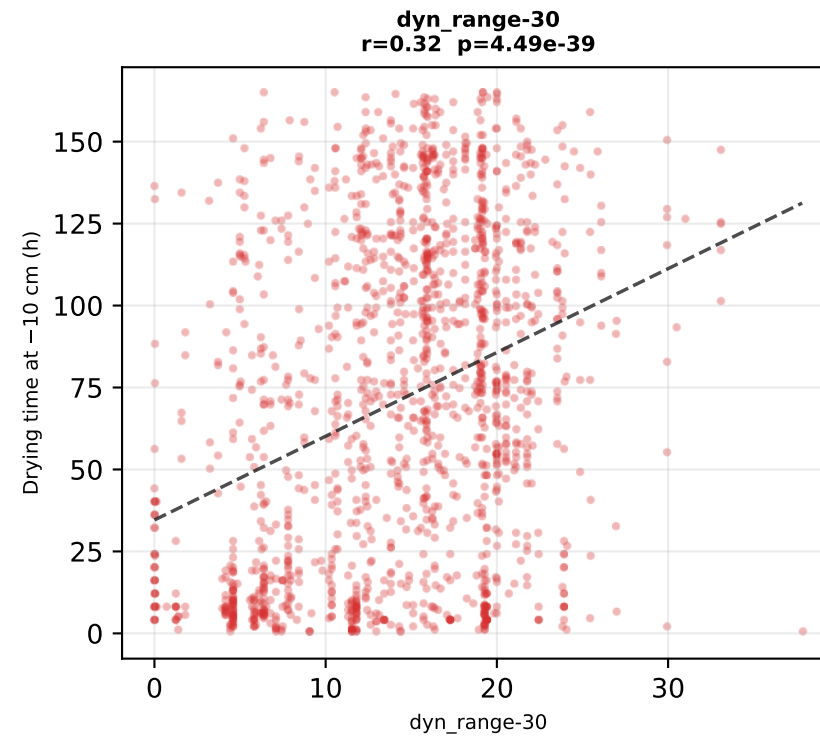
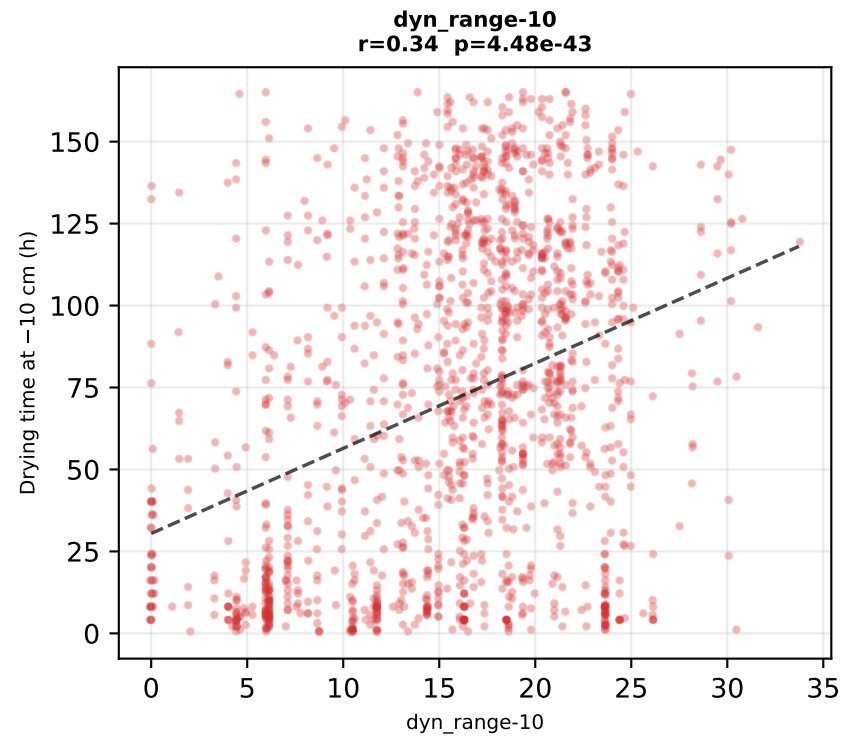
Irrigation strong
avg $R^2 = 0.446$



Top feature scatter plots — BSC 00XX + Huff Q3/Q4 (back-loaded rain)
Target: Drying time at -10 cm (n = 1,636)



Top feature scatter plots — Irrigation
Target: Drying time at −10 cm (n = 1,579)



Top feature scatter plots — Irrigation strong
Target: Drying time at −10 cm (n = 502)

